ENT ATLAS

CLINICAL

# Larynx & airway

14 articles · imaging key facts · November 2026

# Contents


01 Croup and epiglottitis

02 Foreign bodies of the airway

03 Vocal fold paralysis

04 Benign vocal fold lesions and hoarseness

05 Cancer of the larynx

06 Applied anatomy and physiology of the larynx

07 Laryngotracheal trauma and stenosis

08 Chronic laryngitis, reflux and specific infections

09 Congenital lesions of the larynx

10 Hoarseness, voice and speech disorders

11 Tracheostomy and airway endoscopy

12 Lasers, radiofrequency, cryosurgery and hyperbaric oxygen

13 The hoarse child and the child who cannot swallow

14 Anaesthesia and the shared airway

· Imaging for this region

CLINICAL · LARYNX & AIRWAY

# Croup and epiglottitis

Acute inflammation of the larynx ranges from the hoarse voice that follows a cold to the child sitting bolt upright, drooling and silent, whose swollen epiglottis may close the airway within the hour. The difference between croup and epiglottitis is one of the oldest lessons in paediatrics and still one of the most important: the barking child can usually be soothed; the drooling child must not be upset, examined or sent to radiology without someone who can secure the airway.

**AT A GLANCE**

- Croup barks; epiglottitis drools.
- Dexamethasone for croup.
- Don't examine the throat in epiglottitis.
- Secure the airway in theatre.

## 1 CHAPTER ONE Acute laryngitis

The commonest.

### 1.1 Causes

- **Infective:** usually viral after a cold (parainfluenza, influenza, rhinovirus), followed by bacterial invasion (pneumococcus, *H. influenzae*, *Moraxella*, streptococci, staphylococci); with measles, chickenpox and whooping cough; candida in inhaled steroid users.
- **Non-infective:** vocal abuse (shouting, cheering), allergy, inhaled irritants, thermal and chemical burns, reflux, and intubation.

### 1.2 Clinical features

Abrupt hoarseness, sometimes complete loss of voice; pain in the throat after talking; a dry, irritating cough, worse at night; and the features of a cold. Early the epiglottis, aryepiglottic folds, arytenoids and false folds are red and swollen while the true folds stay white, standing out against the redness; later the folds become

red and swollen, with sticky mucus between them and subglottic swelling. Vocal abuse may cause submucosal haemorrhage in a fold. Breathlessness is *not* a feature of simple laryngitis and suggests another diagnosis.

### 1.3 Treatment

- **Voice rest:** the most important measure; forcing the voice delays recovery and risks haemorrhage and nodules.
- No smoking or alcohol; hydration; steam inhalation.
- Cough suppressants, mucolytics and analgesics.
- Antibiotics only for secondary bacterial infection with fever or purulent sputum.
- Proton pump inhibitors where reflux is suspected.
- Steroids after burns, or for professional voice users who must perform.

Most settle in one to two weeks; hoarseness persisting beyond three weeks needs laryngoscopy.

### 1.4 Acute membranous laryngitis

Pyogenic membrane on the larynx, often spreading from the pharynx; it must be distinguished from diphtheria by swab and culture.

## 2 CHAPTER TWO Acute epiglottitis

Supraglottitis.

### 2.1 Definition and cause

Acute inflammation of the supraglottis (epiglottis, aryepiglottic folds and arytenoids), whose loose mucosa swells until it obstructs the airway; the swollen epiglottis acts like a ball valve. Classically in children of two to seven years from *H. influenzae* type b; since Hib vaccination it is rare in children and now seen mostly in adults, with streptococci, staphylococci and other organisms.

## 2.2 Clinical features

- **Abrupt onset,** rapid progression over hours, high fever and toxicity (septicaemia).
- **Children:** stridor and difficulty breathing are the presenting features; the child sits upright, leaning forward with the chin out (tripod position), drooling because swallowing hurts, with a muffled voice and usually no cough.
- **Adults:** severe sore throat and painful swallowing out of proportion to a normal-looking oropharynx, a muffled "hot-potato" voice, and tenderness over the hyoid.

**DO NOT UPSET THE CHILD**

No tongue depressor, no throat examination, no cannula, no lying down, no unaccompanied trip to radiology. Keep the child with the parent, give oxygen without a struggle, and take them to theatre where an experienced anaesthetist and an ENT surgeon ready for tracheostomy can secure the airway under gaseous induction.

## 2.3 Investigations

- **Lateral soft tissue radiograph** of the neck (only if the patient is stable and accompanied): a swollen epiglottis, the *thumb sign*, with ballooned hypopharynx.
- **Flexible endoscopy** in adults is usually safe and shows a cherry-red, swollen epiglottis and arytenoids.
- Blood cultures (more often positive than surface swabs) and epiglottic swabs after the airway is secure; CT for a suspected epiglottic abscess.

## 2.4 Treatment

1. **Airway:** intubation (or tracheostomy if intubation fails) in theatre for children and for adults with stridor or rapid progression; selected adults presenting later without stridor can be watched closely in high-dependency care.
2. **Intravenous antibiotics** at once: a third-generation cephalosporin (ceftriaxone or cefotaxime), or ampicillin in sensitive strains.
3. **Steroids** to reduce oedema; humidified oxygen; intravenous fluids.
4. Rifampicin prophylaxis for household contacts of Hib cases, and vaccination.

Intubation is usually needed for 24 to 72 hours, until the epiglottis has shrunk.

# 3 CHAPTER THREE Croup

Acute laryngotracheobronchitis.

## 3.1 Cause and pathology

The commonest cause of acute stridor in children, between six months and three years, more in boys and in autumn and winter. Usually *parainfluenza* viruses (types 1 and 2), also RSV, influenza and adenovirus, with secondary bacterial infection. The loose areolar tissue of the subglottis swells inside the rigid cricoid ring; thick secretions and crusts add to the obstruction.

## 3.2 Clinical features

A day or two of a cold, then hoarseness, a harsh *barking (seal-like) cough* and inspiratory or biphasic stridor, worse at night; fever is usually low or moderate and the child is not toxic. Increasing obstruction shows as suprasternal, intercostal and subcostal recession, tachypnoea and agitation; restlessness, a rising pulse, pallor and cyanosis and finally exhaustion or drowsiness are late, ominous signs.

CROUP AND EPIGLOTTITIS IN CHILDREN

| FEATURE | CROUP (LARYNGOTRACHEOBRONCHITIS) | ACUTE EPIGLOTTITIS |
|---|---|---|
| Organism | Parainfluenza virus | *H. influenzae* type b |
| Age | Six months to three years | Two to seven years |
| Site | Subglottis | Supraglottis |
| Prodrome | Cold for a day or two | None |
| Onset | Gradual | Sudden |
| Fever | Low or moderate | High |
| General state | Not toxic | Toxic |
| Cough | Barking | Usually absent |
| Voice | Hoarse | Muffled |

| FEATURE | CROUP (LARYNGOTRACHEOBRONCHITIS) | ACUTE EPIGLOTTITIS |
|---|---|---|
| Drooling and painful swallowing | Absent | Present |
| Posture | Any | Upright, leaning forward |
| Radiograph | Steeple sign on an AP view | Thumb sign on a lateral view |
| Treatment | Steroids, nebulised adrenaline, humidified oxygen | Airway in theatre, intravenous antibiotics |

## 3.3 Treatment

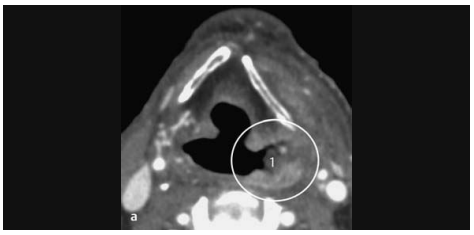
- **Mild:** a single dose of oral dexamethasone (or nebulised budesonide) and observation; most go home.
- **Moderate to severe:** humidified oxygen, steroids and *nebulised adrenaline*, which works within minutes but wears off in about two hours, so observe for rebound; admit.
- **Severe or failing:** intubation in theatre with a tube smaller than usual for age; tracheostomy rarely now.
- Antibiotics only for suspected secondary bacterial infection.

**RECURRENT CROUP**

A child with repeated attacks, attacks under six months, or croup that will not settle needs microlaryngoscopy and bronchoscopy when well, to look for subglottic stenosis, a subglottic cyst or haemangioma, or a laryngeal cleft.

**IMAGING IN THE APP** 1 case

Supraglottic swelling on imaging.



Supraglottic and Pharyngeal Tumors
Tumours · 4 images

### 3.4 Spasmodic croup

Sudden nocturnal attacks of barking cough and stridor in an otherwise well child without fever, settling by morning; often allergic or reflux-related.

## 4 CHAPTER FOUR Bacterial tracheitis

Croup that is not croup.

A rare, serious bacterial infection (*Staphylococcus aureus*, streptococci, pneumococcus), often following viral croup: high fever, toxicity, stridor and a cough with thick secretions, failing to respond to croup treatment. Bronchoscopy shows purulent tracheitis with oedema, sloughed mucosa and membranes that must be removed, often repeatedly; intubation and intravenous antibiotics; pneumonia frequently follows.

## 5 CHAPTER FIVE Laryngeal diphtheria

The membrane in the airway.

Usually secondary to faucial diphtheria in unimmunised children under ten. An insidious onset with low fever but marked toxaemia (tachycardia, thready pulse); hoarseness, croupy cough, inspiratory stridor and progressive obstruction; an adherent grey-white membrane on the tonsils, palate and pharynx that bleeds when lifted and extends into the

larynx and trachea; a bull neck. The exotoxin causes myocarditis and palatal, pharyngeal, ocular and diaphragmatic paralysis.

**LARYNGEAL DIPHTHERIA**

1. Treat on clinical suspicion; take smears and cultures.
2. Antitoxin, 20,000 to 100,000 units intravenously after a test dose, according to severity and duration.
3. Benzylpenicillin or erythromycin for 7 to 10 days.
4. Airway: intubation after laryngoscopy and removal of membrane, or tracheostomy.
5. Strict bed rest for two to four weeks because of myocarditis; cardiac monitoring.
6. Isolation, immunisation of contacts.

CHAPTER SIX

# 6 Laryngeal oedema

Many causes, one danger.

Oedema collects where the mucosa is loose, in the supraglottis and subglottis, and rarely in the vocal folds.

## 6.1 Causes

- **Infection:** epiglottitis, croup, tuberculosis, syphilis; nearby quinsy, retropharyngeal abscess and Ludwig's angina.
- **Trauma:** surgery of the tongue and floor of mouth, laryngeal trauma, foreign bodies, endoscopy (especially in children), intubation, burns, caustics and irritant gases.
- **Neoplasm:** ulcerating cancers of the larynx and laryngopharynx.
- **Allergy:** angioneurotic oedema, anaphylaxis; ACE inhibitor and hereditary angioedema.
- **Radiotherapy.**
- **Systemic:** nephritis, heart failure, myxoedema.

### 6.2 Management

Inspiratory stridor and respiratory distress call for an airway: intubation or tracheostomy. Lesser degrees are treated according to cause: intramuscular adrenaline (0.5 mL of 1:1000 in adults, repeated after five minutes if needed), antihistamines and steroids for allergy; steroids for infective, traumatic and post-radiation oedema; C1 esterase inhibitor or icatibant for hereditary angioedema; stopping the ACE inhibitor.

## 7 CHAPTER SEVEN For the examination

What the examiner wants to hear.

### 7.1 Viva essentials

**WHAT THE EXAMINER IS LOOKING FOR**

- **Croup versus epiglottitis:** the table.
- **How not to examine** a child with suspected epiglottitis, and how to secure the airway.
- **Treatment of croup:** dexamethasone and nebulised adrenaline.
- **Bacterial tracheitis** as croup that fails to respond.
- **Laryngeal diphtheria.**
- **Causes and treatment** of laryngeal oedema.

### 7.2 Common pitfalls

**COMMON PITFALLS**

- Using a tongue depressor on a drooling, stridulous child.
- Sending a child with epiglottitis to radiology alone.
- Discharging a child soon after nebulised adrenaline.

- Missing epiglottitis in an adult with severe throat pain and a normal oropharynx.

---

Written for this app from Oxford Textbook of Otolaryngology ch. 74, Oxford Specialist Handbook ch. 23, Oxford Handbook of ENT ch. 17, Dhingra ch. 58 and K. J. Lee ch. 50, 53. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · LARYNX & AIRWAY

# Foreign bodies of the airway

A toddler with a peanut in his mouth laughs, gasps and coughs; the coughing stops, and for days or weeks he seems well except for a wheeze that is treated as asthma. Inhaled foreign bodies are a leading cause of accidental death in small children, and a missed one slowly destroys a lung. Suspect it from the story, look for the triad of cough, wheeze and reduced air entry, do not trust a normal chest film, and remember the rule of airway foreign bodies: a negative bronchoscopy is far better than a missed foreign body.

**AT A GLANCE**

- Choking history is enough.
- A normal film doesn't exclude it.
- Rigid bronchoscopy is the treatment.

## 1 CHAPTER ONE Overview

Who, what and where.

### 1.1 Who

Children, more than half under four, who inhale suddenly while laughing, crying, playing or running with something in the mouth; infants have no molars to grind nuts and a weak cough. In adults, aspiration occurs in coma, sleep, intoxication, seizures, under anaesthesia (teeth, dentures), with neurological disease, and in the elderly.

### 1.2 What

- **Vegetable (organic):** peanuts above all, also seeds (watermelon, pumpkin), beans, peas, grains, nuts, pieces of carrot or apple; regional patterns follow diet (areca nut in parts of India, watermelon seeds in Egypt). They absorb water and swell, and their oils provoke an intense *vegetal bronchitis* with oedema, granulation and suppuration.

- **Non-vegetable:** plastic toys and whistles, beads, pins and needles, nails, wires, ball bearings, coins, pen caps and springs; inert ones may stay silent for long periods.
- **Food boluses** that obstruct the larynx in adults (the "café coronary").

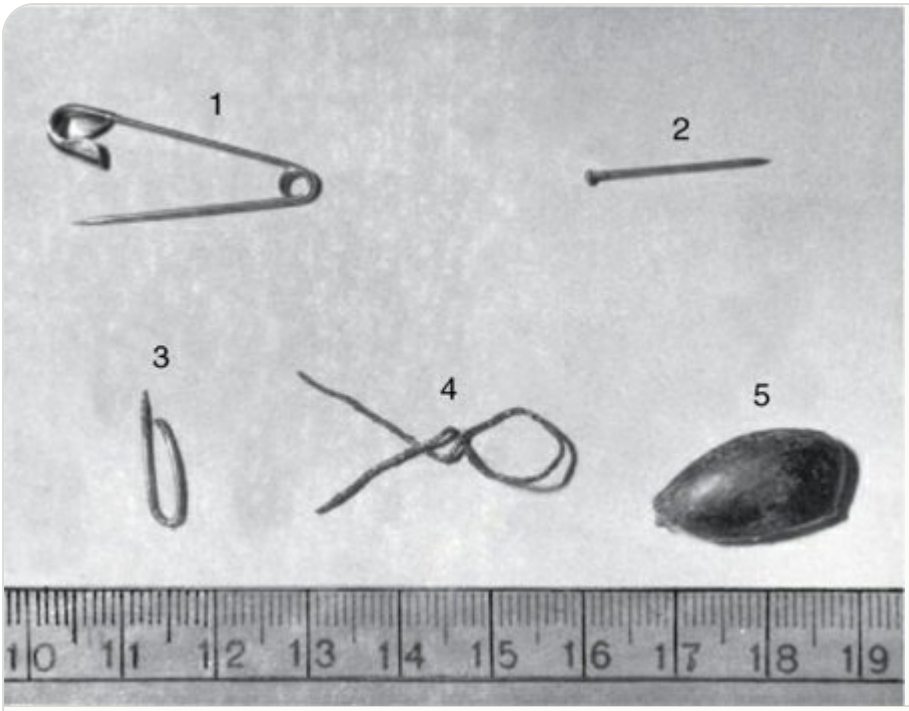



**FIGURE 1** Laryngeal foreign bodies: (1) safety pin, (2) pin, (3, 4) twisted wire, (5) a fruit seed. Dhingra, fig. 65.1

### 1.3 Where

Large objects lodge above the glottis; small ones pass to the trachea and bronchi, most often into the *right main bronchus*, which is wider, shorter and more vertical than the left (in young children the difference is less). Sharp objects stick anywhere.

## 2 CHAPTER TWO Clinical features

Three stages.

### 2.1 Stage 1: the event

Sudden choking, gagging, coughing and wheezing, sometimes cyanosis, lasting minutes. The object may be coughed out or lodge. Often the event is not witnessed,

or forgotten.

## 2.2 Stage 2: the silent interval

The mucosa adapts and the cough settles, for hours to weeks, depending on the object.

## 2.3 Stage 3: complications

Obstruction, inflammation and infection, according to the site.

FEATURES BY SITE

| SITE | FEATURES |
|---|---|
| Larynx | Complete obstruction: aphonia, inability to cough or breathe, rapid death. Partial: pain, hoarseness, croupy cough, stridor, dyspnoea, haemoptysis |
| Trachea | Choking, cough, biphasic stridor and wheeze heard at the open mouth; a mobile object gives an *audible slap* and *palpable thud* as it strikes the undersurface of the folds |
| Bronchus | The triad of *cough, wheeze and reduced air entry* on one side; later collapse, emphysema, pneumonia, bronchiectasis, lung abscess |

## 2.4 Patterns of bronchial obstruction

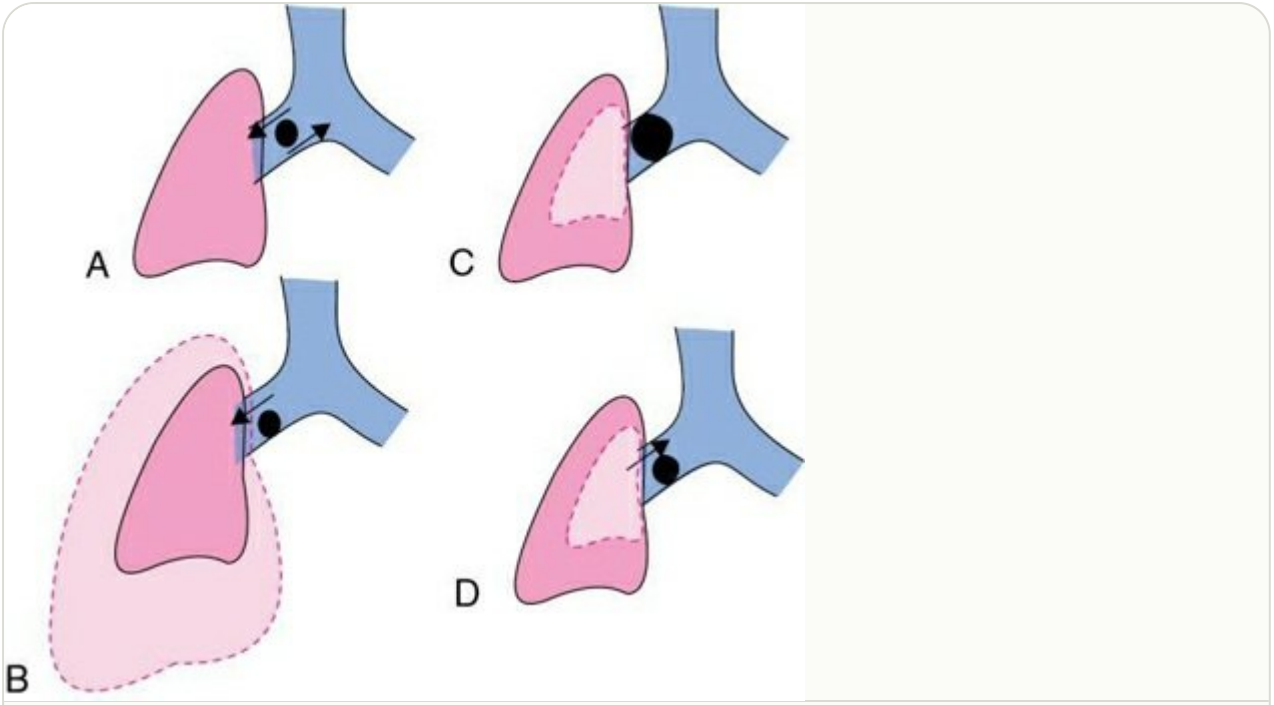



**FIGURE 2** Types of bronchial obstruction: (A) partial (bypass valve) with wheeze; (B) check-valve with obstructive emphysema; (C) complete (stop valve) with collapse; (D) ball valve.

Dhingra, fig. 65.2

- **Bypass valve:** air passes both ways; only a wheeze.
- **Check valve:** air enters but cannot leave; *obstructive emphysema* of the lobe or lung, with hyper-resonance, reduced breath sounds and mediastinal shift to the other side.
- **Stop valve:** complete obstruction; air is absorbed and the segment collapses (*atelectasis*), with dullness and shift towards the affected side.
- **Ball valve:** a mobile object enters on inspiration but blocks on expiration.

An emphysematous bulla may rupture into a pneumothorax, and a mobile object may shift sides, changing the signs.

**THE ASTHMATIC CHILD WHO IS NOT ASTHMATIC**

New wheeze that is one-sided, does not respond to bronchodilators, or follows a choking episode; recurrent pneumonia in the same lobe; a persistent cough after eating nuts: think foreign body and arrange bronchoscopy.

# 3 CHAPTER THREE Diagnosis

History, examination, imaging.

## 3.1 History

The single most important clue: a witnessed choking episode has high sensitivity. Ask directly, and ask other family members.

## 3.2 Examination

Respiratory rate and effort, stridor, voice, unilateral wheeze, reduced air entry, signs of collapse or hyperinflation, and the audible slap of a tracheal object.

## 3.3 Imaging

- **Soft tissue radiographs of the neck (anteroposterior and lateral, neck extended):** radio-opaque objects, sometimes radiolucent ones in the larynx and trachea. A coin in the trachea lies edge-on in the PA view (sagittal), whereas one in the oesophagus lies flat (coronal).
- **Chest radiograph (PA and lateral):** the object if radio-opaque; lobar or segmental *collapse*; unilateral *hyperinflation* with mediastinal shift away; pneumothorax, pneumomediastinum; pneumonia or bronchiectasis in late cases.
- **Inspiratory and expiratory films, or fluoroscopy** (lateral decubitus films in young children): air trapping on the affected side, which does not deflate on expiration, and the mediastinum swings towards the normal side on expiration.
- **CT of the chest** (low dose) in doubtful, stable cases; virtual bronchoscopy.

**A NORMAL CHEST FILM DOES NOT EXCLUDE IT**

Most inhaled foreign bodies in children are radiolucent nuts and seeds, and the film is normal in a large proportion, especially in the first day. If the story is suggestive, proceed to bronchoscopy.

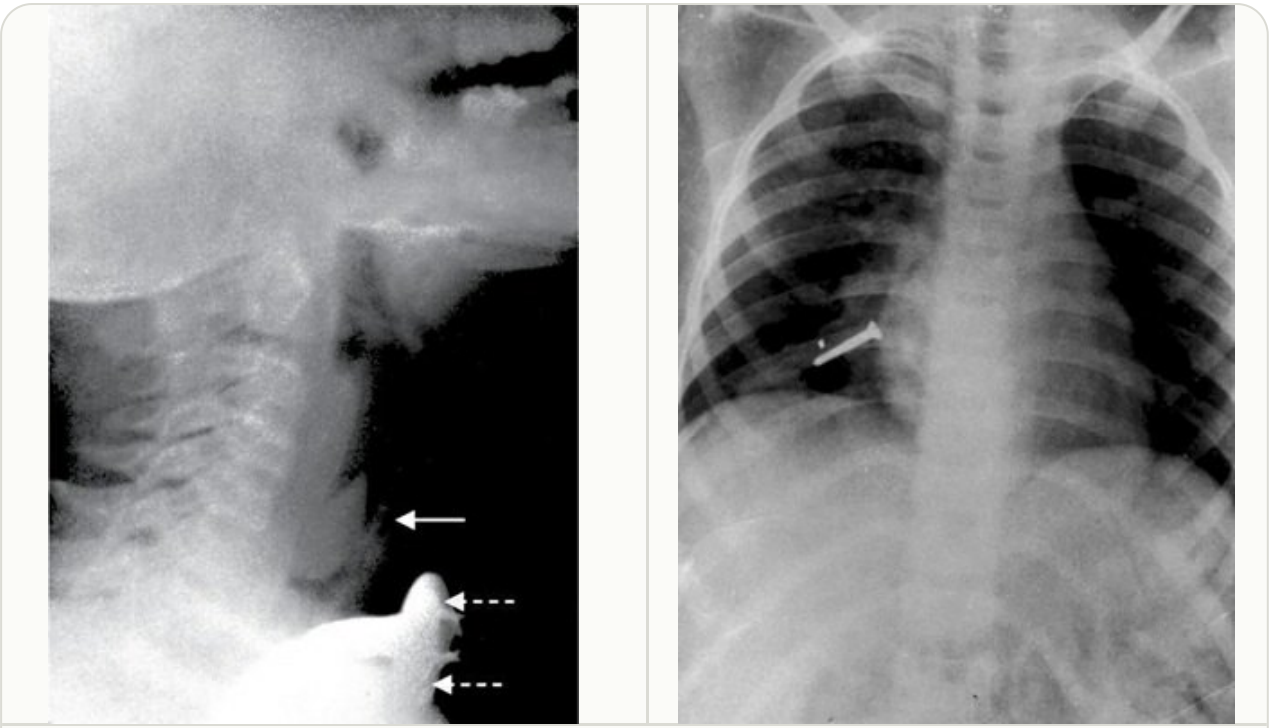


**FIGURE 3** First, a radiolucent fruit seed (chiku) in the subglottis (arrow) in a child with a tracheostomy. Second, a nail in the right bronchus on a PA chest film.

Dhingra, figs 65.3 and 65.4

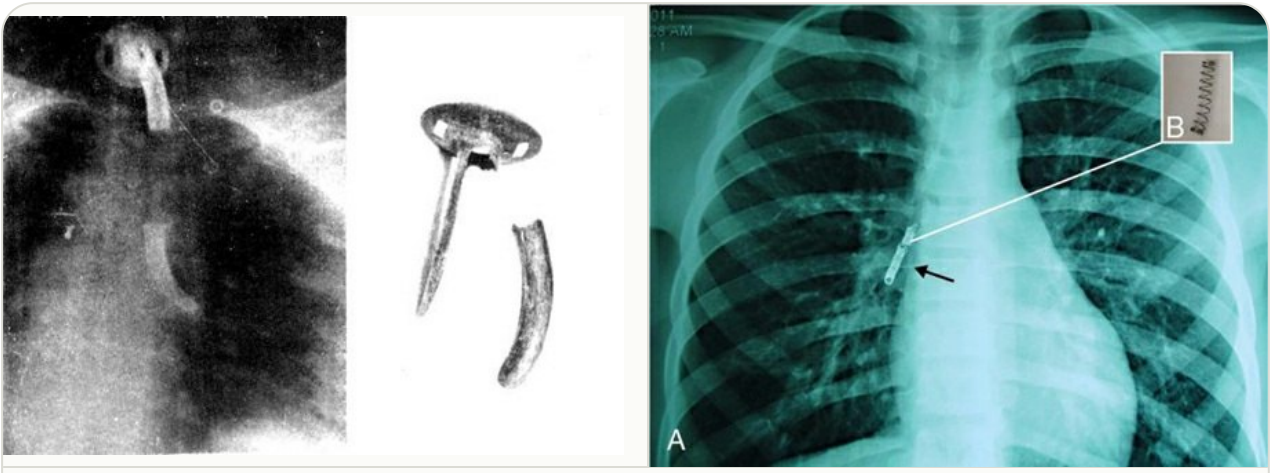



**FIGURE 4** First, a broken piece of a Fuller's tracheostomy tube in the left bronchus. Second, (A) a spring from a ball pen in the right bronchus, and (B) the spring after removal.

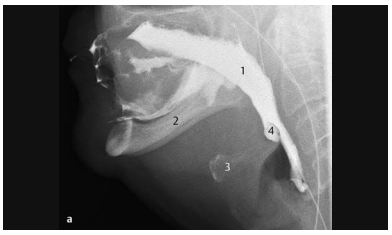
Dhingra, figs 65.5 and 65.6

**IMAGING IN THE APP** 2 cases

Foreign bodies and aspiration on imaging.



**Foreign Bodies**
Trauma · 3 images




**Aspiration**
Infection & inflammation · 3 images

# 4 CHAPTER FOUR Management

First aid, then the bronchoscope.

## 4.1 Complete obstruction: first aid

A patient who cannot speak, breathe or cough needs immediate action:

- **Infants under one year:** five back blows with the infant prone along the forearm, head down, then five chest thrusts; repeat. No abdominal thrusts.
- **Children and adults:** five back blows, then five *abdominal thrusts (Heimlich manoeuvre)*: standing behind, fist below the xiphisternum, sharp inward and upward thrusts that use the residual lung air to expel the object; repeat.
- **If unconscious:** start cardiopulmonary resuscitation; look in the mouth and remove a visible object (no blind finger sweeps).
- If these fail and expertise is available: laryngoscopy and removal with Magill forceps, or cricothyroidotomy (needle in small children) or emergency tracheostomy.




**FIGURE 5** The Heimlich manoeuvre: a sudden thrust upwards and backwards below the epigastrium squeezes air from the lungs to dislodge the object.

Dhingra, fig. 65.7

**PARTIAL OBSTRUCTION**

A patient who can cough and breathe should be encouraged to keep coughing and taken calmly to theatre; back blows and thrusts may convert a partial into a complete obstruction.

## 4.2 Laryngeal foreign bodies

After the emergency, removal by direct laryngoscopy under general anaesthesia, or by laryngofissure for an impacted object; tracheostomy first if the airway is compromised.

## 4.3 Tracheobronchial foreign bodies

Removed by *rigid ventilating bronchoscopy* under general anaesthesia, ideally with spontaneous ventilation, by a prepared team: an anaesthetist and surgeon who agree a plan, bronchoscopes of the right size and one size either side, telescopes and *optical forceps*, two laryngoscopes, a range of forceps, Dormia baskets and

Fogarty balloon catheters, and suction, all checked beforehand. Practise grasping a duplicate object before starting.

- Urgent removal is needed for airway obstruction and for vegetable matter that will swell; otherwise it is done on the next planned list with a full team, not in the middle of the night.
- Small objects are withdrawn through the scope; larger ones are held against the tip and the scope, forceps and object removed together; a sharp point is sheathed in the scope or turned to trail.
- Steroids reduce oedema; physiotherapy and antibiotics treat secondary infection.
- Fluoroscopy (C-arm) for peripheral metallic objects; flexible bronchoscopy in selected adults; tracheostomy and bronchoscopy through the stoma for large objects; *thoracotomy and bronchotomy* for objects that cannot be reached.

**OPERATIVE GUIDES IN THEATRE** 2 operations

Tracheoscopy, bronchoscopy and airway foreign bodies.

**Tracheoscopy, Bronchoscopy, and Airway Foreign Bodies**
Airway foreign body
›

**Bronchoscopy**
Bronchoscopy
›

## 4.4 Complications of removal

Oedema of the subglottis, bronchospasm, bleeding from granulations, pneumothorax and pneumomediastinum, fragmentation of nuts with distal spread, and complete obstruction if the object is dropped in the glottis: push it back into a main bronchus and ventilate the other lung, then retrieve it.

## 4.5 Late complications of a retained object

Recurrent pneumonia, bronchiectasis, lung abscess, granulation tissue and stenosis.

### 4.6 Prevention

No nuts, popcorn, hard sweets or whole grapes for children under four or five; supervise meals and play; age-appropriate toys without small parts; never hold pins or small objects in the mouth.

# 5 CHAPTER FIVE For the examination

What the examiner wants to hear.

### 5.1 Viva essentials

**WHAT THE EXAMINER IS LOOKING FOR**

- **Three stages** of symptoms and the silent interval.
- **Why the right bronchus.**
- **Signs by site:** audible slap and palpable thud; the triad.
- **Valve types** and their radiological signs; inspiratory and expiratory films.
- **First aid:** back blows, chest thrusts in infants, Heimlich in older children and adults.
- **Rigid bronchoscopy:** equipment and technique.

### 5.2 Common pitfalls

**COMMON PITFALLS**

- Excluding a foreign body because the chest film is normal.
- Treating a sudden one-sided wheeze in a toddler as asthma.
- Performing back blows or the Heimlich manoeuvre on a patient who is coughing effectively.
- Attempting removal without the right-sized instruments checked and ready.

Written for this app from Oxford Textbook of Otolaryngology ch. 74, Oxford Specialist Handbook ch. 23, Oxford Handbook of ENT ch. 25, Dhingra ch. 65 and K. J. Lee ch. 50, 53. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · LARYNX & AIRWAY

# Vocal fold paralysis

A paralysed vocal fold is a sign that something has happened along a very long nerve: in the brainstem, at the skull base, in the neck or deep in the chest. One fold paralysed costs the voice and sometimes the swallow; both paralysed near the midline cost the airway while sparing the voice. Find the cause, especially the lung or thyroid cancer that may be hiding behind a hoarse voice, and then choose between voice, airway and safe swallowing.

**AT A GLANCE**

- Unilateral affects voice; bilateral affects airway.
- Image the whole nerve path.
- Voice therapy, injection or medialisation.
- Secure the airway in bilateral palsy.

## 1 CHAPTER ONE Nerve supply

Two nerves, one long way round.

### 1.1 Motor

Every muscle that moves the vocal fold is supplied by the *recurrent laryngeal nerve*, except the *cricothyroid* (a tensor), which is supplied by the *external laryngeal nerve*, a branch of the superior laryngeal nerve. The interarytenoid receives fibres from both sides.

### 1.2 Sensory

Above the vocal folds, the internal laryngeal nerve; below them, the recurrent laryngeal nerve.

## 1.3 Course

- **Right recurrent nerve:** leaves the vagus at the root of the neck, hooks under the subclavian artery and climbs in the tracheo-oesophageal groove.
- **Left recurrent nerve:** leaves the vagus in the mediastinum, hooks under the arch of the aorta beside the ligamentum arteriosum, and climbs a much longer course; hence it is paralysed far more often.
- **Superior laryngeal nerve:** from the inferior ganglion of the vagus, behind the internal carotid artery, dividing at the greater horn of the hyoid into the internal (sensory, through the thyrohyoid membrane) and external (to cricothyroid, beside the superior thyroid artery) branches.

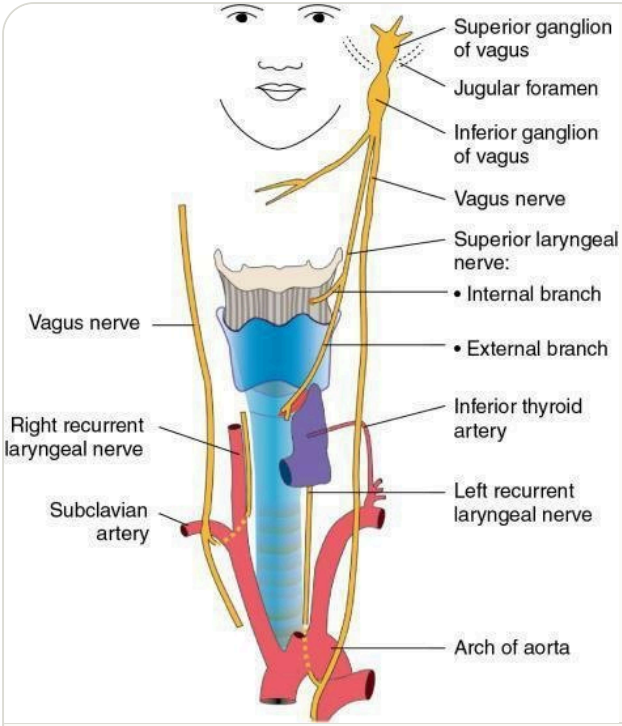
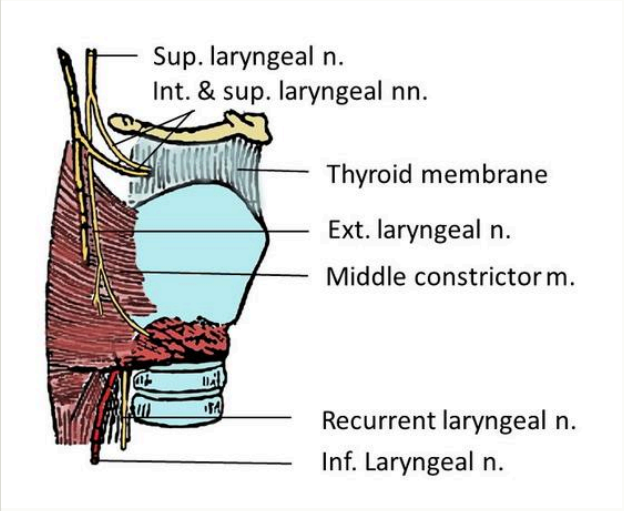



**FIGURE 1** The recurrent and superior laryngeal nerves and their course: the left recurrent nerve loops around the arch of the aorta, the right around the subclavian artery.

Dhingra, fig. 60.1



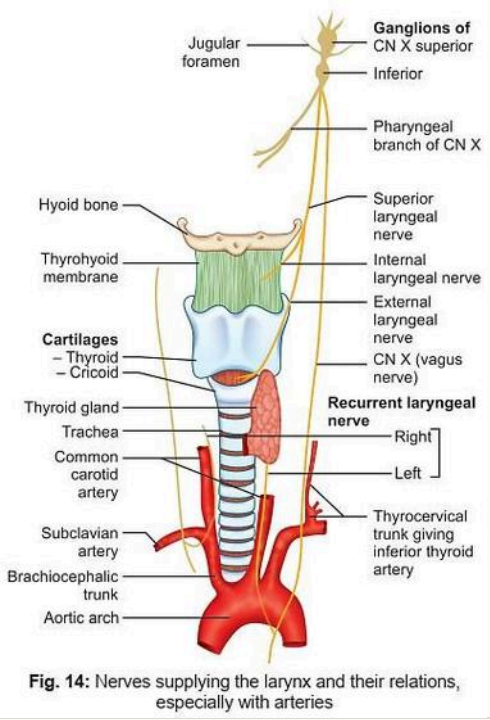




**FIGURE 2** The nerve supply of the larynx, and the nerves with their relations.

Anatomy atlas · Doss p. 175, ENT Steps p. 251

# 2 CHAPTER TWO Positions of the cord

What the folds show.

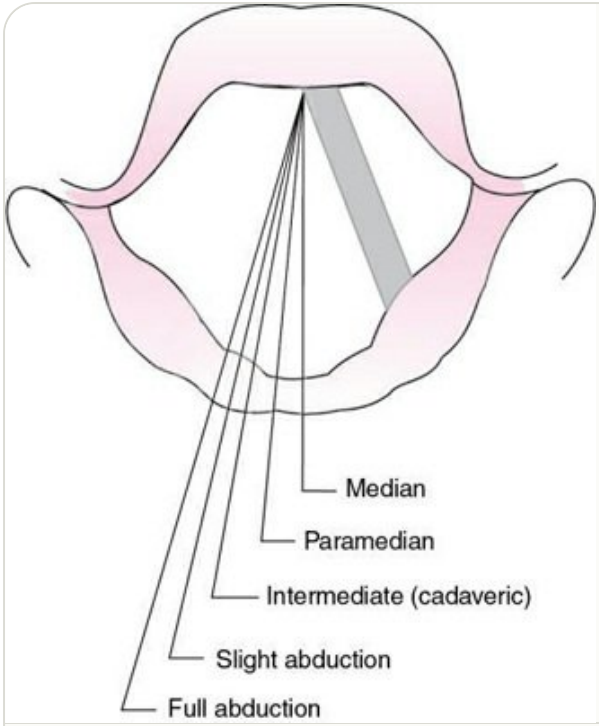



**FIGURE 3** Positions of the vocal folds: median, paramedian, intermediate (cadaveric), slight abduction and full abduction.

Dhingra, fig. 60.2

## POSITIONS OF THE VOCAL FOLD

| POSITION | DISTANCE FROM MIDLINE | IN HEALTH | IN DISEASE |
|---|---|---|---|
| Median | 0 | Phonation | Recurrent nerve paralysis |
| Paramedian | 1.5 mm | Strong whisper | Recurrent nerve paralysis |
| Intermediate (cadaveric) | 3.5 mm | Neutral resting position | Combined recurrent and superior nerve paralysis |
| Gentle abduction | 7 mm | Quiet breathing | Paralysis of adductors |
| Full abduction | 9.5 mm | Deep inspiration | None |

### 2.1 Why the paralysed cord lies near the midline

- **Semon's law:** in progressive organic lesions the abductor fibres, phylogenetically newer, are affected before the adductors, so the fold first lies near the midline.
- **Wagner–Grossman hypothesis:** with the recurrent nerve cut and the superior nerve intact, the cricothyroid still adducts the fold slightly into the paramedian position; with both nerves lost the fold lies in the cadaveric position.

Lee adds that the position also reflects partial or misdirected *regeneration*, which restores tone (and synkinesis) without useful movement; a fold often lies laterally at first and medialises over months.

---

## 3 CHAPTER THREE Causes

Along the whole course.

### 3.1 By level

- **Supranuclear:** rare (bilateral cortical lesions).
- **Nuclear (nucleus ambiguus, medulla):** stroke (lateral medullary syndrome), tumours, motor neurone disease, polio, syringobulbia, multiple sclerosis, Arnold–Chiari malformation; other cranial nerves are involved.
- **High vagal (with superior laryngeal involvement, a combined paralysis):** posterior fossa tumours and basal meningitis; skull base fractures, nasopharyngeal carcinoma, glomus tumours; parapharyngeal tumours, metastatic nodes, lymphoma and penetrating injury.
- **Low vagal and recurrent nerve:** see the table.
- **Systemic:** diabetes, viral neuritis, diphtheria, syphilis, lead poisoning.
- **Idiopathic:** about 20 to 30%, many presumed viral; a diagnosis of exclusion.

CAUSES OF RECURRENT LARYNGEAL NERVE PARALYSIS

| SITE | RIGHT | LEFT | BOTH |
|---|---|---|---|
| Neck | Thyroid surgery, thyroid cancer, cervical oesophageal cancer, nodes, trauma, subclavian aneurysm | The same | Thyroidectomy, thyroid cancer, cervical oesophageal cancer |
| Chest | Carcinoma of the right lung apex, tuberculosis of the pleura | Bronchogenic carcinoma, mediastinal nodes, thoracic oesophageal cancer, aortic aneurysm, enlarged left atrium (Ortner syndrome), cardiac and thoracic surgery, ductus ligation | Rare |

**LEFT VOCAL FOLD PALSY**

Bronchogenic carcinoma must be excluded in every unexplained left-sided paralysis: image the whole course of the nerve from the skull base to the aortic arch with contrast CT (or MRI and CT), and examine the thyroid, oesophagus and neck.

### 3.2 Commonest causes overall

Malignancy (lung, thyroid, oesophagus) and surgery (thyroid, cervical spine through an anterior approach, cardiothoracic) lead the list; thyroidectomy is the commonest cause of bilateral paralysis.

## 4 CHAPTER FOUR Recurrent nerve paralysis

One or both.

### 4.1 Unilateral

All intrinsic muscles on that side are paralysed except cricothyroid, and the fold lies median or paramedian. About a third of patients have no symptoms; others have a weak, breathy voice that tires, diplophonia, loss of projection and range, and

sometimes aspiration of thin liquids. The voice often improves over months as the healthy fold learns to cross the midline.

### 4.2 Bilateral (bilateral abductor paralysis)

Usually after thyroidectomy, or neuritis. Both folds lie median or paramedian, held by the unopposed cricothyroids: *the voice is good but the airway is poor*, with stridor and dyspnoea that worsen on exertion and with any respiratory infection, sometimes requiring an emergency tracheostomy.

> **THE PARADOX**
>
> Bilateral recurrent palsy gives a good voice and a bad airway; bilateral combined palsy gives a bad voice, no airway problem and dangerous aspiration.

## 5 CHAPTER FIVE Superior laryngeal and combined paralysis

Tension and sensation.

### 5.1 Superior laryngeal nerve paralysis

Paralysis of the cricothyroid with numbness of the supraglottis on that side; usually part of a combined lesion, or the external branch injured in thyroid surgery. The voice is weak and cannot be raised in pitch, which singers notice at once. The glottis looks *askew*, with the anterior commissure rotated to the healthy side; the paralysed fold is shorter, lax and wavy, and flaps (sagging on inspiration, bulging on expiration). Electromyography of the cricothyroid confirms it. Voice therapy.

- **Bilateral:** paralysis with anaesthesia of the upper larynx causes repeated aspiration, cough and choking, and a weak, husky voice; it may recover after neuritis; tracheostomy with a cuffed tube and tube feeding, and epiglottopexy for persistent aspiration; cricothyroid approximation raises the pitch.

### 5.2 Combined (complete) paralysis

- **Unilateral:** after thyroidectomy injuring both nerves, or with lesions of the nucleus ambiguus or vagus above the origin of the superior laryngeal nerve. The fold lies in the *cadaveric position*, the healthy fold cannot meet it, and the result is a hoarse, breathy voice, aspiration of liquids and a weak cough.
- **Bilateral:** both folds cadaveric with an insensitive larynx: aphonia, aspiration, inability to cough and recurrent pneumonia. Tracheostomy and gastrostomy, then procedures that protect the lungs: epiglottopexy, vocal fold plication (both reversible), laryngotracheal separation or diversion, and, as a last resort, laryngectomy.

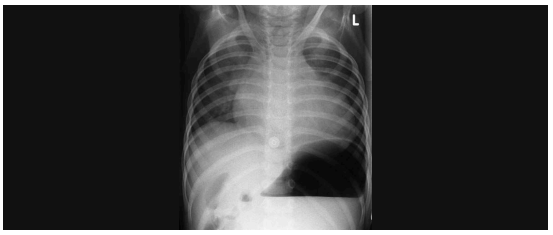
## 6 CHAPTER SIX Evaluation

Paralysis or fixation?

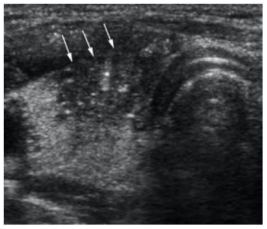
- **Laryngoscopy:** flexible and stroboscopic, observing movement, position, the glottic gap, the level of the folds and arytenoid tilt; confirm with phonation, sniffing and coughing.
- **Fixation versus paralysis:** posterior glottic scarring, cricoarytenoid arthritis or dislocation and tumour can fix a fold; at direct laryngoscopy the arytenoid is *palpated* for passive mobility.
- **Laryngeal electromyography:** distinguishes paralysis from fixation, shows synkinesis and helps predict recovery (absent or synkinetic activity predicts poor recovery).
- **Imaging:** contrast CT from the skull base to the aortic arch, or MRI of the brainstem and skull base for high lesions; chest imaging; thyroid ultrasound.
- **Swallowing assessment:** FEES or videofluoroscopy for aspiration.
- **Blood tests** where indicated (glucose, serology).

**IMAGING IN THE APP** 2 cases

Recurrent laryngeal nerve paralysis and thyroid cancer on imaging.




Paralysis of the Recurrent Laryngeal Nerve
Infection & inflammation · 9 images



Thyroid cancer
Tumours · 10 images

# 7 CHAPTER SEVEN Treatment

Voice, airway, swallow.

## 7.1 Unilateral paralysis

The aim is to close the glottic gap.

- **Voice therapy** early, while awaiting recovery or compensation over six to twelve months.
- **Injection laryngoplasty** (medialisation by injection, through the mouth or the neck, awake or under anaesthesia): temporary fillers (carboxymethylcellulose 2 to 3 months, hyaluronic acid 4 to 9 months, collagen) early, even within weeks, especially if aspiration is a problem; longer-lasting calcium hydroxylapatite (about 18 months) and autologous fat (variable survival). Teflon is abandoned because of granulomas.
- **Type I thyroplasty (medialisation laryngoplasty):** a window in the thyroid lamina and a silastic or Gore-Tex implant in the paraglottic space push the fold medially; done under local anaesthesia so that the voice can be tuned; for permanent paralysis with a gap of a few millimetres and symmetrical arytenoids.
- **Arytenoid adduction:** a suture through the muscular process rotates the arytenoid inwards, mimicking the lateral cricoarytenoid; for a wide posterior gap, a lateralised vocal process or a level difference; usually with thyroplasty.
- **Reinnervation:** ansa cervicalis to recurrent nerve anastomosis restores bulk and tone (not movement); useful when the nerve is known to be cut and in children, who would outgrow an implant.

## 7.2 Bilateral abductor paralysis

The aim is an airway without ruining the voice; any procedure that widens the glottis worsens the voice.

- **Tracheostomy:** the safest, the gold standard; the voice is preserved with finger occlusion or a speaking valve.
- **Endoscopic glottic widening:** *posterior transverse cordotomy* (Kashima), cutting the fold off the vocal process with the laser; *partial medial arytenoidectomy*; suture *lateralisation* of the fold; repeated if needed.
- Older external procedures: arytenoidectomy and arytenoidopexy (Woodman), and type II thyroplasty.
- Botulinum toxin to the adductors (temporary); reinnervation and laryngeal pacing (selected, experimental).
- In the first months after thyroidectomy, a temporary tracheostomy and waiting is wise, since many palsies recover.

**OPERATIVE GUIDES IN THEATRE** 5 operations

Injection, medialisation, intraoperative medialisation, bilateral immobility and protection of the nerve.

**Vocal Fold Injection via Microlaryngoscopy**
Injection
›

**Medialization Laryngoplasty and Arytenoid Adduction**
Medialisation
›

**Intraoperative Medialization Laryngoplasty**
Intraoperative medialisation
›

**Bilateral Vocal Fold Immobility**
Bilateral immobility
›

**Open Thyroidectomy**
Protecting the nerve
›

## 7.3 Isshiki's thyroplasties

- **Type I:** medialisation.

- **Type II:** lateralisation, for the airway.
- **Type III:** shortening and relaxation, to lower the pitch (mutational falsetto, female-to-male voice).
- **Type IV:** lengthening and tensing, to raise the pitch (male-to-female voice, bowed or lax folds).

# 8 CHAPTER EIGHT Congenital paralysis

In infants.

Unilateral (birth trauma, cardiac and great vessel anomalies, cardiac surgery) gives a weak cry and feeding problems; bilateral (Arnold–Chiari malformation, hydrocephalus, birth haemorrhage, meningocele) gives stridor needing tracheostomy. Many recover; definitive surgery is delayed (see congenital lesions of the larynx).

# 9 CHAPTER NINE For the examination

What the examiner wants to hear.

## 9.1 Viva essentials

**WHAT THE EXAMINER IS LOOKING FOR**

- **Nerve supply** and why the left recurrent nerve is more often paralysed.
- **Positions** of the cord, Semon's law and the Wagner–Grossman hypothesis.
- **Causes** by level; exclude carcinoma of the lung.
- **Unilateral versus bilateral** recurrent palsy: voice and airway.
- **Superior laryngeal nerve palsy:** askew glottis, loss of pitch.
- **Treatment:** injection, thyroplasty and its types, arytenoid adduction, cordotomy and tracheostomy.

## 9.2 Common pitfalls

**COMMON PITFALLS**

- Calling a left palsy idiopathic without imaging to the aortic arch.
- Treating a fixed arytenoid as a paralysis without palpation or electromyography.
- Doing an irreversible glottic widening soon after thyroidectomy when the nerve may recover.

---

Written for this app from Oxford Textbook of Otolaryngology ch. 47, 71, Oxford Specialist Handbook ch. 8, Oxford Handbook of ENT ch. 12, Dhingra ch. 60 and K. J. Lee ch. 50. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · LARYNX & AIRWAY

# Benign vocal fold lesions and hoarseness

Most hoarse voices come from the vocal folds themselves, and most of what grows on them is benign: nodules in the teacher, a polyp after a shout, Reinke's oedema in the smoker, a granuloma on the vocal process, papillomas in a child. Benign does not mean harmless. These lesions take voices from those who earn their living by them and airways from small children, and a white patch on a smoker's fold may already be carcinoma in situ. Look carefully, ideally with a stroboscope; give voice therapy its due; and when surgery is needed, preserve the delicate cover of the fold.

**AT A GLANCE**

- Hoarseness beyond a few weeks needs a look.
- Voice therapy first for nodules.
- Stop smoking and treat reflux.
- Keep the superficial lamina propria safe.

## 1 CHAPTER ONE Overview

Lesions and tumours.

**BENIGN LESIONS AND TUMOURS OF THE LARYNX**

| NON-NEOPLASTIC (COMMONER) | NEOPLASTIC |
|---|---|
| Vocal nodules | Squamous papilloma (juvenile and adult) |
| Vocal polyp | Chondroma |
| Reinke's oedema | Haemangioma |
| Contact ulcer and granuloma; intubation granuloma | Granular cell tumour |
| Leukoplakia and keratosis | Pleomorphic adenoma, oncocytoma |
| Amyloid | Neurilemmoma, neurofibroma |
| Cysts: ductal, epidermoid, saccular | Rhabdomyoma, lipoma, fibroma |
| Laryngocele; sulcus vocalis | |

## 1.1 Assessment of the hoarse voice

Any hoarseness lasting more than three weeks, or sooner in a smoker or with pain, dysphagia, otalgia, stridor or a neck mass, needs laryngoscopy. *Videostroboscopy* shows the mucosal wave, which distinguishes superficial lesions (nodules, polyps) from those that stiffen the fold (cysts, scar, sulcus, carcinoma). Perceptual rating, acoustic analysis and voice handicap indices measure the voice before and after treatment.

## 1.2 Surgery

*Phonomicrosurgery* under suspension microlaryngoscopy with cold instruments or laser: a *microflap* is raised through a small incision on the upper surface, the lesion dissected from beneath, and the epithelium laid back, preserving the superficial lamina propria and the vocal ligament. The anterior commissure is never stripped on both sides at once, to avoid a web. Voice therapy before and after prevents recurrence.



**FIGURE 1** The set-up for microlaryngeal surgery.
Dhingra, fig. 61.3

**OPERATIVE GUIDES IN THEATRE** 2 operations

Phonomicrosurgery for benign lesions and office-based laryngeal procedures.

**Phonomicrosurgery for Benign Vocal Fold Lesions**
Phonomicrosurgery
›

**Office-Based Laryngeal Procedures**
Office procedures
›

CHAPTER TWO

# 2 Phonotraumatic lesions

Nodules, polyps and Reinke's oedema.

## 2.1 Vocal nodules

Singers', screamers' or teachers' nodes. *Bilateral, symmetrical* swellings on the free edge at the junction of the anterior third and posterior two-thirds, the midpoint of the membranous fold, where vibration and collision are greatest. They arise from chronic misuse of the voice (shouting, singing too loudly or in the wrong register, speaking at an unnaturally low pitch): submucosal oedema and haemorrhage organise and hyalinise, and the epithelium thickens. Early nodules are soft, red and oedematous; mature ones grey-white and firm. Teachers, singers, actors, vendors and talkative children, especially boys, are typical; in adults, women more often.

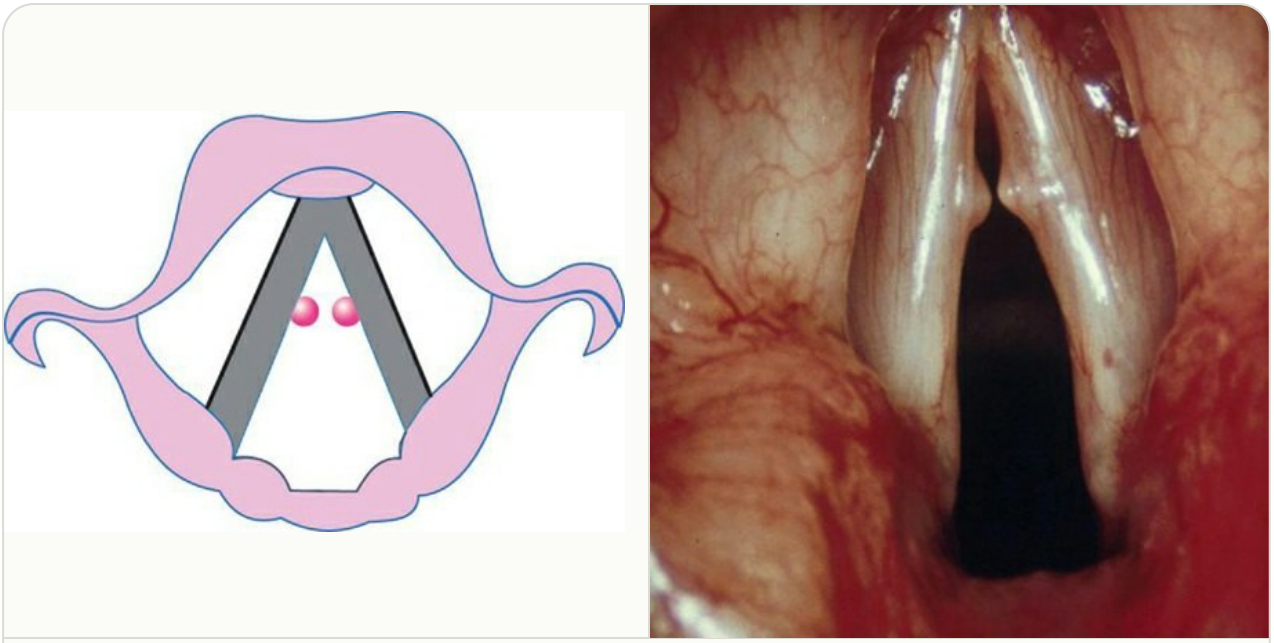


**FIGURE 2** First, the site of vocal nodules at the junction of the anterior third and posterior two-thirds of the folds. Second, vocal nodules on laryngoscopy.
Dhingra, figs 61.1 and 61.2

- **Symptoms:** hoarseness, a breathy voice that tires, loss of the upper range in singers, and neck pain after prolonged speaking.
- **Stroboscopy:** an hourglass closure with a preserved mucosal wave.
- **Treatment:** *voice therapy* is the cornerstone; most children's nodules disappear, and many adults'. Lee goes further: lesions that do not shrink after good, compliant therapy are by definition not nodules. Microsurgical excision for large, fibrotic nodules persisting after therapy, with care not to injure the vocal ligament, and continued therapy afterwards; intralesional steroid may help.

## 2.2 Vocal polyp

Usually a *unilateral*, soft, smooth, often pedunculated swelling at the same site, after sudden shouting causing haemorrhage into the fold, with smoking and allergy contributing; mostly men of thirty to fifty. It may be haemorrhagic, gelatinous or fibrous. Hoarseness, *diplophonia* (two pitches from two differently vibrating folds), and if large and pedunculated, choking or stridor as it flops through the glottis. Treatment is microsurgical excision followed by voice therapy.

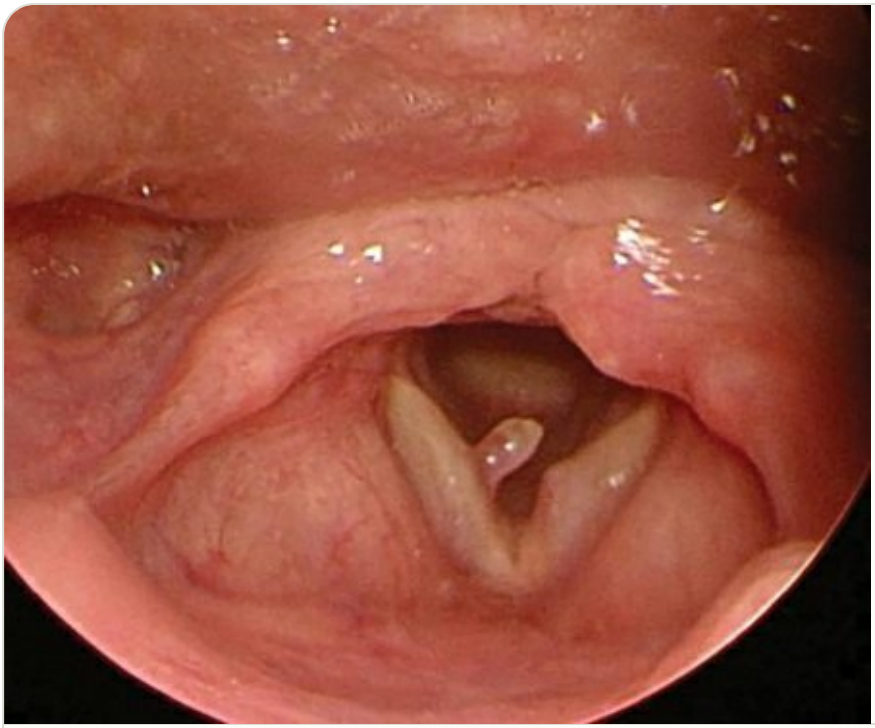


**FIGURE 3** A pedunculated polyp on the right vocal fold.
Dhingra, fig. 61.4

## 2.3 Reinke's oedema (polypoid degeneration)

Bilateral, diffuse, symmetrical, pale, translucent swelling of the whole membranous folds from gelatinous fluid in Reinke's space; smoking is almost always responsible, with voice abuse, reflux and hypothyroidism. The voice becomes deep and rough, more noticeable in women, who may be mistaken for men on the telephone; large swellings obstruct breathing. Stop smoking; treat reflux and thyroid; voice therapy; *decortication* by a lateral incision, suction of the gel and trimming of excess epithelium, traditionally one side at a time.

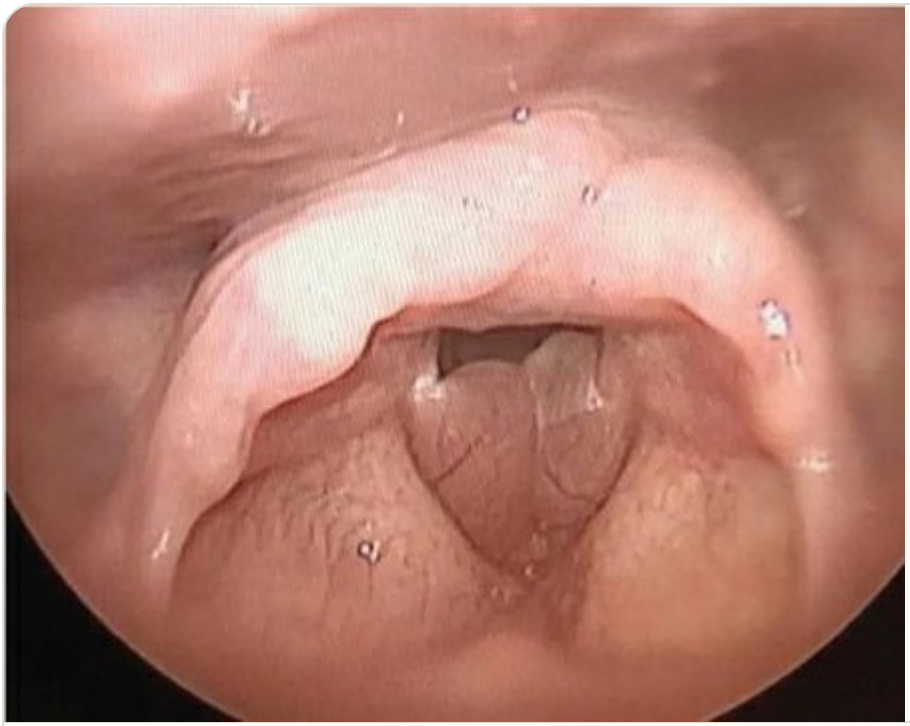


**FIGURE 4** Bilateral Reinke's oedema in a 50-year-old woman who smoked.
Dhingra, fig. 61.5

# 3 CHAPTER THREE Granulomas and keratosis

The back of the fold and the white patch.

## 3.1 Contact ulcer and granuloma

On the *vocal processes* of the arytenoids, where they hammer together with hard glottal attacks, forceful low-pitched speech and habitual throat clearing; laryngopharyngeal reflux is a major factor, and intubation another. Hoarseness, pain on speaking radiating to the ear, a foreign-body feeling and constant throat clearing. Ulcers or granulomas sit on one or both vocal processes with red arytenoid mucosa.

- Anti-reflux treatment (twice-daily proton pump inhibitor), even without heartburn.
- Voice therapy to stop throat clearing and correct pitch.
- Inhaled or injected steroid.
- Botulinum toxin to the adductors (lateral cricoarytenoid and thyroarytenoid) for recurrent granulomas.

- Surgery only for airway obstruction, failure of treatment or suspicion of tumour; recurrence after excision is common. Send tissue for histology to exclude carcinoma and granular cell tumour.

## 3.2 Intubation granuloma

Injury to the vocal processes from rough intubation, a large tube or prolonged intubation; usually bilateral on the posterior folds, with hoarseness and, if large, dyspnoea. Many mature into a pedunculated mass and drop off; treatment is as for contact granuloma.

## 3.3 Leukoplakia and keratosis

White plaques or warty thickening on the upper surface of one or both folds with normal mobility, in smokers. Histology ranges from simple keratosis through dysplasia to carcinoma in situ and invasive carcinoma, so it is a *precancerous condition*. Biopsy or *excisional microflap stripping* for histology, stop smoking, treat reflux, and follow up for recurrence and new lesions; avoid repeated deep stripping that scars the voice.

**THE WHITE PATCH ON A SMOKER'S FOLD**

A keratotic or red lesion with a reduced mucosal wave on stroboscopy, irregular surface, or impaired mobility is suspicious for invasive carcinoma: biopsy rather than repeated observation.

## 3.4 Amyloid

Men of fifty to seventy: a yellow or orange submucosal mass in the false fold, ventricle, true fold or subglottis, causing hoarseness or dyspnoea; apple-green birefringence with Congo red. Exclude myeloma and systemic amyloidosis; remove localised deposits endoscopically; the outlook is good.

# 4 CHAPTER FOUR Cysts and sulcus

Hidden in the fold.

## 4.1 Ductal (retention) cysts

From blocked seromucinous glands in the vallecula, aryepiglottic fold, false fold, ventricle and piriform fossa: silent when small; hoarseness, cough, throat pain and dyspnoea when large.

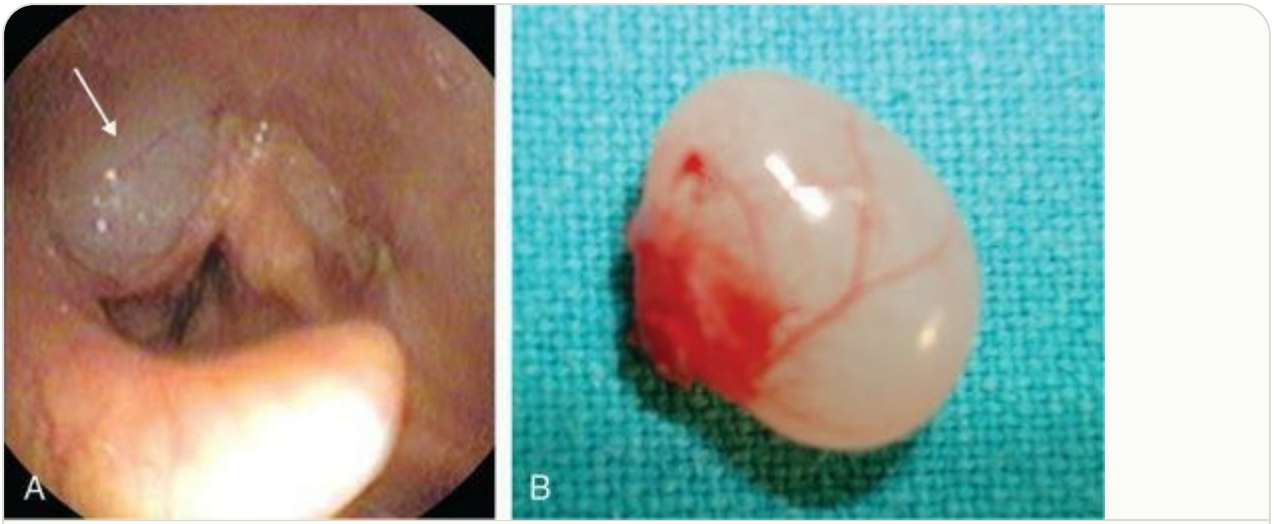



**FIGURE 5** (A) An aryepiglottic cyst causing intermittent laryngeal obstruction. (B) The cyst after removal.
Dhingra, fig. 61.6

## 4.2 Vocal fold cysts

- **Epidermoid cyst:** an inclusion cyst of keratin, congenital or from phonotrauma, white and deep in the fold, often against the ligament.
- **Mucous retention cyst:** smooth, glistening and fluid-filled.

Usually unilateral, sometimes with a reactive nodule on the opposite fold, and easily confused with nodules; stroboscopy shows a stiff segment with reduced mucosal wave, and they do not resolve with voice therapy. Microflap excision, ideally intact; residual wall recurs.



**FIGURE 6** An epidermoid cyst of the left vocal fold, full of cheesy material at excision.
Dhingra, fig. 61.7

## 4.3 Saccular cysts and laryngocele

Obstruction of the saccule's mouth fills it with mucus (a *saccular cyst*: anterior, bulging from the ventricle; or lateral, into the false fold and aryepiglottic fold and even the neck). Distension with air forms a *laryngocele*: internal, external (through the thyrohyoid membrane, a reducible neck swelling that grows with coughing or Valsalva) or combined, linked to raised pressure in wind players, glass blowers and weight lifters. Diagnosed by laryngoscopy, radiographs with Valsalva and CT. Endoscopic marsupialisation or excision for internal lesions; external excision for external and combined lesions. In an adult, a ventricular carcinoma obstructing the saccule must be looked for and biopsied.

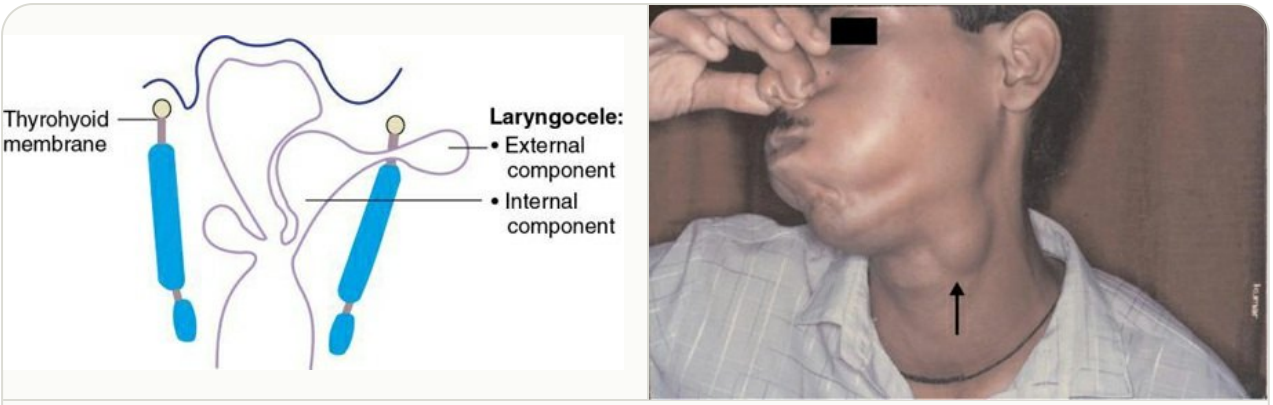
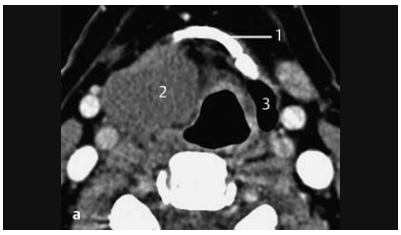



**FIGURE 7** First, a laryngocele of the mixed type with internal and external components. Second, a left laryngocele as seen on Valsalva (arrow).

Dhingra, figs 61.8 and 61.9

**IMAGING IN THE APP** 1 case

Laryngocele on imaging.




**Laryngocele**

Infection & inflammation · 10 images

## 4.4 Sulcus vocalis

A furrow along the free edge where the lamina propria is deficient and the epithelium dips to the ligament or muscle: congenital, or from phonotrauma, perhaps an opened epidermoid cyst. A breathy, weak voice with incomplete closure, a spindle-shaped gap and a reduced mucosal wave; confirmed by palpation at microlaryngoscopy. Prolonged voice therapy first; treat reflux and allergy. Surgery has limited success: steroid or filler injection, fat or fascia implantation, excision of the sulcus, slicing or multiple relaxing incisions, undermining, and medialisation thyroplasty for closure.

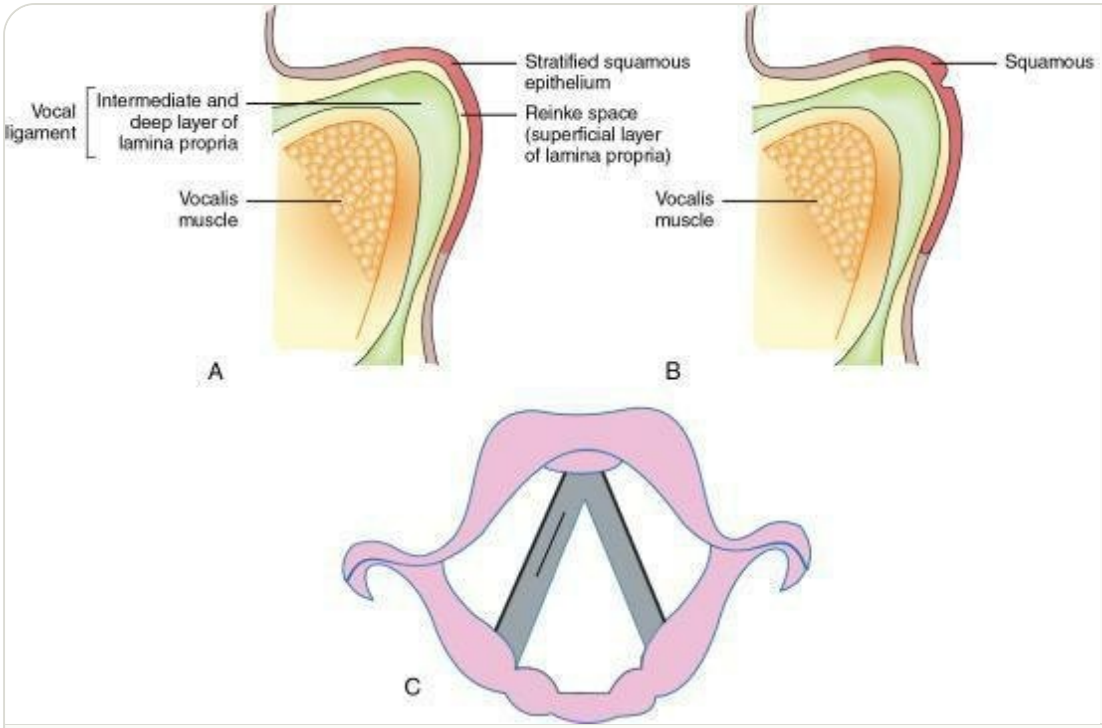



**FIGURE 8** (A) The layers of a normal vocal fold. (B) A sulcus vocalis in coronal section. (C) Its appearance on laryngoscopy.

Dhingra, fig. 61.10

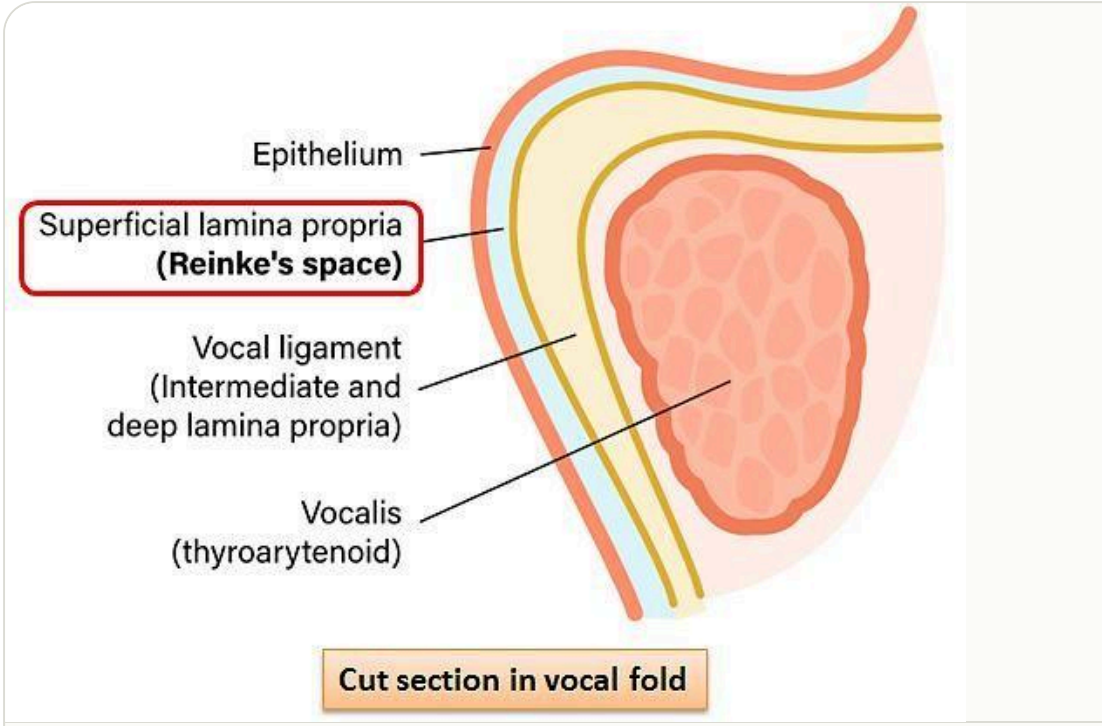



**FIGURE 9** The layers of the vocal fold: Reinke's space and the vocal ligament.

Anatomy atlas · ENT Steps p. 246

# 5 CHAPTER FIVE Neoplasms

Papillomas first.

## 5.1 Recurrent respiratory papillomatosis

The commonest benign tumour of the larynx (about 80% of benign neoplasms) and of the child's larynx, caused by *HPV 6 and 11*.

- **Juvenile onset:** acquired from mothers with genital warts at birth; first-born children of young mothers by vaginal delivery are at greatest risk (caesarean section does not reliably prevent it). Presents at two to five years with hoarseness, a weak cry or aphonia and then stridor; often misdiagnosed as asthma or croup. Multiple, aggressive and recurrent, at junctions of squamous and ciliated epithelium (supraglottis, glottis, subglottis), spreading to the trachea and bronchi, especially after tracheostomy.
- **Adult onset:** men of thirty to fifty, often single, at the anterior fold or commissure, less aggressive; malignant change is rare (more with HPV 16 and 18, smoking and irradiation).

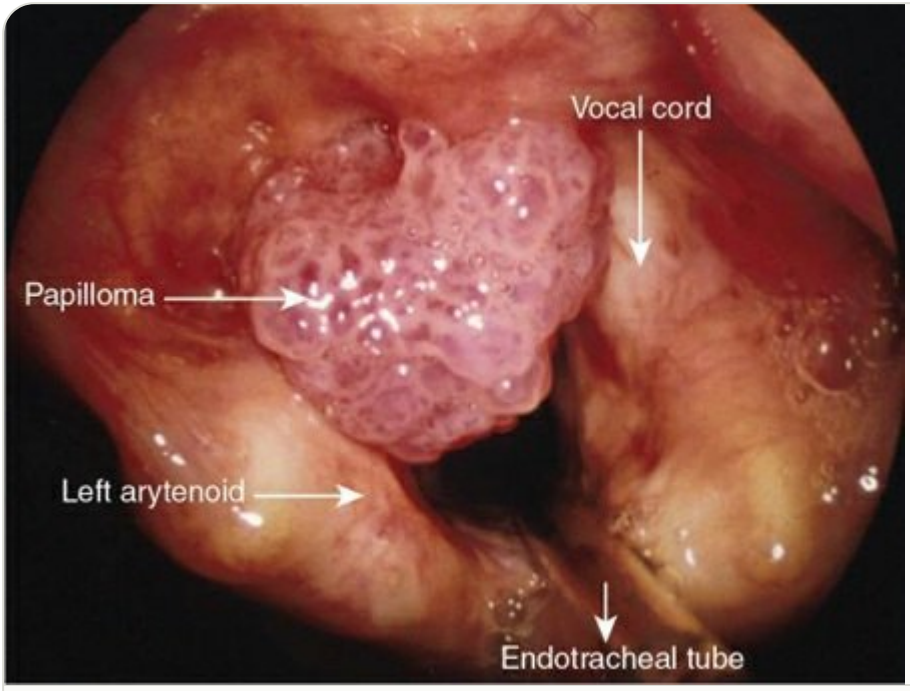



**FIGURE 10** Supraglottic papillomatosis.
Dhingra, fig. 61.11

### 5.1.1 Treatment

The aim is a safe airway and a useful voice with as few scars as possible; surgery debulks, it does not cure.

- **Surgery:** microdebrider, CO2 laser or pulsed KTP laser under suspension laryngoscopy, avoiding injury to the vocal ligament and the anterior commissure; repeated as often as needed. Office KTP laser in adults.
- **Adjuvants** for frequent recurrence or distal spread: intralesional *cidofovir*, systemic or intralesional *bevacizumab*, indole-3-carbinol, and (older) interferon alpha and retinoids; the HPV vaccine as therapy is under study, and as prevention will reduce maternal transmission.
- **Tracheostomy** is avoided if possible, because it encourages tracheal spread.

**OPERATIVE GUIDES IN THEATRE** 1 operation

Recurrent respiratory papillomatosis.

**Recurrent Respiratory Papillomatosis**
Papillomatosis
›

## 5.2 Chondroma

Usually from the posterior lamina of the cricoid, also the thyroid, arytenoid or epiglottis; men of forty to sixty. A slow, hard, submucosal mass causing dyspnoea when subglottic, or a lump feeling and dysphagia when growing back into the hypopharynx. CT shows a calcified cartilaginous mass; biopsy (with the laser through hard tissue) to exclude chondrosarcoma. Excision through a laryngofissure or lateral pharyngotomy, preserving as much cricoid as possible; laryngectomy for large or recurrent tumours.

## 5.3 Haemangioma

- **Infantile:** subglottic, capillary, presenting with stridor in the first six months, half with skin haemangiomas; involutes, and is treated with propranolol (see congenital lesions).
- **Adult:** cavernous, on the fold or supraglottis; left alone if silent, treated with steroids, laser or excision if symptomatic.

### 5.4 Granular cell tumour

From Schwann cells, submucosal, usually on the posterior fold; the overlying *pseudoepitheliomatous hyperplasia* can be mistaken for well-differentiated carcinoma. Complete excision prevents recurrence.

### 5.5 Others

Pleomorphic adenoma, oncocytoma, neurilemmoma and neurofibroma, rhabdomyoma, lipoma and fibroma are rare and excised endoscopically or through an external approach.

## 6 CHAPTER SIX For the examination

What the examiner wants to hear.

### 6.1 Viva essentials

**WHAT THE EXAMINER IS LOOKING FOR**

- **Site of nodules** and why there.
- **Nodule versus polyp versus cyst:** bilaterality, stroboscopy and response to therapy.
- **Reinke's oedema** and smoking.
- **Contact granuloma:** reflux and voice misuse.
- **Leukoplakia** as a premalignant lesion.
- **Laryngocele:** types and the hidden carcinoma.
- **Recurrent respiratory papillomatosis:** HPV types, behaviour, treatment and adjuvants, and why tracheostomy is avoided.

### 6.2 Common pitfalls

**COMMON PITFALLS**

- Operating on nodules before a proper course of voice therapy.

- Stripping both sides of the anterior commissure at once.
- Repeatedly observing a white patch in a smoker without a biopsy.
- Diagnosing a child with papillomas as asthma.

---

Written for this app from Oxford Textbook of Otolaryngology ch. 47, 71, Oxford Specialist Handbook ch. 8, Oxford Handbook of ENT ch. 12, Dhingra ch. 61 and K. J. Lee ch. 50. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · LARYNX & AIRWAY

# Cancer of the larynx

Laryngeal cancer is a smoker's disease with an unusual mercy: when it starts on the vocal fold it makes the voice hoarse while it is still small and has almost no lymphatics to spread through, so it is often found early and cured. Above or below the folds it is silent, spreads to the neck, and is found late. The rule is simple: hoarseness for more than three weeks in an adult, above all a smoker, means the larynx must be seen. Treatment then balances cure against voice, swallowing and the stoma.

**AT A GLANCE**

- Hoarse smoker gets a scope.
- Glottic tumours speak early; others spread early.
- Laser or radiotherapy for early disease.
- Chemoradiation or laryngectomy for advanced.
- Image before biopsy; NICE: offer T3 a choice of chemoradiotherapy or surgery; consider surgery first for T4a.

## 1 CHAPTER ONE Overview

Who and why.

### 1.1 Epidemiology

The second commonest cancer of the upper aerodigestive tract and about 1 to 5% of all malignancies; men are affected four to eight times as often as women, mostly in the sixth and seventh decades. In India it accounts for about 4% of male cancers. Glottic cancer is the commonest in most series (about 60%), supraglottic about 35 to 40%, and subglottic only 1 to 2%; supraglottic cancer predominates in some populations.

### 1.2 Causes

- **Tobacco:** the dominant cause; current smokers carry a ten- to twentyfold risk, which falls substantially after ten years of stopping.

- **Alcohol:** especially for supraglottic cancer; with tobacco the effects multiply (about fifteenfold).
- **Others:** previous radiation to the neck, papillomatosis (HPV 6 and 11, rarely malignant), occupational exposures (asbestos, nickel, mustard gas, wood dust, paints, diesel fumes), reflux, and genetic susceptibility.

### 1.3 Premalignant lesions

Hyperplasia, keratosis, mild, moderate and severe dysplasia and carcinoma in situ form a spectrum. Mild to moderate dysplasia progresses in about 7 to 11%, severe dysplasia in 18 to 30%; severe dysplasia and carcinoma in situ behave alike. Excisional biopsy (microflap or laser) is the diagnosis and usually the treatment; radiotherapy for multifocal disease or the frail; lifelong follow-up.

## 2 CHAPTER TWO Pathology and regions

Where it starts determines how it behaves.

### 2.1 Histology

Squamous cell carcinoma in 85 to 95%; glottic tumours tend to be well differentiated, supraglottic ones less so. Others: *verrucous carcinoma* (1 to 2%; exophytic, well-differentiated, with pushing margins; does not metastasise; treated by excision rather than radiation), spindle cell and basaloid carcinoma, neuroendocrine tumours, salivary carcinomas, chondrosarcoma (usually the cricoid) and lymphoma.

## 2.2 Regions

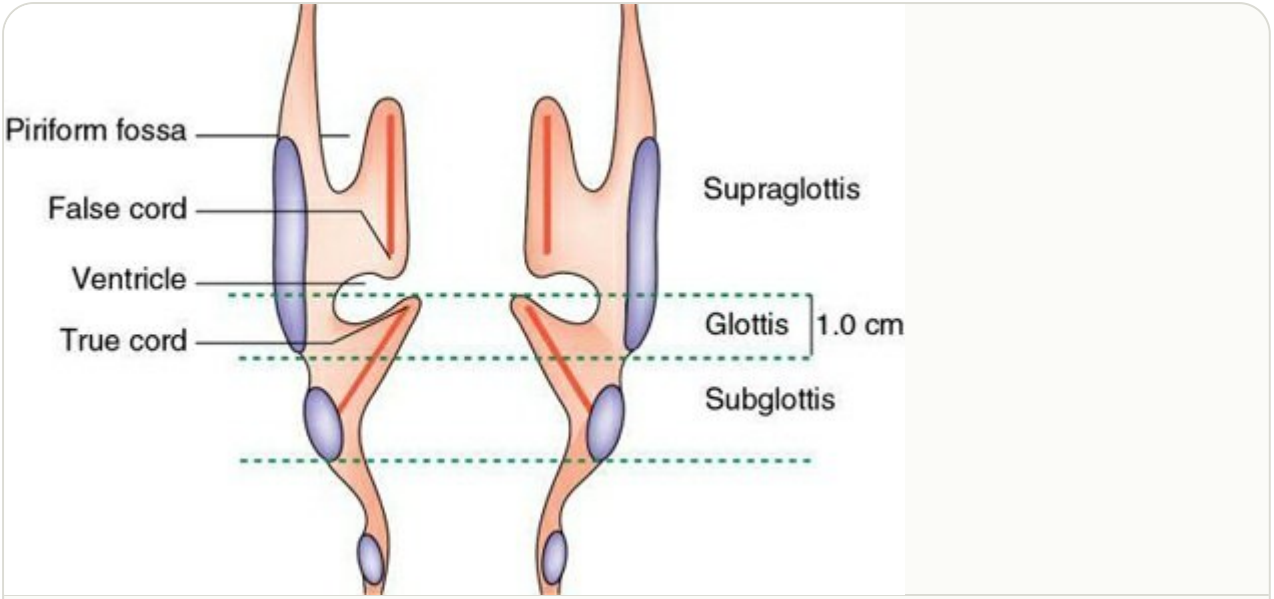



**FIGURE 1** The regions of the larynx (AJCC): the glottis extends from the ventricle to 1 cm below the free edge of the fold; the subglottis from there to the lower border of the cricoid.

Dhingra, fig. 62.1

- **Supraglottis:** suprahyoid and infrahyoid epiglottis, aryepiglottic folds, arytenoids, false folds and ventricles. Embryologically from the third and fourth arches without midline fusion, with rich, bilateral lymphatics.
- **Glottis:** the true folds with the anterior and posterior commissures, extending 1 cm below the free edge. From the sixth arch, with scanty lymphatics.
- **Subglottis:** to the lower border of the cricoid.

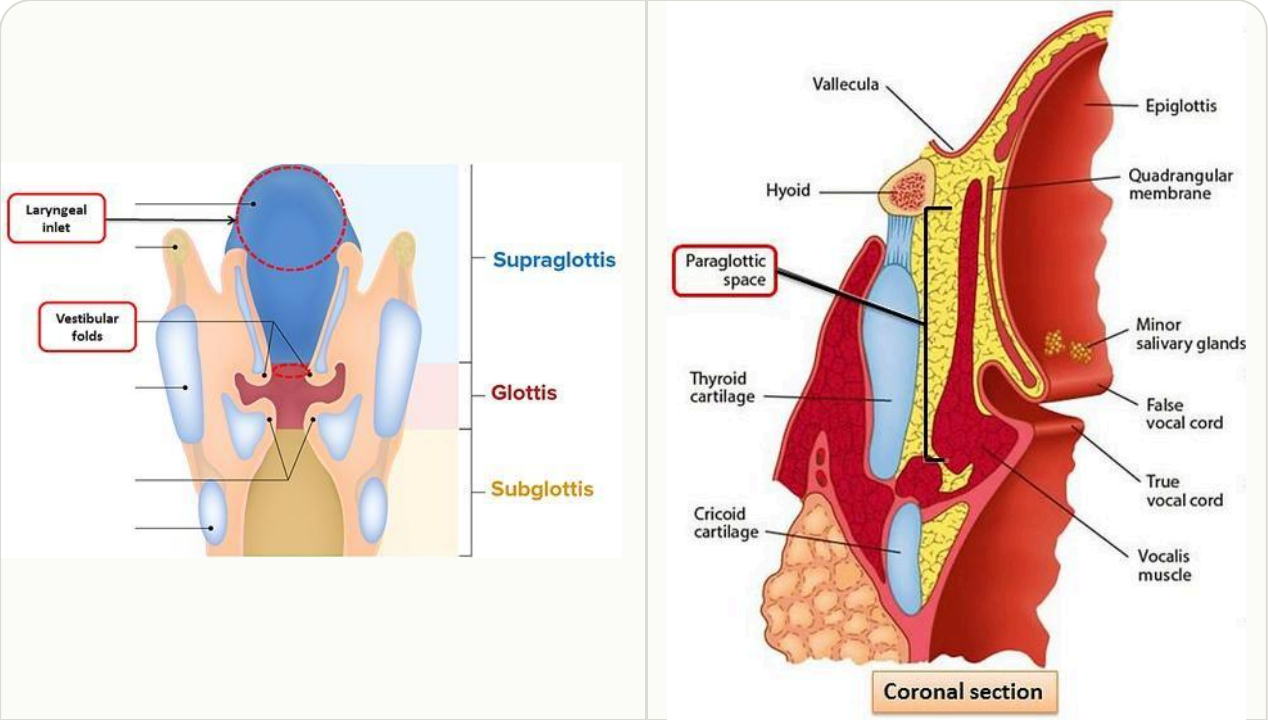



**FIGURE 2** Supraglottis, glottis and subglottis, and the paraglottic space.
Anatomy atlas · ENT Steps p. 239, ENT Steps p. 244

## 2.3 Barriers and pathways

The conus elasticus and the quadrangular membrane resist spread; the *pre-epiglottic space* (reached through pits in the epiglottis) and the *paraglottic space* (lateral to the membranes) carry tumour silently across the ventricle (*transglottic* cancer) and up to the thyroid cartilage. At the anterior commissure, *Broyles' ligament* inserts into thyroid cartilage without perichondrium, a route into the cartilage.

# 3 CHAPTER THREE Supraglottic cancer

Silent and node-seeking.

## 3.1 Sites

Epiglottis, false folds and aryepiglottic folds, in that order. Suprahyoid epiglottic tumours tend to be exophytic, infrahyoid ones ulcerative.

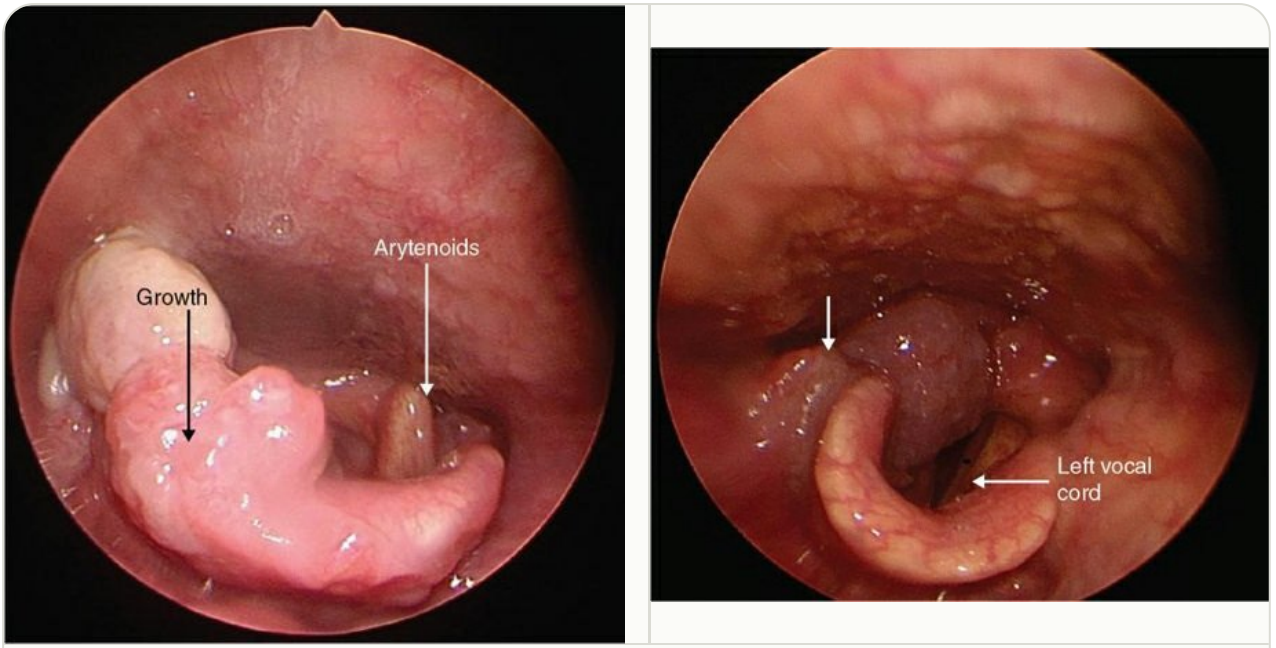



**FIGURE 3** First, a supraglottic cancer of the epiglottis and right aryepiglottic fold. Second, a supraglottic cancer also invading the right piriform fossa (arrow).
Dhingra, figs 62.2 and 62.3

## 3.2 Spread

To the vallecula, base of tongue, piriform fossa and pre-epiglottic space, and through the thyroid cartilage. Nodes (levels II to IV) appear early: 25 to 50% at presentation, often bilateral, with occult nodes in 20 to 40% of N0 necks.

## 3.3 Symptoms

Often silent for a long time: discomfort or pain in the throat, a lump feeling, painful swallowing, referred otalgia, a neck mass; hoarseness, dyspnoea, halitosis and weight loss come late.

# 4 CHAPTER FOUR Glottic cancer

Loud and contained.

## 4.1 Site

The free edge and upper surface of the anterior and middle thirds of the fold.

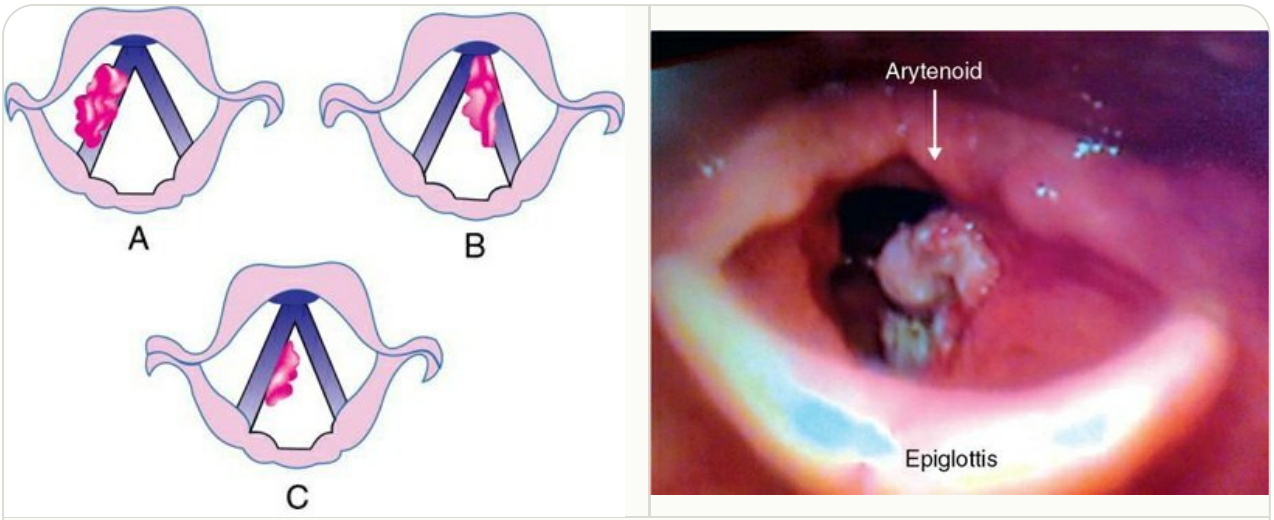



**FIGURE 4** First, cancer of the larynx: (A) supraglottic, (B) glottic, (C) subglottic. Second, a cancer involving the supraglottic, glottic and subglottic areas on the left, on rigid endoscopy.
Dhingra, figs 62.4 and 62.5

## 4.2 Spread

Forwards to the anterior commissure and opposite fold; backwards to the vocal process and arytenoid; upwards to the ventricle and false fold; downwards to the subglottis; deeply into the thyroarytenoid muscle and paraglottic space. *Impaired mobility* means deep invasion of muscle, and *fixation* means invasion of the paraglottic space, cricoarytenoid joint or nerve: a bad prognostic sign. Because the folds have almost no lymphatics, node metastases are rare (under 5% in T1 and T2) until the tumour leaves the membranous fold.

### 4.3 Symptoms

*Hoarseness* early, because even a tiny lesion alters vibration; stridor when the tumour enlarges or fixes the fold.

---

CHAPTER FIVE

# 5 Subglottic cancer

Rare and late.

It starts on one side, spreads around the front and down into the trachea, through the cricothyroid membrane into the thyroid gland and strap muscles; nodes go to the prelaryngeal (Delphian), pretracheal, paratracheal (level VI), lower jugular and superior mediastinal groups. Stridor and dyspnoea are often the first symptoms; hoarseness is late, meaning spread to the undersurface of the fold, the thyroarytenoid or the recurrent nerve. It is usually advanced and poorly differentiated, and prone to stomal recurrence.

CHAPTER SIX

# 6 Diagnosis and staging

Seeing and mapping.

### 6.1 History and examination

Any patient of cancer age with hoarseness for more than two to three weeks needs laryngoscopy. Also ask about sore throat, otalgia, dysphagia, aspiration, dyspnoea, haemoptysis, weight loss, smoking and drinking, and lung function.

- **Laryngoscopy:** mirror, flexible or rigid video laryngoscopy shows the appearance (a nodule, ulcer, thickening or granular mass), mobility of each fold, and extent to the vallecula, base of tongue and piriform fossae. Narrow-band imaging helps separate malignant from benign lesions; office biopsy is possible with flexible scopes. *Stroboscopy* shows a stiff, non-vibrating segment in invasive cancer.
- **Neck:** extralaryngeal spread (a midline swelling over the cricothyroid membrane, tender cartilage from perichondritis, thyroid and strap muscle involvement, widening of the thyroid cartilage) and nodes (number, size, side, fixity).



**FIGURE 5** Flexible fibreoptic laryngoscopy in the clinic.
Dhingra, fig. 62.6

## 6.2 Imaging

- **CT** of the neck with contrast: pre-epiglottic and paraglottic space invasion, cartilage destruction, extralaryngeal spread and nodes.
- **MRI:** more sensitive for cartilage invasion and for recurrence after radiotherapy.
- **CT of the chest:** lung metastases and second primaries (a solitary lung nodule is more likely a second primary).
- **PET-CT** for advanced disease and after chemoradiation (at 10 to 12 weeks) to detect residual disease.

## 6.3 Endoscopy under anaesthesia

Direct laryngoscopy shows the *hidden areas* (infrahyoid epiglottis, anterior commissure, ventricle, subglottis), maps the extent and allows biopsy; microlaryngoscopy for small fold lesions; panendoscopy for second primaries. Toluidine blue staining once guided biopsy of keratotic patches.

## 6.4 T staging (AJCC 8th edition)

### T STAGING OF LARYNGEAL CANCER

| CATEGORY | SUPRAGLOTTIS | GLOTTIS | SUBGLOTTIS |
|---|---|---|---|
| T1 | One subsite, normal fold mobility | Folds only, normal mobility (T1a one fold, T1b both) | Limited to the subglottis |
| T2 | More than one subsite of supraglottis, or glottis, or outside (base of tongue, vallecula, medial piriform wall), without fixation | Extends to supraglottis or subglottis, or impaired mobility | Extends to the folds, with normal or impaired mobility |
| T3 | Fold fixation, or postcricoid area, pre-epiglottic space, paraglottic space, or inner cortex of thyroid cartilage | Fold fixation, paraglottic space, or inner cortex of thyroid cartilage | Fold fixation |
| T4a | Through the thyroid cartilage or beyond the larynx (trachea, soft tissues, deep tongue muscle, strap muscles, thyroid, oesophagus) | As supraglottis | Cricoid or thyroid cartilage, or beyond the larynx |
| T4b | Prevertebral space, encasing the carotid, or mediastinum | As supraglottis | As supraglottis |

N staging follows the general head and neck rules, now including *extranodal extension*, which upstages nodes (N2a and N3b); Dhingra prints the earlier (6th edition) tables, based on size alone.

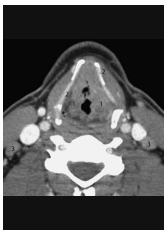
**IMAGING IN THE APP** 5 cases

Glottic and supraglottic tumours, pharyngeal tumours and synchronous malignancies on imaging.



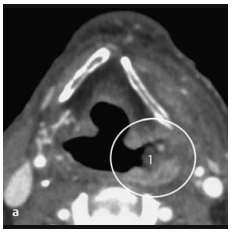
Glottic Tumor

Tumours · 5 images



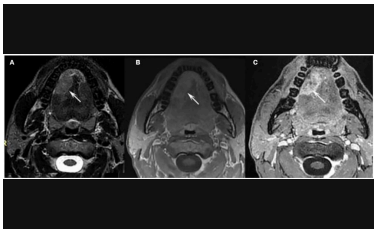
Supraglottic Tumor

Tumours · 9 images



Supraglottic and Pharyngeal Tumors

Tumours · 4 images



Squamous cell carcinoma: oral cavity, oropharynx and larynx

After surgery · 14 images



Multiple Synchronous Malignancies

Tumours · 3 images

# 7 CHAPTER SEVEN Treatment

Cure, then function.

## 7.1 Principles

Early cancers (stage I and II) are treated by a single modality, radiotherapy or surgery, with similar control; the choice turns on voice, swallowing, access, fitness and preference. Advanced cancers (stage III and IV) need combined treatment, with organ preservation where it is safe and total laryngectomy where it is not.

## 7.2 Radiotherapy

For early tumours with mobile folds and no cartilage invasion or nodes: about 60 to 70 Gy over six weeks. T1 glottic cancer is controlled in about 90% or more with good voice; T2 in about 85% with normal mobility but much less (about 63 to 70%) when mobility is impaired. It spares the voice and avoids surgery but causes mucositis, oedema, dryness and occasional chondronecrosis, makes recurrence harder to detect, and cannot be repeated. Poor results with fixed folds, subglottic extension, cartilage invasion and nodes.

## 7.3 Transoral surgery

*Transoral laser microsurgery* (CO2 laser under the microscope), and more recently transoral robotic surgery, has largely replaced open partial laryngectomy for T1, T2 and selected T3 tumours: cordectomy for glottic cancer, supraglottic resection for supraglottic cancer. It is cheaper and quicker, avoids tracheostomy, keeps radiotherapy in reserve, and gives similar control to radiotherapy, but the voice may be poorer after wide cordectomy. It needs good exposure, and is contraindicated by fold fixation, interarytenoid involvement and poor lung function.

## 7.4 Open conservation surgery

Now used mainly in selected patients and after radiotherapy failure:

- **Cordectomy through a laryngofissure.**
- **Vertical partial (frontolateral, hemi-) laryngectomy:** one fold with the anterior commissure and adjacent thyroid cartilage.
- **Supraglottic (horizontal partial) laryngectomy:** removes the epiglottis, pre-epiglottic space, false folds and aryepiglottic folds above the ventricles; for T1 and T2 supraglottic cancer with mobile folds, clear of the anterior commissure, with good lung function, because every patient aspirates at first and must relearn swallowing.
- **Supracricoid partial laryngectomy:** removes both folds, the paraglottic spaces and the thyroid cartilage, rebuilt by cricohyoidopexy or cricohyoidoepiglottopexy; for selected T2 and T3 tumours with at least one mobile arytenoid.

## 7.5 Total laryngectomy

Removal of the whole larynx with the hyoid, pre-epiglottic space and upper trachea; the pharynx is closed and the trachea brought out as a permanent end stoma. Dhingra's indications:

- T3 and T4 tumours (fold fixed, cartilage invaded, extralaryngeal spread).
- Invasion of the thyroid or cricoid cartilage; both arytenoids; posterior commissure.
- Transglottic tumours with fold fixation.
- Failure after radiotherapy or conservation surgery.
- A non-functioning larynx with intractable aspiration.

Combined with neck dissection and postoperative radiotherapy as indicated; contraindicated by distant metastases.

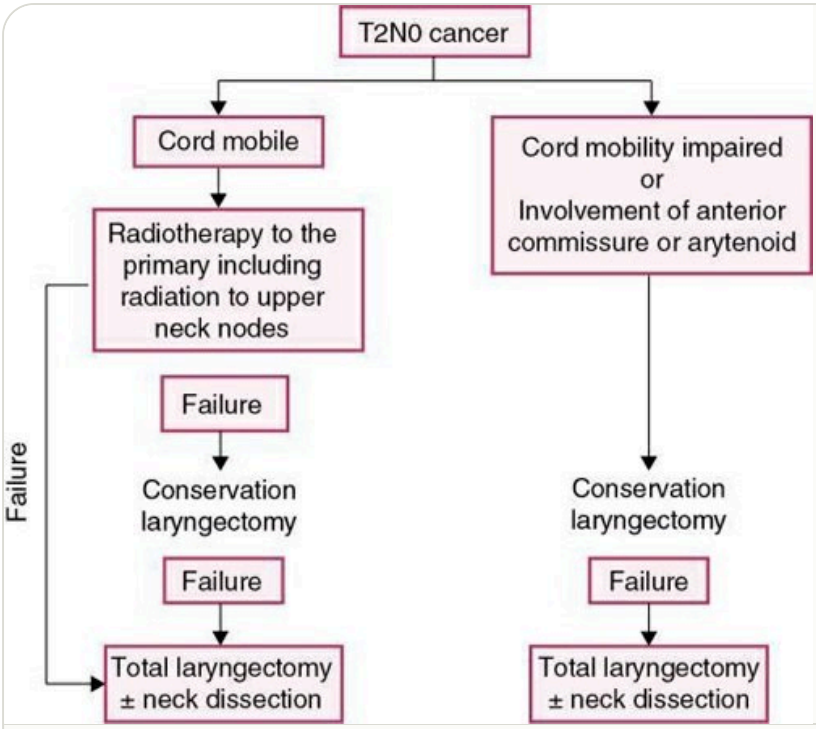



**FIGURE 6** Treatment of T2N0 glottic cancer: radiotherapy when the fold is mobile and the commissure and arytenoid are spared; conservation laryngectomy when mobility is impaired or the commissure or arytenoid is involved; total laryngectomy for failure.
Dhingra, fig. 62.7

## 7.6 Organ preservation

The *VA Larynx trial* (1991) showed that induction chemotherapy (cisplatin and fluorouracil) followed by radiotherapy in responders gave the same survival as laryngectomy, with two-thirds keeping their larynx. *RTOG 91-11* (2003) showed that concurrent cisplatin with radiotherapy gave the best laryngeal preservation and locoregional control, although at ten years overall survival did not differ. Concurrent chemoradiation is now standard for most T3 and node-positive T2 cancers, with salvage laryngectomy for failure; cetuximab or carboplatin for those unfit for cisplatin. T4a tumours through cartilage, a non-functioning larynx and severe airway obstruction are better treated by laryngectomy and adjuvant therapy, which gives better control.

**OPERATIVE GUIDES IN THEATRE** 8 operations

Laser microsurgery, partial and total laryngectomy, laryngopharyngectomy and voice restoration.

**Transoral CO2 Laser Microsurgery for Cancer of the Larynx**
Laser
›

**Horizontal Partial Laryngectomy**
Horizontal partial
›

**Vertical Partial Laryngectomy**
Vertical partial
›

**Supracricoid Partial Laryngectomy**
Supracricoid
›

**Supraglottic Transoral Robotic Surgery**
TORS supraglottic
›

**Total Laryngectomy**
Total laryngectomy
›

**Total Laryngopharyngectomy**
Laryngopharyngectomy
›

**Voice Restoration After Total Laryngectomy**
Voice restoration
›

## 7.7 By site

- **Carcinoma in situ:** transoral laser excision (or stripping) with histology; radiotherapy if multifocal or invasive.
- **Glottic T1:** radiotherapy or laser cordectomy; with anterior commissure involvement, radiotherapy or a frontolateral laryngectomy.
- **Glottic T2:** radiotherapy if the fold is mobile and the commissure and arytenoid are free; otherwise conservation surgery; total laryngectomy if it recurs. The N0 neck is not treated (occult nodes under 5 to 25%).
- **Glottic T3 and T4:** chemoradiation for selected T3; total laryngectomy with neck dissection and postoperative radiotherapy for T4a.
- **Supraglottic T1 and T2:** radiotherapy, laser resection or supraglottic laryngectomy, always with *bilateral elective treatment of levels II to IV*.

- **Subglottic:** radiotherapy for early lesions; total laryngectomy with hemithyroidectomy or total thyroidectomy and level VI and paratracheal dissection, and radiotherapy including the superior mediastinum, for advanced lesions.

### 7.8 The neck

Clinically positive nodes need neck dissection or radiotherapy. After chemoradiation, a complete response on PET-CT at about twelve weeks allows surveillance instead of planned neck dissection. Postoperative chemoradiation is given for positive margins or extranodal extension.

## 8 CHAPTER EIGHT After laryngectomy

Voice, smell and follow-up.

### 8.1 Voice restoration

COMMUNICATION AFTER TOTAL LARYNGECTOMY

| METHOD | HOW IT WORKS | COMMENTS |
|---|---|---|
| Tracheo-oesophageal speech | A one-way valved prosthesis (Blom–Singer, Provox) in a puncture between trachea and oesophagus diverts lung air into the pharynx when the stoma is occluded | The best voice; most widely used; valve care and replacement needed |
| Oesophageal speech | Air is swallowed or injected into the upper oesophagus and released to vibrate the pharyngo-oesophageal segment | No device; rough but loud; hard to learn; six to ten words per breath |
| Electrolarynx | A battery-powered vibrator held against the neck (or a tube into the mouth) supplies sound that the mouth shapes | Easy to learn; mechanical voice |
| Writing and lip speech | | Temporary |

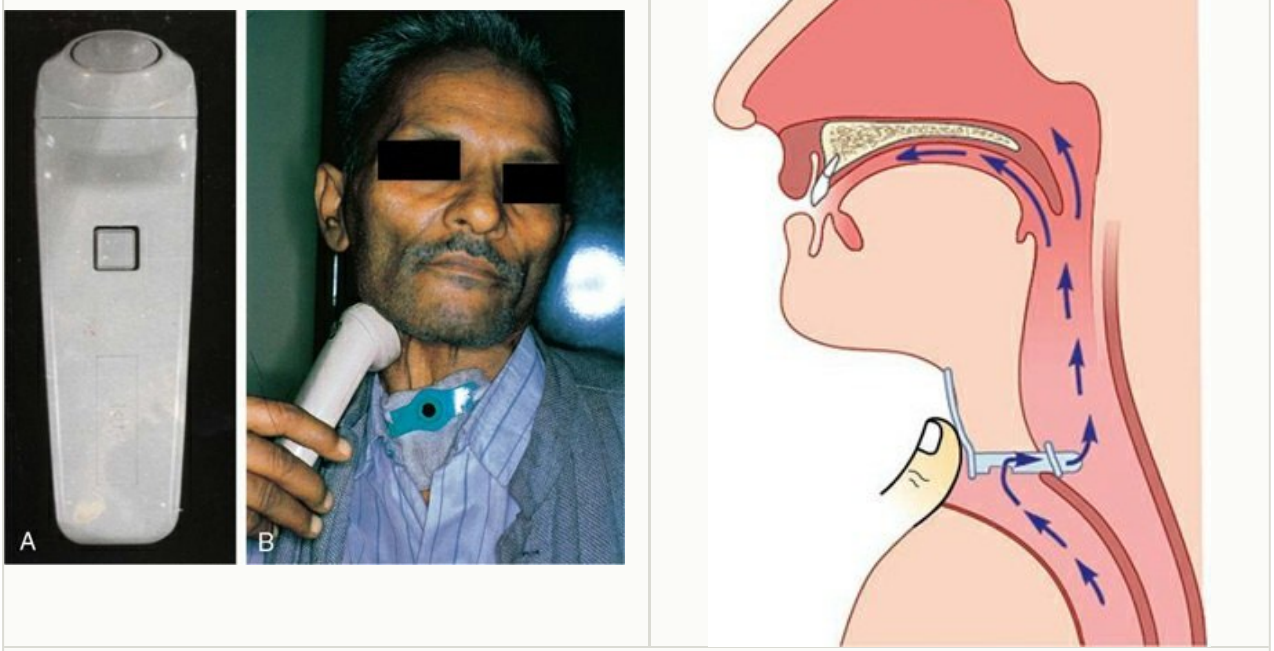



**FIGURE 7** First, (A) an electrolarynx and (B) a laryngectomised patient using it. Second, tracheo-oesophageal speech: the thumb closes the stoma and air passes through the prosthesis into the oesophagus.
Dhingra, figs 62.8 and 62.9

## 8.2 Other rehabilitation

- **Smell:** lost because air no longer passes through the nose; the *polite yawning* manoeuvre (lowering the jaw and tongue with the lips closed) draws air through the nose.
- **Stoma care:** humidifying heat-moisture exchangers, protection from water.
- **Swallowing** and **thyroid function** (hypothyroidism in a fifth to two-thirds after treatment).

## 8.3 Complications

Pharyngocutaneous fistula (more after radiotherapy; reduced by a pectoralis major or free flap), stomal stenosis and stomal recurrence, pharyngeal stricture, chondronecrosis after radiotherapy, and a non-functioning larynx after chemoradiation.

## 8.4 Follow-up and recurrence

Every one to three months in the first year, less often thereafter, lifelong, looking for recurrence and second primaries. Recurrence shows as new oedema, pain, worse voice or reduced mobility; PET-CT and biopsy confirm it; about half are salvageable, usually by total laryngectomy (selected small recurrences by conservation surgery).

## 8.5 Prognosis

Five-year survival is about 90% for T1 glottic cancer, falling with stage; supraglottic and subglottic cancers do worse than glottic. Nodes, extranodal extension, cartilage invasion and subglottic extension are adverse.

### NICE PATHWAY AND THE DECLINE IN SURVIVAL (OXFORD TEXTBOOK)

In the UK about 2,400 people a year develop laryngeal cancer, many from deprived communities; tobacco and alcohol account for about three-quarters of head and neck cancer. When cancer is suspected and the airway is safe, **cross-sectional imaging is done before biopsy**, because post-biopsy swelling can distort staging; ultrasound assesses the neck and allows node sampling, and panendoscopy looks for a synchronous primary (1 to 6%). Speech and language therapy assesses voice and swallow before treatment decisions.

**Early disease (T1–T2, N0)** is treated by transoral laser surgery or radiotherapy with similar control; overall survival ranges from about 45 to 77%. Laser is completed in a day and is cheaper, whereas radiotherapy takes three to seven weeks; attempts at a randomised trial have failed to recruit. Nodal spread is rare, so the neck is not routinely treated.

**Advanced disease (T3–T4)** is about 40% of cases. NICE advises offering T3 patients a choice of **chemoradiotherapy** or **surgery with adjuvant (chemo)radiotherapy**, and considering surgery first for T4a. The VA and RTOG 91-11 trials established larynx preservation with equivalent locoregional control, but population survival has fallen since practice shifted to non-surgical treatment, and patients who fail chemoradiotherapy face salvage laryngectomy with high complication rates and poor quality of life. Factors favouring primary laryngectomy are a poorly functioning larynx (aspiration, weight loss, tube feeding), a tracheostomy, T4 disease and patient preference. Quality of life is similar overall after surgical and non-surgical treatment, so

decisions should centre on the individual patient's values about voice and swallowing, made in the multidisciplinary team.

# 9 CHAPTER NINE For the examination

What the examiner wants to hear.

## 9.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Regions and subsites,** and why glottic cancer rarely spreads to nodes.
- **Pre-epiglottic and paraglottic spaces,** transglottic spread, Broyles' ligament.
- **Symptoms** by site; the hoarseness rule.
- **T staging** and the significance of fold mobility.
- **Treatment:** radiotherapy versus laser for early cancer; organ preservation (VA and RTOG 91-11); indications for total laryngectomy.
- **Voice rehabilitation** after laryngectomy.

## 9.2 Common pitfalls

COMMON PITFALLS

- Treating persistent hoarseness in a smoker with antibiotics and voice rest.
- Missing the anterior commissure, ventricle or subglottis without direct laryngoscopy.
- Forgetting bilateral neck treatment in supraglottic cancer.
- Offering organ preservation to a T4a tumour through cartilage.

Written for this app from Oxford Textbook of Otolaryngology ch. 55, Oxford Specialist Handbook ch. 8, Oxford Handbook of ENT ch. 12, Dhingra ch. 62 and K. J. Lee ch. 44. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · LARYNX & AIRWAY

# Applied anatomy and physiology of the larynx

The larynx is a valve before it is a voice. It evolved to guard the lungs, closing in three tiers when we swallow and coughing out whatever gets past; phonation came later, borrowing the same folds. Its framework of cartilages, its membranes and spaces, and its two nerves explain nearly everything in laryngology: why glottic cancer rarely spreads to nodes, why a supraglottic one does, why a long left recurrent nerve is so often injured, and why a baby's airway closes so quickly.

**AT A GLANCE**

- Posterior cricoarytenoid is the only abductor.
- Cricothyroid is supplied by the external laryngeal nerve.
- The glottis has almost no lymphatics; supraglottis drains richly and bilaterally.
- In infants the subglottis is the narrowest part.

## 1 CHAPTER ONE Framework

Cartilages, joints and membranes.

### 1.1 Position

The larynx lies in front of the hypopharynx, opposite the third to sixth cervical vertebrae in the adult, ending at the lower border of C6. It rises and moves forwards in swallowing and can be moved from side to side over the spine with a grating feel, the *laryngeal crepitus*, which is lost when a postcricoid tumour fills the gap.

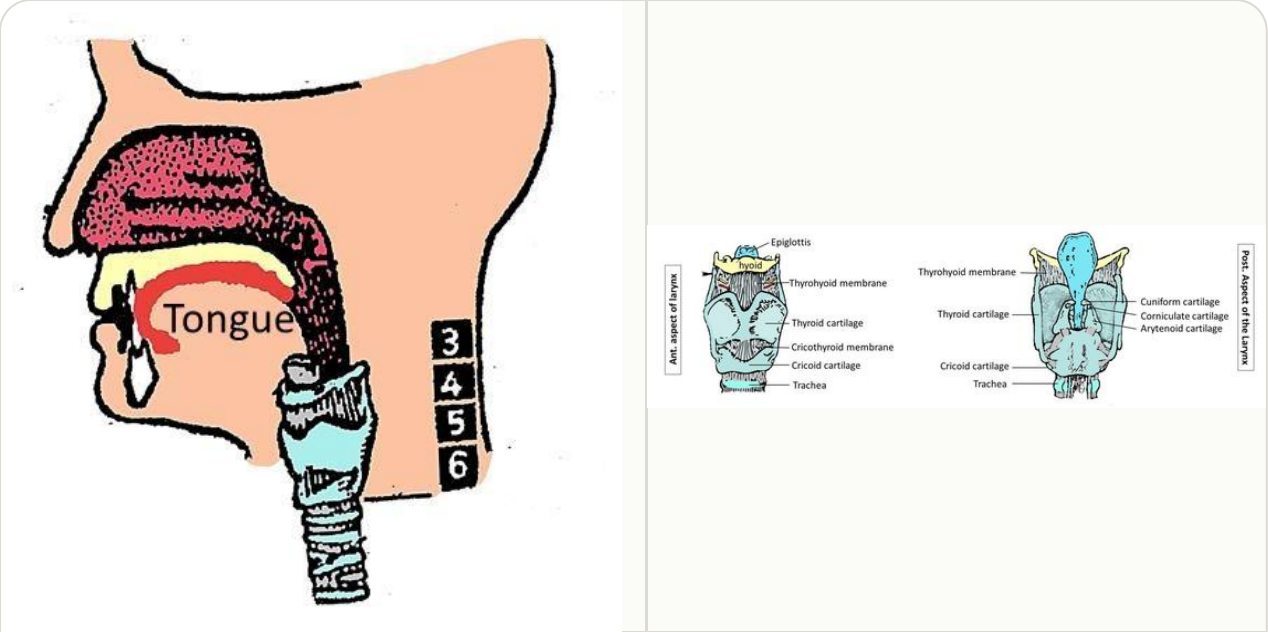



**FIGURE 1** The site of the larynx from C3 to C6, and its anterior and posterior aspects.
Anatomy atlas · Doss p. 170

## 1.2 Cartilages

Three unpaired and three paired.

- **Thyroid:** the largest; two laminae meeting in front at about 90° in men (the Adam's apple) and 120° in women, with superior and inferior horns. The vocal folds attach to the inner surface of the angle at about its midpoint, so an airway can be made below them through the cricothyroid membrane.
- **Cricoid:** the only complete ring, a narrow arch in front and a tall lamina behind; it holds the airway open, and its loss or injury causes stenosis.
- **Epiglottis:** a leaf of elastic cartilage attached by its stalk (petiole) to the thyroid angle above the vocal folds, and by the hyoepiglottic ligament to the hyoid, which divides it into suprahyoid and infrahyoid parts. Its posterior bulge, the *tubercle*, hides the anterior commissure on mirror examination; pits and perforations in it let supraglottic cancer reach the pre-epiglottic space. It is not essential for swallowing and can be removed with little aspiration.
- **Arytenoids:** pyramids sitting on the cricoid lamina, each with a *vocal process* forwards (for the vocal ligament) and a *muscular process* laterally (for the cricoarytenoid muscles).
- **Corniculate (Santorini)** on the arytenoid apex, and **cuneiform (Wrisberg)** in the aryepiglottic folds.

Thyroid, cricoid and most of the arytenoid are hyaline and ossify from about 25 years; ossified cartilage can be mistaken for a foreign body on radiographs. The epiglottis, corniculate, cuneiform and arytenoid tips are elastic and do not ossify.

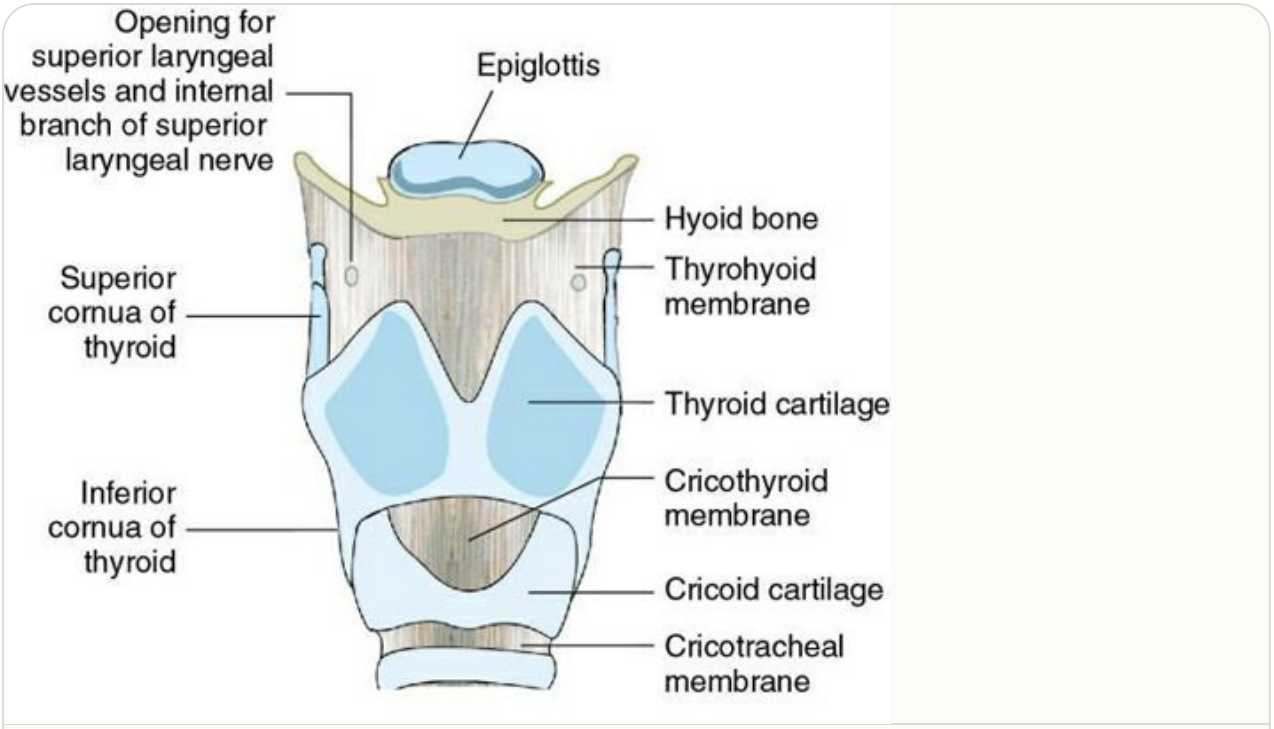



**FIGURE 2** The laryngeal framework: thyroid, cricoid, epiglottis, hyoid and the thyrohyoid, cricothyroid and cricotracheal membranes.

Dhingra, fig. 56.1

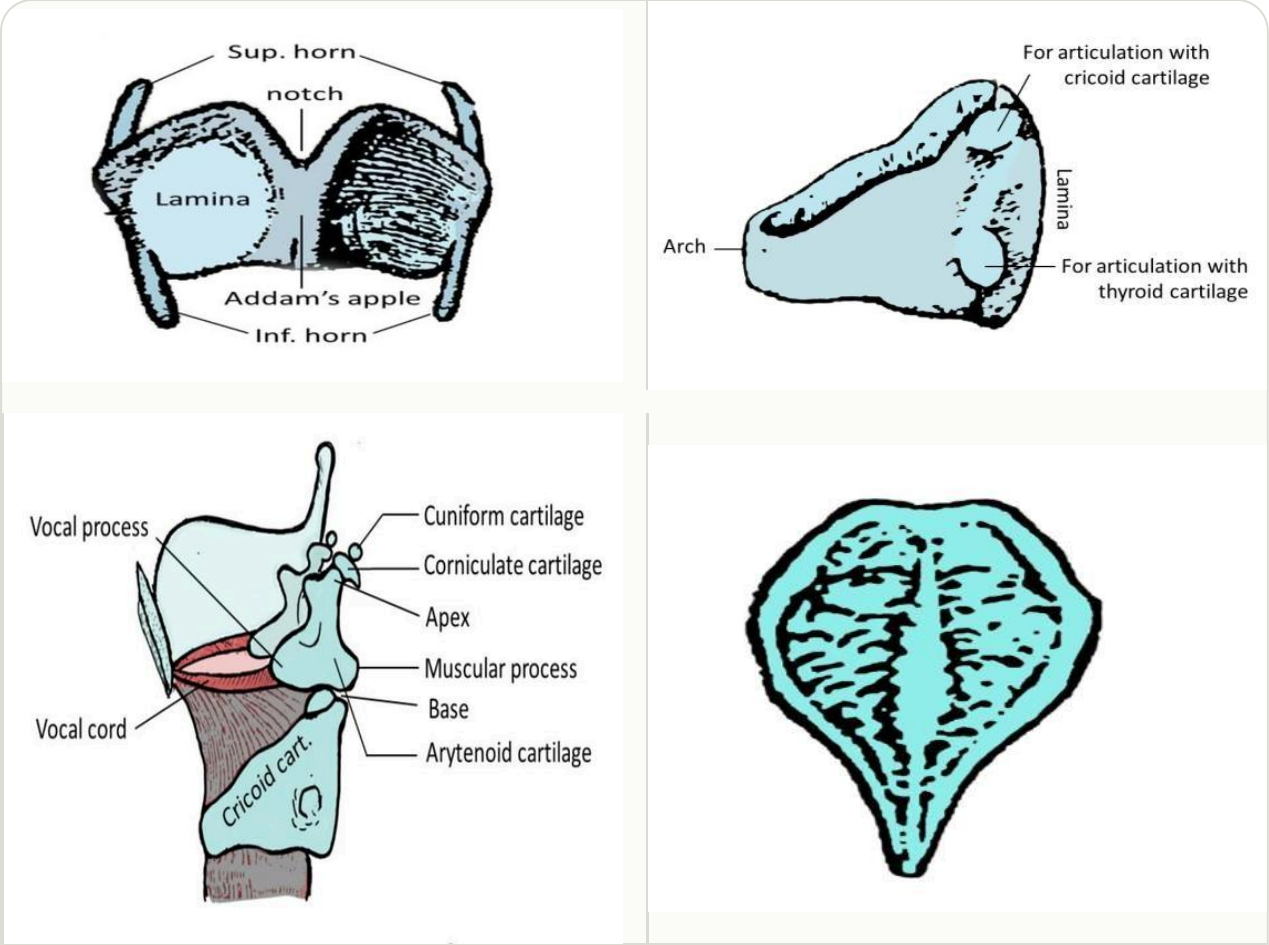



**FIGURE 3** The thyroid, cricoid, arytenoid and epiglottic cartilages.

Anatomy atlas · Doss p. 171

## 1.3 Joints

- **Cricoarytenoid:** synovial; the arytenoid rotates about a vertical axis (abducting and adducting the vocal fold) and glides towards or away from its fellow (closing or opening the posterior glottis). Rheumatoid arthritis and intubation injury can fix it.
- **Cricothyroid:** synovial; the cricoid tilts on a transverse axis, lengthening and tensing the folds.

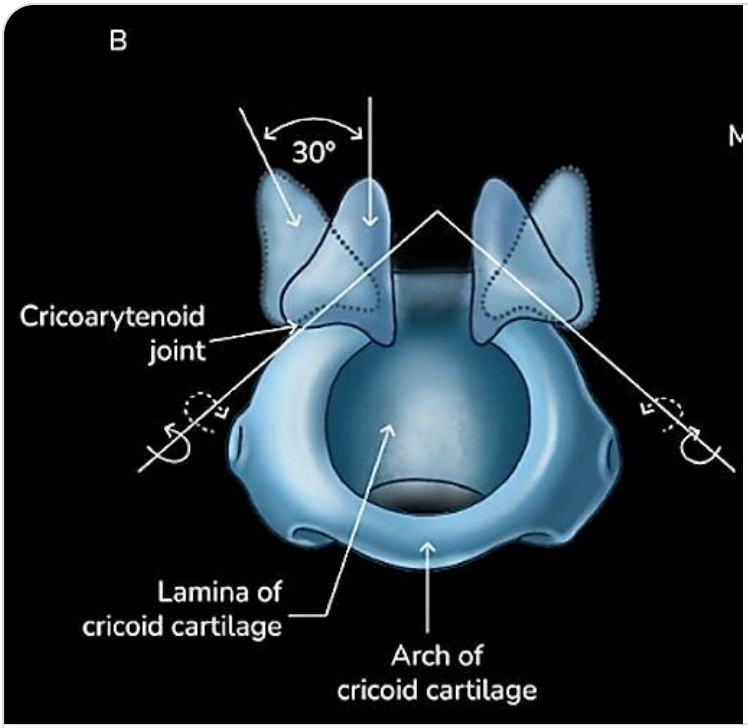



**FIGURE 4** The cricoarytenoid joint.
Anatomy atlas · ENT Steps p. 234

## 1.4 Membranes and ligaments

- **Extrinsic:** the *thyrohyoid membrane* (pierced by the superior laryngeal vessels and internal laryngeal nerve), the cricotracheal membrane and the hyoepiglottic ligament.
- **Intrinsic:** the *cricovocal membrane (conus elasticus)*, whose free upper edge is the vocal ligament and whose two halves funnel up from the cricoid, a place where subglottic foreign bodies lodge; the *quadrangular membrane* between epiglottis and arytenoid, whose lower edge is the vestibular ligament; the *cricothyroid ligament* (the thickened front of the conus); and the thyroepiglottic ligament.

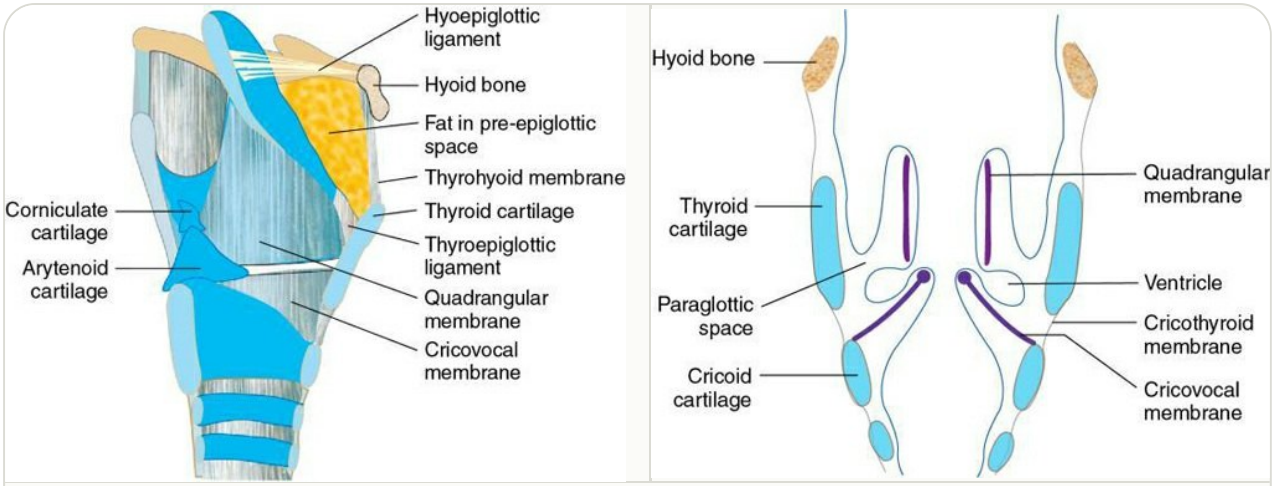



**FIGURE 5** First, a sagittal section showing the cricovocal and quadrangular membranes and the pre-epiglottic space. Second, a coronal section: the quadrangular membrane ends in the false cord, the cricovocal membrane in the vocal ligament, forming the conus elasticus.

Dhingra, figs 56.2 and 56.3

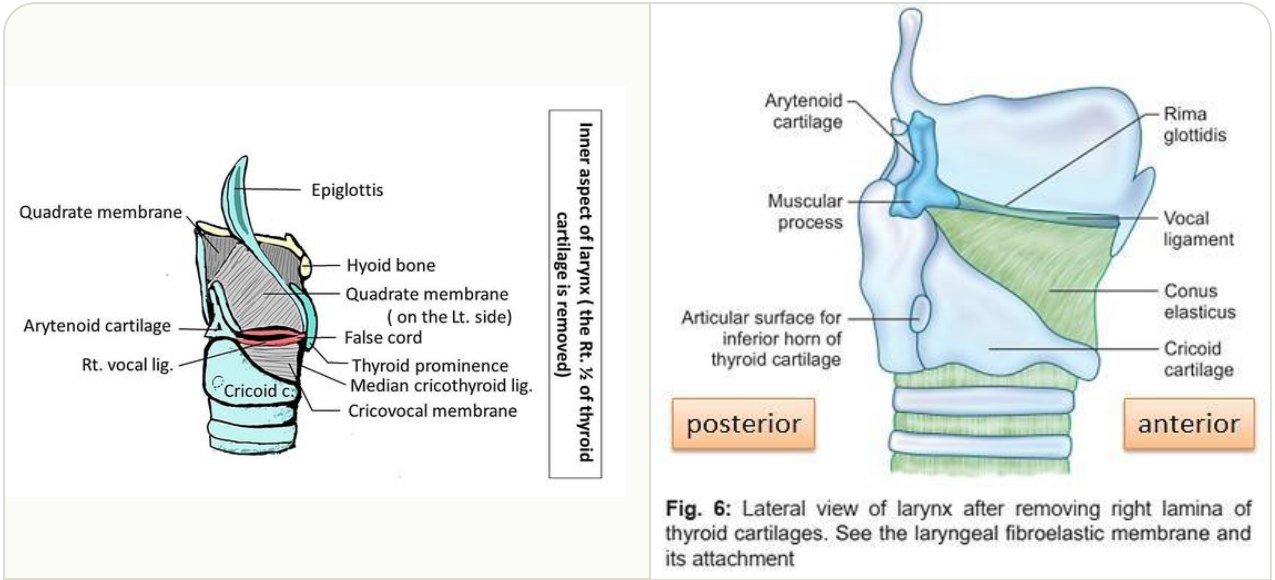



**FIGURE 6** The membranes of the larynx, and the conus elasticus seen after removing a thyroid lamina.

Anatomy atlas · Doss p. 172, ENT Steps p. 237

# 2 CHAPTER TWO Muscles

Opening, closing, tensing.

## 2.1 Intrinsic muscles

### INTRINSIC LARYNGEAL MUSCLES

| ACTION | MUSCLES |
|---|---|
| Abduct the vocal folds | Posterior cricoarytenoid (the only abductor) |
| Adduct the vocal folds | Lateral cricoarytenoid, interarytenoid (transverse arytenoid), external thyroarytenoid |
| Tense the vocal folds | Cricothyroid (lengthens), vocalis (internal thyroarytenoid; fine tension) |
| Relax the vocal folds | Thyroarytenoid |
| Open the inlet | Thyroepiglottic |
| Close the inlet | Oblique interarytenoid and aryepiglottic |

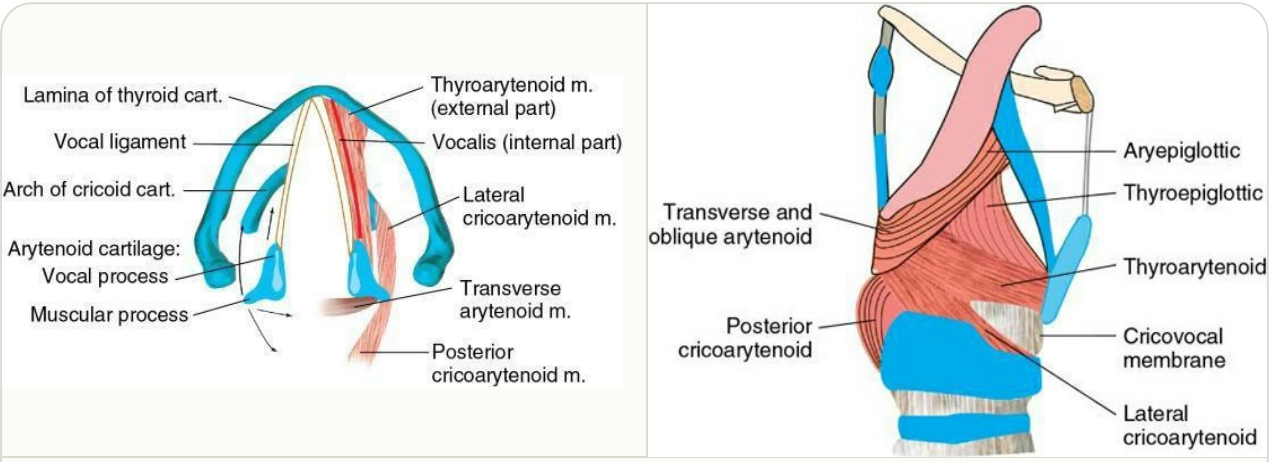



**FIGURE 7** First, the laryngeal muscles and their actions on the arytenoid. Second, the intrinsic muscles seen from the side.

Dhingra, figs 56.4 and 56.5

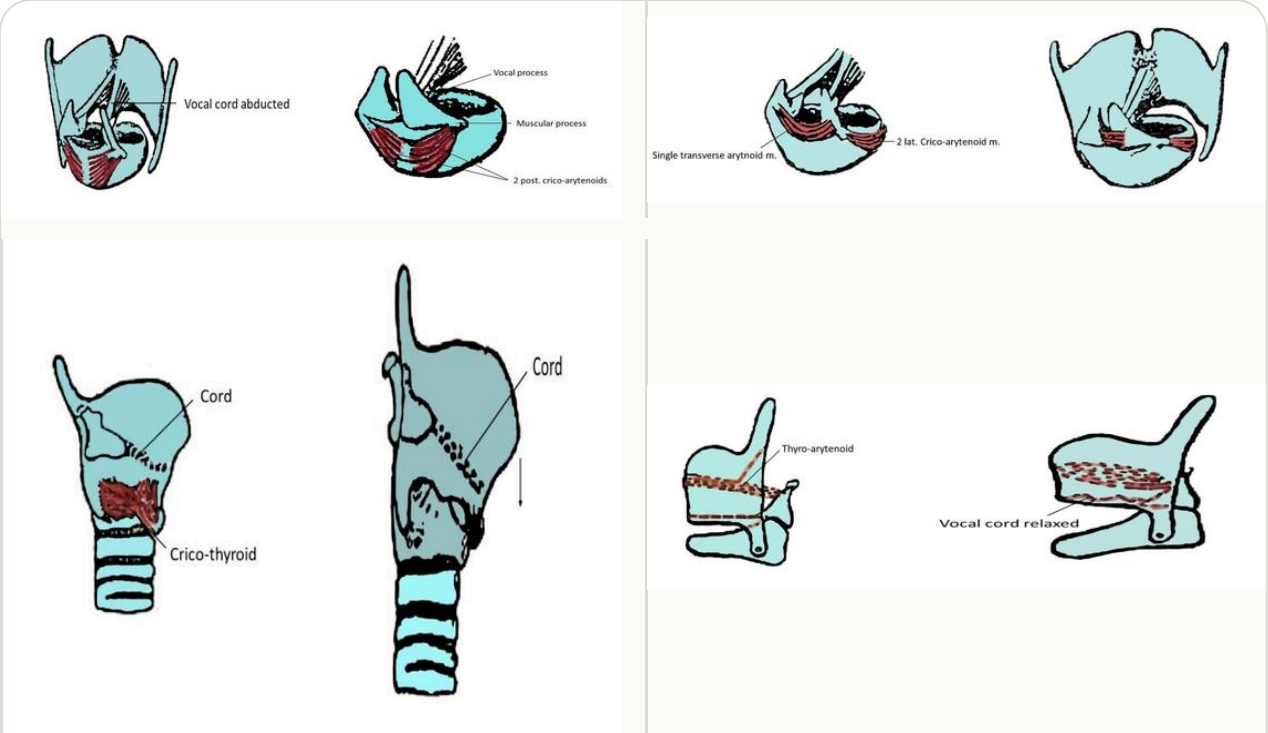



**FIGURE 8** The posterior cricoarytenoid abducts; the lateral cricoarytenoid and interarytenoid adduct; the cricothyroid stretches and the thyroarytenoid relaxes the folds.
Anatomy atlas · Doss p. 174, Doss p. 173

**THE ONE ABDUCTOR**

The posterior cricoarytenoid alone opens the glottis. Lose it on both sides and the folds lie near the midline with a good voice and a poor airway.

## 2.2 Extrinsic muscles

- **Primary elevators** (attached to the thyroid cartilage): stylopharyngeus, salpingopharyngeus, palatopharyngeus and thyrohyoid.
- **Secondary elevators** (through the hyoid): mylohyoid, digastric, stylohyoid and geniohyoid.
- **Depressors:** sternohyoid, sternothyroid and omohyoid.

# 3 CHAPTER THREE Cavity and spaces

Above, at and below the folds.

## 3.1 The cavity

From the *inlet* (bounded by the epiglottis, aryepiglottic folds and interarytenoid fold) to the lower border of the cricoid. Two pairs of folds divide it:

- **Vestibule:** from the inlet down to the vestibular (false) folds.
- **Ventricle:** the recess between false and true folds; from its front a pouch, the *saccule*, rises between the false fold and the thyroid lamina, lubricating the folds with mucus. Its pathological enlargement is a *laryngocele*.
- **Subglottis (infraglottic space):** from the vocal folds to the lower border of the cricoid.

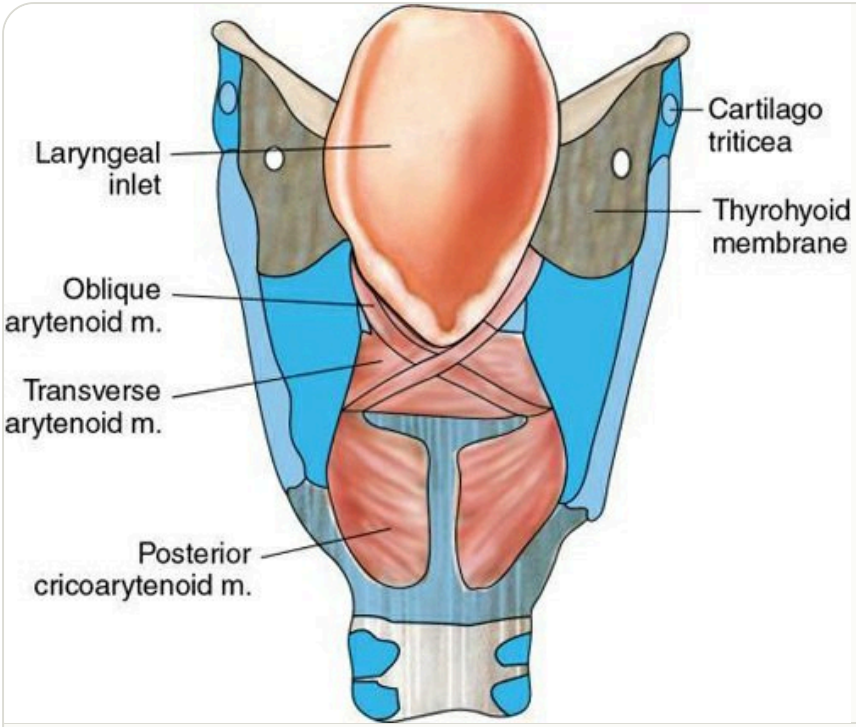



**FIGURE 9** The laryngeal inlet and intrinsic muscles seen from behind.
Dhingra, fig. 56.6

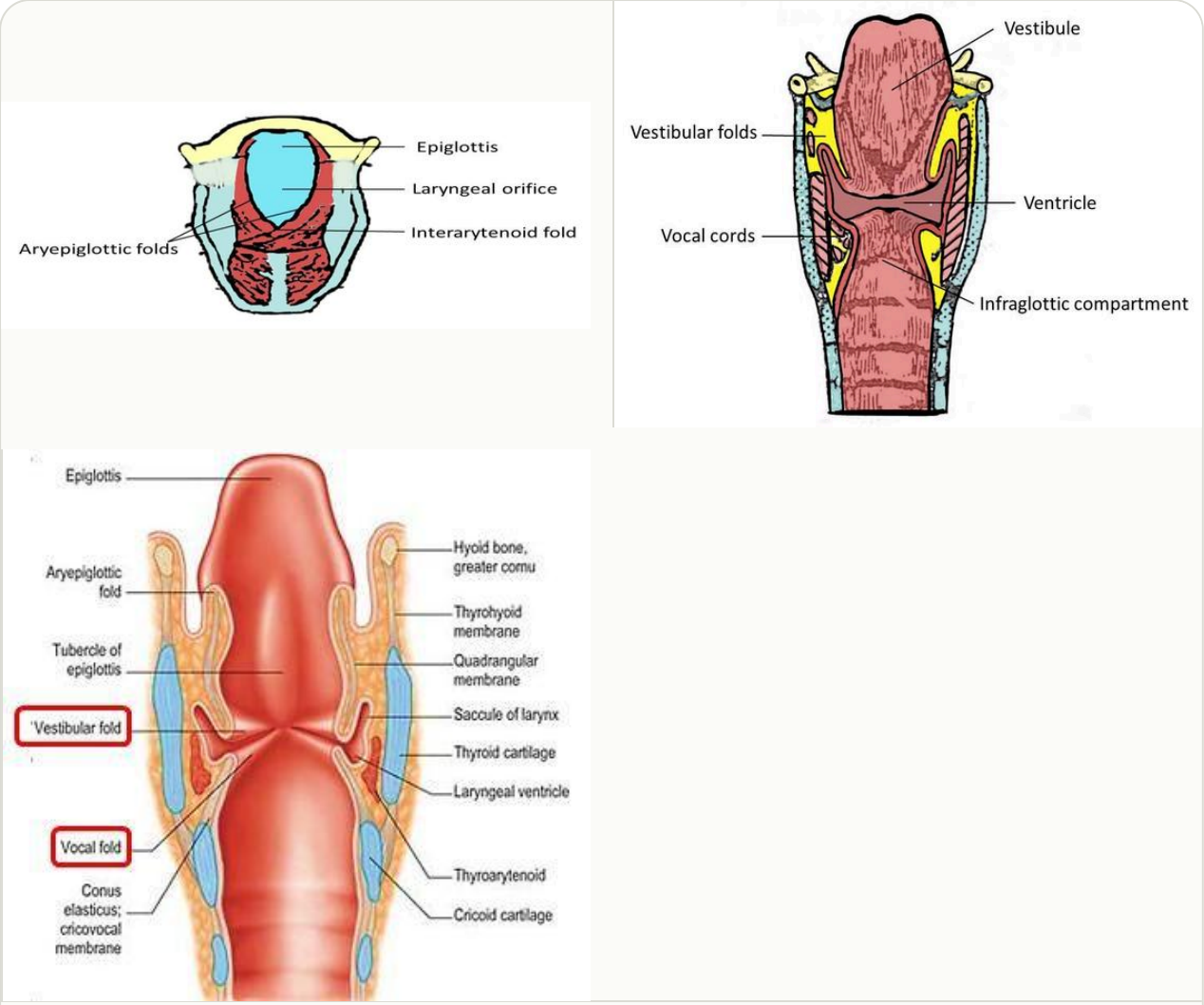



**FIGURE 10** The laryngeal inlet, the cavity with its vestibule, ventricle and infraglottic part, and a coronal section with the saccule.

Anatomy atlas · Doss p. 175, Doss p. 174, ENT Steps p. 238

## 3.2 The vocal folds and glottis

The true folds run from the thyroid angle to the vocal processes: pearly white because they are covered by closely bound squamous epithelium without glands. The *glottis (rima glottidis)* is the gap between them, about 24 mm long in men and 16 mm in women, the narrowest part of the adult airway. Its anterior two-thirds, between the membranous folds, is the *phonatory glottis*; its posterior third, between the arytenoids, the *respiratory glottis*.

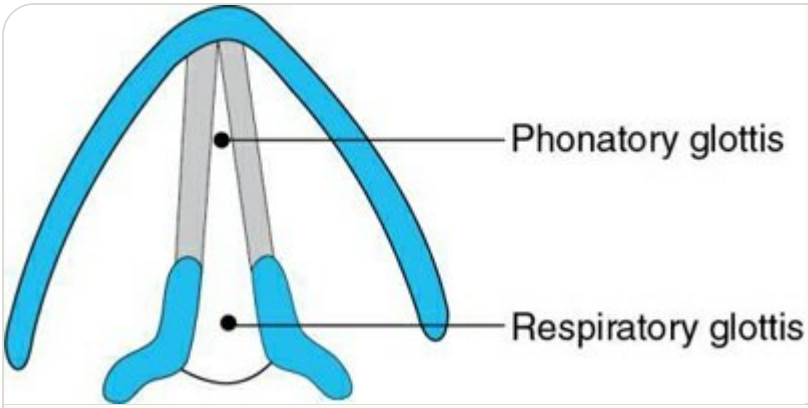



**FIGURE 11** The rima glottidis: the anterior two-thirds is the phonatory glottis, the posterior third (between the arytenoids) the respiratory glottis.

Dhingra, fig. 56.7

## 3.3 Layers of the vocal fold

- Squamous epithelium.
- Lamina propria: a *superficial layer* (*Reinke's space*, loose and rich in hyaluronic acid), an intermediate layer (elastic) and a deep layer (collagen); the intermediate and deep layers form the *vocal ligament*.
- The vocalis muscle.

The epithelium and superficial lamina propria form the *cover*, which ripples over the stiffer *body* as the mucosal wave; scarring of the cover ruins the voice, and fluid in Reinke's space gives Reinke's oedema.

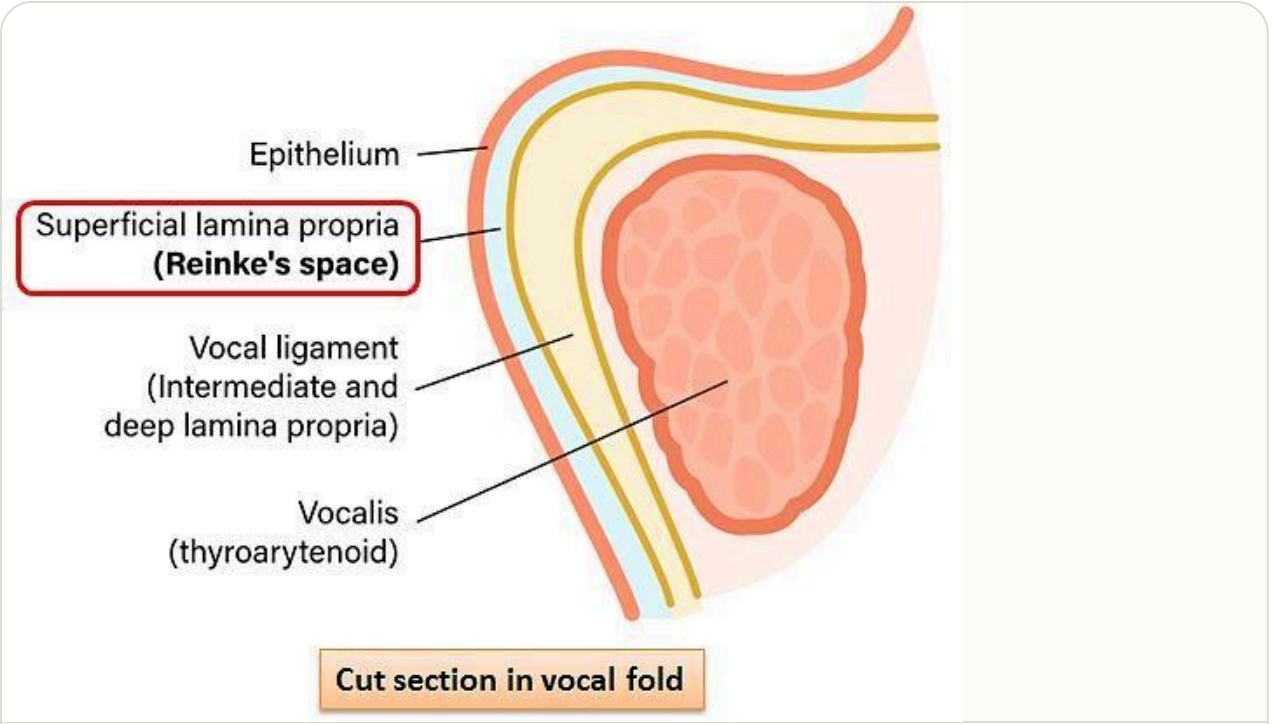



**FIGURE 12** The layers of the vocal fold: Reinke's space and the vocal ligament.
Anatomy atlas · ENT Steps p. 246

## 3.4 Regions

- **Supraglottis:** epiglottis, aryepiglottic folds, arytenoids, false folds and ventricles.
- **Glottis:** the vocal folds with the anterior and posterior commissures, extending about 1 cm below the free edge.
- **Subglottis:** from there to the lower border of the cricoid.

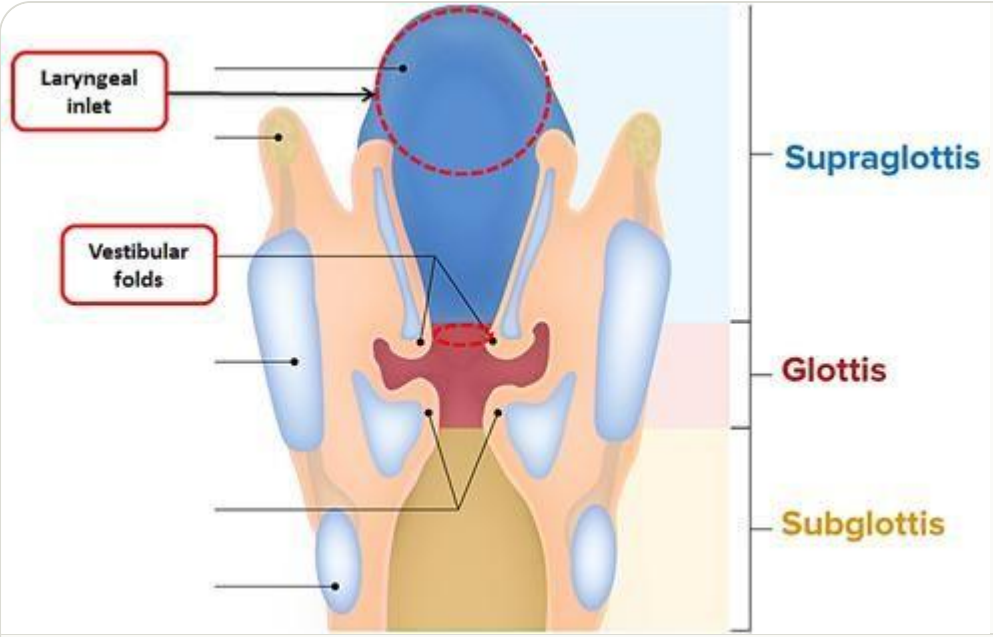



**FIGURE 13** Supraglottis, glottis and subglottis.
Anatomy atlas · ENT Steps p. 239

## 3.5 Spaces

- **Pre-epiglottic space (of Boyer):** fat in front of the infrahyoid epiglottis, behind the thyrohyoid membrane and thyroid cartilage, below the hyoepiglottic ligament; invaded by supraglottic and tongue base cancers.
- **Paraglottic space:** between the thyroid lamina laterally and the ventricle, quadrangular membrane and conus elasticus medially, the piriform sinus behind; continuous with the pre-epiglottic space. Cancers spreading in it are transglottic and can leave the larynx through the cricothyroid membrane.
- **Reinke's space:** under the epithelium of the membranous fold, between the superior and inferior arcuate lines.

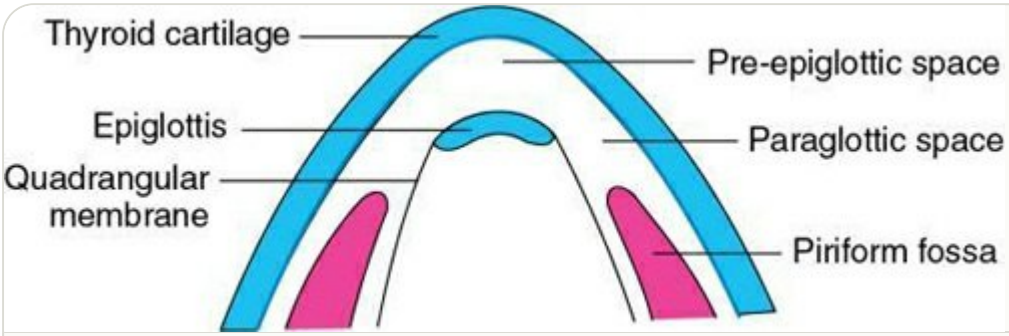



**FIGURE 14** The paraglottic and pre-epiglottic spaces communicate with each other.
Dhingra, fig. 56.8

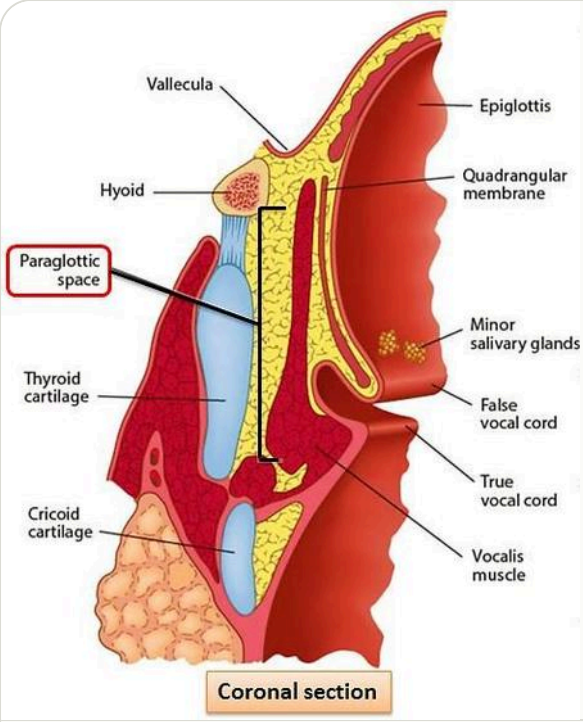



**FIGURE 15** The paraglottic space in coronal section.
Anatomy atlas · ENT Steps p. 244

### 3.6 Mucosa

Ciliated columnar epithelium except over the vocal folds and upper vestibule (squamous). It is loosely attached, and swells readily, everywhere except over the posterior epiglottis, vocal folds and the corniculate and cuneiform cartilages. Glands are plentiful on the posterior epiglottis, aryepiglottic folds and saccule, and absent from the vocal folds.

## 4 CHAPTER FOUR Nerves, vessels and lymph

The supply.

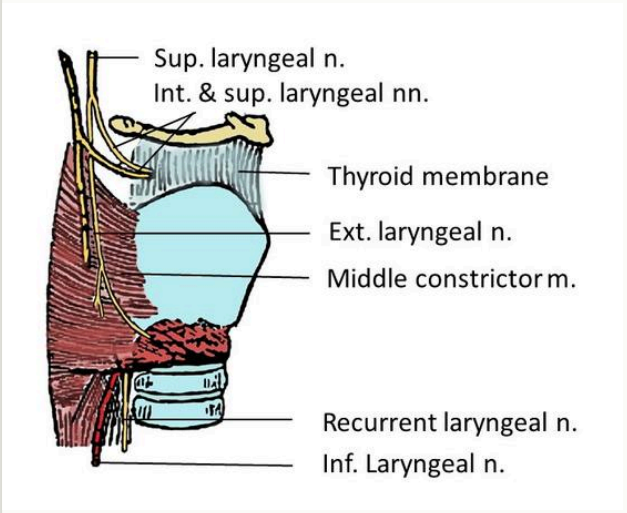
### 4.1 Superior laryngeal nerve

From the inferior (nodose) ganglion of the vagus, it descends behind the internal carotid artery and divides at the level of the greater horn of the hyoid:

- **Internal branch:** sensory to the mucosa above the vocal folds, piercing the thyrohyoid membrane with the superior laryngeal artery; it runs under the mucosa of the piriform fossa.
- **External branch:** motor to the *cricothyroid*, running with the superior thyroid artery, where it is at risk in thyroidectomy (loss of high pitch and vocal stamina).

## 4.2 Recurrent laryngeal nerve

Motor to every intrinsic muscle except the cricothyroid (the interarytenoid receives both sides), and sensory below the vocal folds. The *left* nerve leaves the vagus in the chest, hooks under the arch of the aorta beside the ligamentum arteriosum and climbs in the tracheo-oesophageal groove; the *right* hooks around the subclavian artery. Both enter the larynx behind the cricothyroid joint. The left, with its long course through the chest, is paralysed far more often. A *non-recurrent* right nerve accompanies an aberrant retro-oesophageal right subclavian artery.



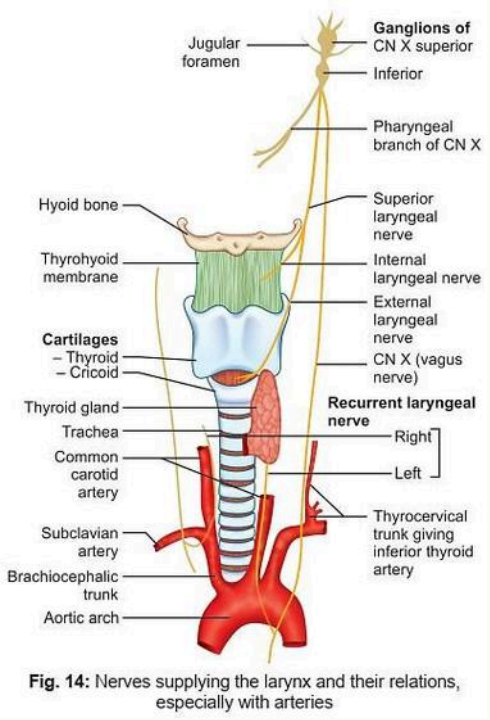




**Fig. 14:** Nerves supplying the larynx and their relations, especially with arteries

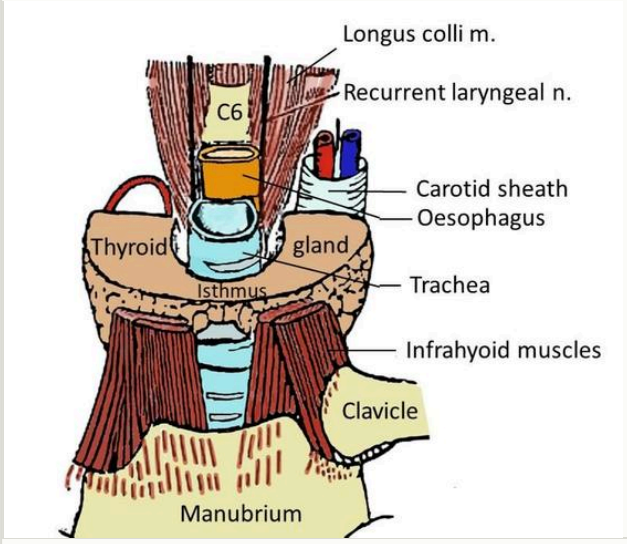



**FIGURE 16** The nerve supply of the larynx, the nerves and their relations, and the cervical trachea with the recurrent laryngeal nerve.

Anatomy atlas · Doss p. 175, ENT Steps p. 251, Doss p. 176

## 4.3 Embryology

The larynx develops from the laryngotracheal groove of the foregut; the cartilages and muscles come from the fourth and sixth arch mesenchyme. The *superior laryngeal nerve* is the nerve of the fourth arch (hence the cricothyroid), and the *recurrent laryngeal nerve* that of the sixth (all other intrinsic muscles); the sixth arch artery on the left becomes the ductus arteriosus, which is why the left nerve hooks

around it. The lumen is obliterated in the sixth week and recanalises by the tenth: failure gives webs and atresia.

## 4.4 Vessels

The superior laryngeal artery (from the superior thyroid) with the internal laryngeal nerve, and the inferior laryngeal artery (from the inferior thyroid) with the recurrent nerve; matching veins drain to the internal jugular and brachiocephalic veins.

## 4.5 Lymphatic drainage

- **Supraglottis:** rich lymphatics through the thyrohyoid membrane to the upper and middle deep cervical nodes, often both sides.
- **Glottis:** almost none; hence early glottic cancer rarely metastasises.
- **Subglottis:** through the cricothyroid membrane to the prelaryngeal (Delphian), pretracheal and paratracheal nodes, and to the lower deep cervical and mediastinal nodes.

# 5 CHAPTER FIVE The child's larynx

Not a small adult larynx.

- **High:** opposite C3 or C4 at rest, rising to C1 or C2 in swallowing; the epiglottis meets the soft palate, so an infant can breathe through the nose while milk passes either side of the epiglottis. The larynx descends to C5 to C6 by adulthood.
- **Soft:** the cartilages collapse easily; the epiglottis is omega-shaped and the arytenoids are relatively large.
- **Overlapping:** the hyoid overlaps the thyroid and the thyroid overlaps the cricoid, so the cricothyroid and thyrohyoid membranes are narrow and hard to find; cricothyroidotomy is avoided in young children.
- **Funnel-shaped:** the *subglottis (cricoid ring)* is the narrowest part, about 4 to 5 mm in the newborn; one millimetre of oedema there halves the airway. A diameter under 4 mm in a term baby is subglottic stenosis.
- **Loose submucosa** swells quickly with infection and trauma.

The larynx grows rapidly in the first three years, which allows some congenital narrowings to be watched, and again at puberty, when the male vocal folds lengthen and the voice breaks. The folds measure about 6 to 8 mm in childhood and 15 to 23 mm in adults.

# 6 CHAPTER SIX Physiology

Four functions.

## 6.1 Protection of the lower airway

The oldest function. During a swallow three sphincters close in turn: the *inlet* (aryepiglottic folds, epiglottic tubercle and arytenoids), the *false folds*, and the *true folds*, the last barrier; breathing stops reflexly; the larynx rises and moves forward under the base of tongue as the epiglottis tilts back. Anything that gets through provokes a *cough*: the folds close, pressure builds below them, and a sudden release blows the material out. The larynx is the watchdog of the lungs.

## 6.2 Phonation

The *myoelastic–aerodynamic theory*: the adducted folds are blown apart by subglottic pressure from the lungs and fall back by their elasticity and the Bernoulli effect, releasing puffs of air that vibrate the cover as a mucosal wave. Loudness depends on subglottic pressure; pitch on the frequency of vibration, set by length, tension and mass of the folds. The pharynx, mouth, nose and chest resonate the sound, and the lips, tongue, palate and teeth articulate it into speech.

## 6.3 Respiration

The folds abduct on inspiration and partly adduct on expiration, regulating airflow and providing back pressure to the lungs.

## 6.4 Fixation of the chest

A closed glottis fixes the chest so that the arm, trunk and abdominal muscles can work against it in lifting, climbing, coughing, vomiting, defecation, micturition and

childbirth.

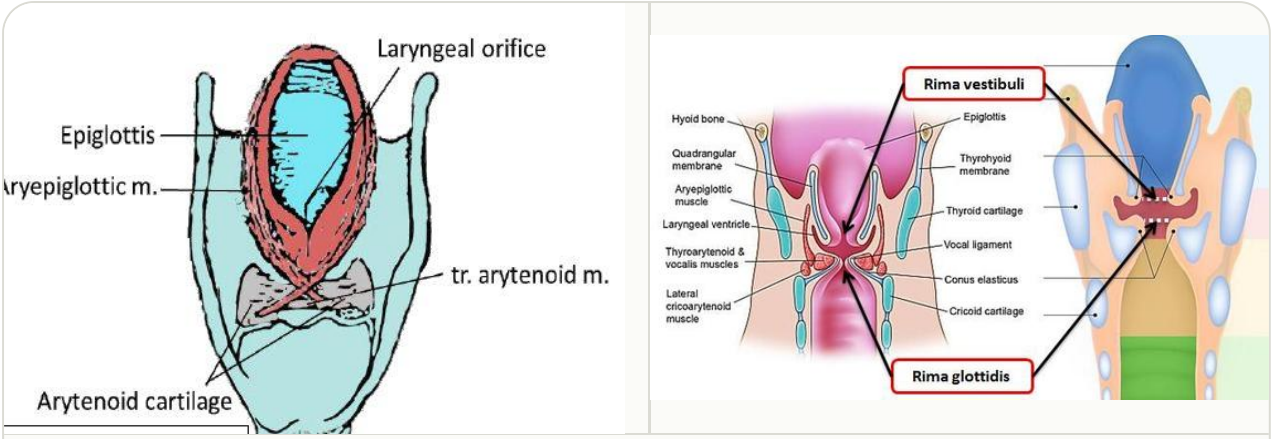



**FIGURE 17** The muscles that close the laryngeal inlet, and the rima vestibuli and rima glottidis.
Anatomy atlas · Doss p. 173, ENT Steps p. 241

# 7 CHAPTER SEVEN For the examination

What the examiner wants to hear.

## 7.1 Viva essentials

**WHAT THE EXAMINER IS LOOKING FOR**

- **Cartilages,** which are paired, and which ossify.
- **Intrinsic muscles** and their actions; the only abductor.
- **Conus elasticus,** quadrangular membrane, pre-epiglottic and paraglottic spaces.
- **Layers of the vocal fold** and Reinke's space.
- **Nerve supply** and the course of the recurrent laryngeal nerves.
- **Lymphatic drainage** by region and its importance in cancer.
- **The infant larynx.**
- **Functions** of the larynx.

## 7.2 Common pitfalls

**COMMON PITFALLS**

- Forgetting that the cricothyroid is supplied by the external laryngeal nerve.
- Calling the glottis the narrowest part of the infant airway; it is the subglottis.
- Mistaking ossified laryngeal cartilage for a foreign body on a radiograph.

---

Written for this app from Oxford Textbook of Otolaryngology ch. 47, Oxford Specialist Handbook ch. 8, Dhingra ch. 56, 60 and K. J. Lee ch. 50. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · LARYNX & AIRWAY

# Laryngotracheal trauma and stenosis

A blow to the front of the neck can break the larynx without breaking the skin. Nearly half of those with a laryngeal fracture die at the scene; of the rest, many look well for a few hours while oedema and haematoma quietly close the airway. Secure the airway without a tube that may make a false passage, then restore the framework early, before scar turns a repairable injury into a lifelong stenosis.

**AT A GLANCE**

- Loss of thyroid prominence, emphysema and crepitus.
- Tracheostomy under local anaesthesia, not blind intubation.
- Repair displaced fractures and tears within days.
- Intubation is the commonest cause of subglottic stenosis.
- Penetrating neck injury: hard signs mean theatre; otherwise CT angiography and selective exploration.

## 1 CHAPTER ONE Overview

Blunt, penetrating and from within.

### 1.1 Causes

- **Blunt:** road traffic accidents (the neck striking the steering wheel or dashboard with the head extended), blows and kicks, clothesline injuries from a stretched wire or cable, and strangulation.
- **Penetrating:** knives, glass and gunshot wounds.
- **Internal (iatrogenic):** intubation, endoscopy, inhalation and caustic burns, and foreign bodies.

### 1.2 Why the larynx is usually spared

It is shielded by the mandible above, the sternum below and the spine behind, and it is mobile. Fractures need a direct anterior blow with the head extended, and become commoner after forty, when the cartilages calcify and become brittle; in

children the elastic cartilages bend rather than break, but soft tissue injury and airway swelling are more dangerous.

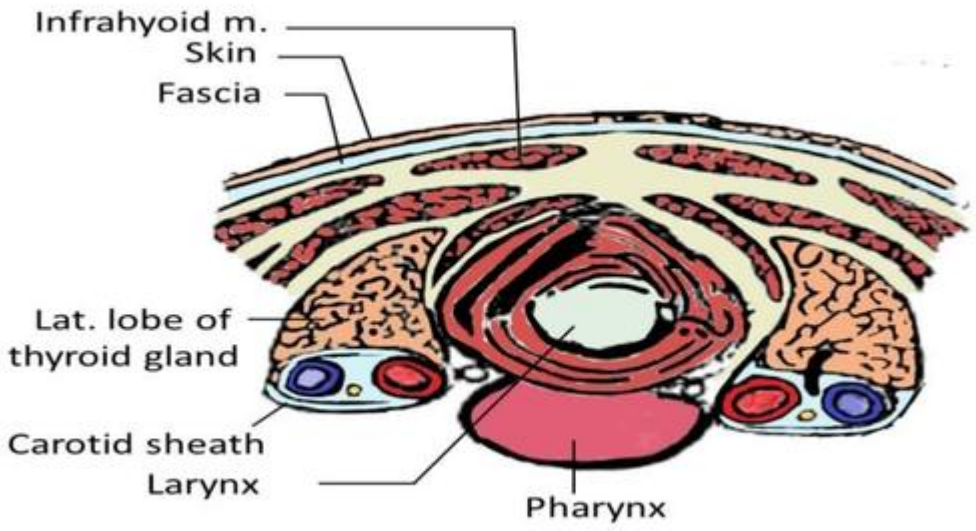



**FIGURE 1** The relations of the larynx in cross-section.
Anatomy atlas · Doss p. 170

## 2 CHAPTER TWO Pathology

What breaks.

- Haematoma and oedema of the supraglottis or subglottis.
- Mucosal tears of the larynx or pharynx, with surgical emphysema.
- Dislocation or avulsion of an arytenoid at the cricoarytenoid joint.
- Dislocation of the cricothyroid joint, injuring the recurrent laryngeal nerve that runs just behind it.
- Fracture of the hyoid.
- Fractures of the thyroid cartilage, vertical or transverse: an upper fracture can avulse the epiglottis and false folds; a lower one displaces or tears the true folds and anterior commissure.
- Fracture of the cricoid, the ring that holds the airway open.
- Fractures of the upper tracheal rings.

- *Laryngotracheal separation*: the trachea parts from the cricoid and retracts into the mediastinum, usually with bilateral recurrent nerve injury.

**THE LUCID INTERVAL**

A patient with a laryngeal fracture may talk and breathe normally on arrival, then obstruct suddenly hours later as haematoma and oedema accumulate. Every significant anterior neck injury is observed in hospital with the airway assessed by endoscopy.

# 3 CHAPTER THREE Clinical features

Voice, breath, swallow, blood.

## 3.1 Symptoms

- Respiratory distress and stridor.
- Hoarseness or aphonia.
- Painful, difficult swallowing, with aspiration.
- Pain in the larynx, worse with speaking and swallowing.
- Haemoptysis from mucosal tears.

## 3.2 Signs

- Bruising and abrasions of the neck.
- Tenderness over the larynx.
- *Surgical emphysema*, increasing with coughing.
- Loss of the thyroid prominence and flattening of the neck contour; the thyroid notch may be impalpable.
- A palpable step or gap between fragments, and crepitus.
- Separation of the cricoid from the larynx or trachea.

## 3.3 Associated injuries

Cervical spine (assume it is injured), great vessels, oesophagus and pharynx, chest

(pneumothorax), head and abdomen.

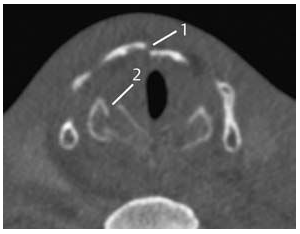
# 4 CHAPTER FOUR Evaluation

Only when the airway allows.

- **Flexible nasendoscopy:** the most useful first examination in a stable patient: oedema, haematoma, lacerations, exposed cartilage, arytenoid displacement, vocal fold movement, and the size of the airway.
- **CT of the larynx (with 3D reconstruction):** cartilage fractures, joint dislocation, displacement; also the cervical spine and vessels. Only when the airway is safe.
- **Rigid endoscopy** (laryngoscopy, oesophagoscopy, bronchoscopy) under anaesthesia, after the airway has been secured, to map injuries.
- **Chest radiograph** for pneumothorax and mediastinal air; **contrast swallow** for pharyngo-oesophageal tears; **angiography** if vascular injury is suspected.

**IMAGING IN THE APP** 1 case

Laryngeal trauma on imaging.




Laryngeal Trauma

Trauma · 10 images

### SCHAEFER–FUHRMAN GRADING OF LARYNGEAL INJURY

| GROUP | FINDINGS | MANAGEMENT |
|---|---|---|
| 1 | Minor haematoma or laceration, no fracture | Observe, humidify, steroids |
| 2 | Oedema, haematoma, minor mucosal disruption, no exposed cartilage, non-displaced fracture | Observe; tracheostomy if the airway is threatened |
| 3 | Massive oedema, mucosal tears, exposed cartilage, displaced fracture, cord immobility | Tracheostomy and open repair |

| GROUP | FINDINGS | MANAGEMENT |
|---|---|---|
| 4 | As group 3 with a disrupted anterior commissure or unstable framework | Tracheostomy, open repair and stenting |
| 5 | Complete laryngotracheal separation | Tracheostomy and repair by anastomosis |

# 5 CHAPTER FIVE Management

Airway, then framework.

## 5.1 The airway

- **Acute distress:** awake *tracheostomy under local anaesthesia*, then laryngoscopy under general anaesthesia. Orotracheal intubation is risky: a tube can enter a false passage through torn mucosa or complete a laryngotracheal separation. If intubation is attempted, it is by the most experienced operator, fibreoptically and with the neck prepared for a surgical airway.
- **Stable airway:** flexible endoscopy, then CT.

## 5.2 Conservative treatment

For minor injuries with a normal or near-normal endoscopy and no displaced fracture: admission and observation, voice rest, humidified air, head elevation, systemic steroids early and in full dose (to reduce oedema and scarring), antibiotics (to prevent perichondritis), and anti-reflux treatment.

## 5.3 Surgical repair

Indicated for displaced or unstable fractures, exposed cartilage, large mucosal tears, disruption of the anterior commissure, arytenoid dislocation and cord immobility, and laryngotracheal separation. Repair early, within the first few days and ideally before ten days, before granulation and scar form.

- **Exposure:** a midline thyrotomy (laryngofissure) to inspect and close the mucosa.

- **Mucosa:** all lacerations sutured; exposed cartilage covered with local flaps or grafts; loose fragments removed.
- **Framework:** fractures reduced and fixed with wire or titanium miniplates.
- **Epiglottis:** reattached, or excised if avulsed.
- **Arytenoid:** reduced if dislocated; removed if completely avulsed.
- **Anterior commissure:** a silastic keel prevents a web.
- **Stents:** a soft laryngeal stent or silicone tube for comminuted fractures, left for two to six weeks; stents themselves provoke granulation.
- **Laryngotracheal separation:** end-to-end anastomosis, with assessment of both recurrent nerves.

**OPERATIVE GUIDES IN THEATRE** 1 operation

Repair of laryngeal trauma.

**Laryngeal Trauma**
Laryngeal trauma
›

## 5.4 Complications

Laryngeal stenosis (supraglottic, glottic or subglottic), perichondritis and abscess, chondronecrosis, vocal fold paralysis or fixation, webs, and a poor voice.

# 6 CHAPTER SIX Laryngotracheal stenosis

The late result.

## 6.1 Causes

Prolonged or traumatic *intubation* (the commonest cause: pressure necrosis of the posterior glottis and subglottis by the tube and cuff), external trauma, high tracheostomy and cricothyroidotomy, burns and caustics, radiotherapy, granulomatosis with polyangiitis, relapsing polychondritis, sarcoidosis, reflux, and idiopathic subglottic stenosis in women.

## 6.2 Presentation

Progressive dyspnoea and stridor, often biphasic, sometimes with hoarseness; many patients are misdiagnosed with asthma.

## 6.3 Evaluation

History of intubation or injury; office endoscopy of the supraglottis and glottis and vocal fold movement; CT of the subglottis and trachea; and direct laryngoscopy and bronchoscopy to measure the length, grade and site and to palpate the arytenoids (fixation versus paralysis).

**COTTON–MYER GRADING OF SUBGLOTTIC STENOSIS**

Grade I, up to 50% obstruction; grade II, 51 to 70%; grade III, 71 to 99%; grade IV, no detectable lumen.

## 6.4 Treatment

- **Supraglottic:** endoscopic excision of scar; recurrence is common after radiotherapy.
- **Glottic:** an anterior web affects voice more than breathing and is divided with a keel; posterior glottic scarring that fixes the folds is treated by division of the interarytenoid scar, or by procedures that enlarge the glottis at the expense of voice (as for bilateral paralysis).
- **Subglottic:** endoscopic radial incisions with laser and balloon dilatation (with topical mitomycin or steroid injection) for thin, soft stenoses; *laryngotracheal reconstruction* with cartilage grafts, or *cricotracheal resection* with anastomosis, for firm, long or high-grade stenoses.
- **Tracheal:** resection and end-to-end anastomosis is definitive; balloon dilatation for thin webs or unfit patients.
- **Complex, multilevel stenosis:** a Montgomery T-tube stents the airway while preserving voice. Expanding metal stents are avoided in benign disease because of granulation and erosion.

**OPERATIVE GUIDES IN THEATRE** 4 operations

Acquired laryngeal stenosis, tracheal stenosis, laryngotracheal reconstruction and tracheal resection.

**Acquired Laryngeal Stenosis**
Acquired laryngeal stenosis
›

**Management of Tracheal Stenosis**
Tracheal stenosis
›

**Laryngotracheal**
Laryngotracheal reconstruction
›

**Tracheal Resection**
Tracheal resection
›

## 6.5 Prevention

Use the right-sized tube, keep cuff pressure below about 25 to 30 cm of water, avoid prolonged translaryngeal intubation where tracheostomy is better, and avoid placing a tracheostomy through the first ring or cricoid.

PENETRATING NECK INJURY (OXFORD TEXTBOOK)

Penetrating injuries are those that breach the **platysma**; they carry a mortality of up to about 6%, and half of deaths are from bleeding. The anterior neck is divided into three zones. **Zone I** runs from the clavicles and sternal notch to the cricoid and contains the great vessels at the thoracic inlet, trachea, oesophagus, lung apices and thoracic duct. **Zone II**, from the cricoid to the angle of the mandible, is the commonest site and the easiest to expose, with the best prognosis. **Zone III**, from the angle of the mandible to the skull base, contains the distal carotid and vertebral arteries and cranial nerves IX to XII and is hard to reach surgically.

Management follows trauma life support principles, with control of the airway (often by early intubation or a surgical airway) and of bleeding by direct pressure. **Hard signs**, such as an expanding haematoma, active bleeding, air bubbling through the wound, shock not responding to fluids, absent pulse, bruit or evolving stroke, mean immediate exploration in theatre. Stable patients with **soft signs** (haemoptysis, dysphonia, dysphagia, surgical emphysema, a non-expanding haematoma) or none are investigated: **CT angiography** has replaced catheter angiography as the first test in all three zones, supplemented by endoscopy or contrast swallow for the aerodigestive tract. The old rule of mandatory exploration of every zone II wound has given way to

this **selective approach**, which reduces negative explorations. Carotid injuries found at exploration are generally repaired rather than ligated.

# 7 CHAPTER SEVEN For the examination

What the examiner wants to hear.

## 7.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Mechanism** of laryngeal fracture and why it is commoner after forty.
- **Signs:** loss of thyroid prominence, emphysema, crepitus.
- **Airway first:** tracheostomy under local anaesthesia, not intubation.
- **Indications and timing** of open repair.
- **Laryngotracheal stenosis:** causes, grading and treatment.

## 7.2 Common pitfalls

COMMON PITFALLS

- Discharging a talking patient with a laryngeal injury who later obstructs.
- Intubating blindly through a disrupted larynx.
- Delaying repair beyond the first week or two.
- Treating post-intubation subglottic stenosis as asthma.

Written for this app from Oxford Textbook of Otolaryngology ch. 53, Oxford Specialist Handbook ch. 8, Oxford Handbook of ENT ch. 25, Dhingra ch. 57 and K. J. Lee ch. 50. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · LARYNX & AIRWAY

# Chronic laryngitis, reflux and specific infections

A voice that has been hoarse for months, in a smoker who clears his throat all day, is the classic picture of chronic laryngitis. Lee warns that the term is not a diagnosis but a description: behind it lie reflux, smoking, voice misuse, infection, allergy and, occasionally, systemic disease, and on the vocal folds some of these can look exactly like cancer. Find the cause, and biopsy anything that does not behave.

**AT A GLANCE**

- Smoking, reflux and voice misuse dominate.
- Laryngopharyngeal reflux: posterior oedema; twice-daily PPI for months.
- Tuberculosis favours the posterior larynx; lupus the anterior.
- Biopsy persistent lesions.

## 1 CHAPTER ONE Overview

Prolonged laryngeal inflammation.

### 1.1 Causes

- **Irritants:** smoking (the main cause), alcohol, dust and fumes in mines and industry, polluted air.
- **Laryngopharyngeal reflux.**
- **Voice misuse:** prolonged loud or forced voice, habitual throat clearing.
- **Infection nearby or below:** chronic sinusitis with postnasal drip, dental and tonsillar sepsis, chronic bronchitis and bronchiectasis.
- **Incomplete resolution** of acute laryngitis.
- **Allergy** as part of the unified airway.
- **Specific infections:** tuberculosis, syphilis, leprosy, scleroma, fungi, and MRSA.
- **Systemic disease:** sarcoidosis, rheumatoid arthritis, granulomatosis with polyangiitis, relapsing polychondritis, amyloidosis, pemphigus and pemphigoid.
- **Radiotherapy.**

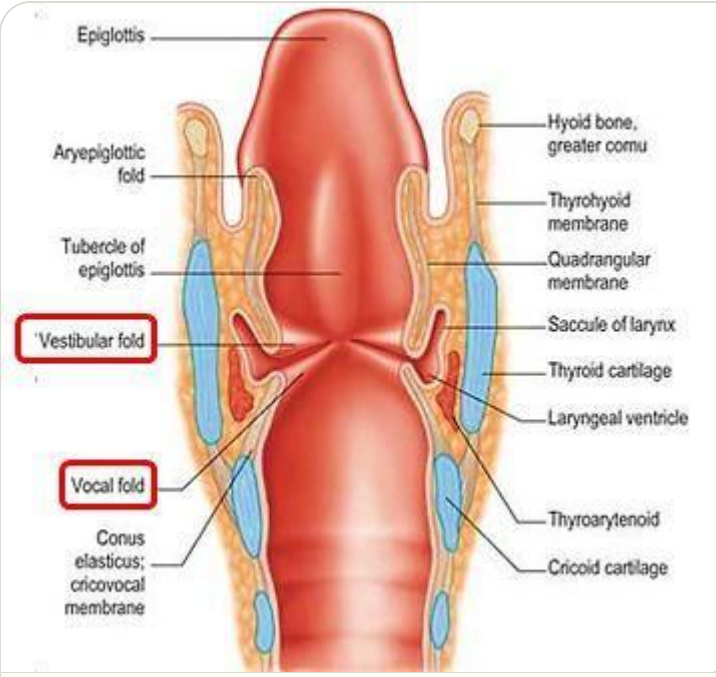



**FIGURE 1** Coronal section of the larynx: vestibular and vocal folds, ventricle and saccule.
Anatomy atlas · ENT Steps p. 238

## 1.2 Approach

Stroboscopic or high-quality endoscopic examination of the larynx; treat reflux, irritants and voice use; biopsy any lesion that persists, is asymmetric, ulcerated or exophytic, or occurs in a smoker, because tuberculosis, granulomas and carcinoma can look alike.



**FIGURE 2** Flexible laryngoscopy in clinic.
Dhingra, fig. 62.6

CHAPTER TWO

# 2 Chronic non-specific laryngitis

Hyperaemic and hypertrophic.

## 2.1 Chronic hyperaemic laryngitis

A diffuse, symmetrical inflammation of the whole larynx. Hoarseness is the commonest complaint: the voice tires and fails by evening. Constant hawking and clearing of the throat, dryness and tickling, discomfort and a dry cough. The folds look dull red and rounded with flecks of sticky mucus on them and between the arytenoids.

## 2.2 Chronic hypertrophic (hyperplastic) laryngitis

Mostly men of thirty to fifty. Diffuse thickening that begins at the glottis and spreads to the false folds, epiglottis and subglottis: the respiratory epithelium changes to squamous, the vocal fold epithelium becomes hyperplastic and keratinised, glands first enlarge then atrophy. The mucosa is dusky red and thick,

the folds red, rounded and later bulky and nodular, the false folds swollen (they may be mistaken for prolapse of the ventricle), and fold movement may become sluggish. It can be localised, as vocal nodules, polyps, Reinke's oedema and contact ulcers (described with benign lesions).

### 2.3 Treatment

1. Treat infection in the sinuses, teeth, tonsils and chest.
2. Stop smoking and alcohol; avoid dust and fumes.
3. Voice rest for weeks, then voice therapy.
4. Steam inhalations and expectorants.
5. Treat reflux.
6. *Stripping* of the hyperplastic mucosa under the microscope in selected cases, one fold at a time and preserving the vocal ligament, with histology to exclude dysplasia.

## 3 CHAPTER THREE Laryngopharyngeal reflux

Acid at the top.

### 3.1 Features

Hoarseness, globus, chronic cough, throat clearing, excess mucus and dysphagia; only a quarter to a third have heartburn, because the larynx is injured by small amounts of acid and pepsin that the oesophagus tolerates. Symptoms are worse after late meals, alcohol and exertion; a bitter taste on waking suggests nocturnal reflux.

### 3.2 Signs

Oedema and redness of the posterior larynx and interarytenoid area, *subglottic oedema* (pseudosulcus), diffuse vocal fold oedema, thick endolaryngeal mucus, and granulomas or contact ulcers.

### 3.3 Diagnosis

Usually clinical, confirmed by response to treatment. Combined multichannel *pH and impedance* monitoring (detecting acid and non-acid reflux) is the most accurate test; oesophagoscopy excludes Barrett oesophagus.

### 3.4 Treatment

Diet and behaviour changes (no late meals, less fat, coffee, chocolate and alcohol, weight loss, head of the bed raised), a *twice-daily proton pump inhibitor* half an hour before breakfast and dinner for at least two to three months, an H2 blocker or alginate at night, and fundoplication for proven refractory reflux.

## 4 CHAPTER FOUR Special forms

Pachydermia, Reinke's oedema and sicca.

### 4.1 Polypoid degeneration (Reinke's oedema)

Bilateral, symmetrical swelling of the whole membranous vocal folds from fluid in Reinke's space, in middle-aged smokers (women more often seek help because their voice drops), with voice misuse, reflux, chronic sinusitis and myxoedema. The voice is low-pitched and rough; the folds are pale, translucent fusiform swellings that flop with breathing. Stop smoking, treat reflux and hypothyroidism, voice therapy; *microlaryngeal decortication* by a lateral incision with aspiration of the gel and trimming of excess epithelium, traditionally one side at a time.

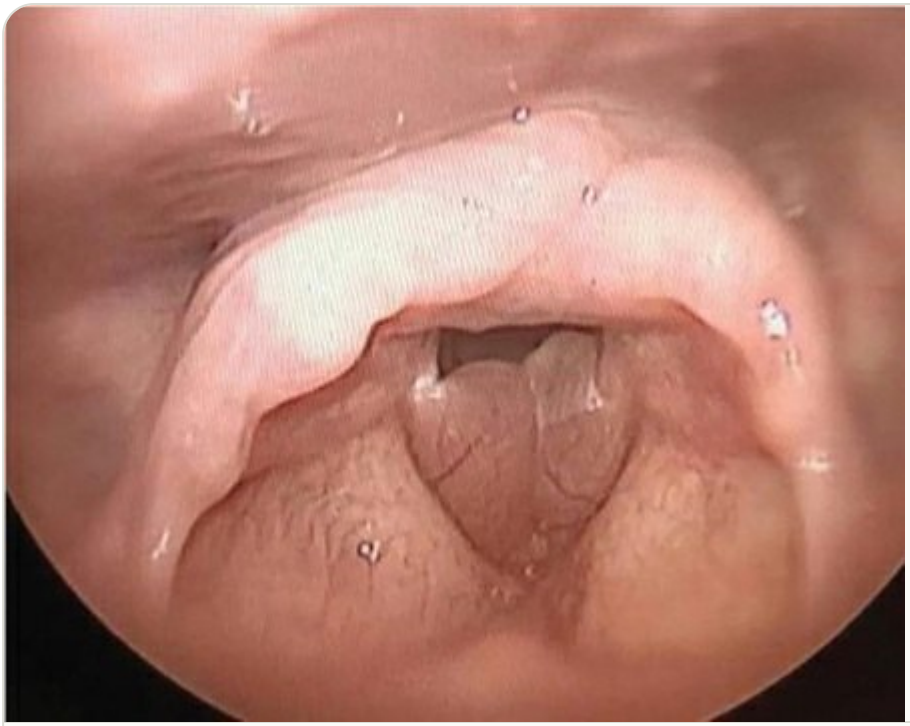


**FIGURE 3** Bilateral Reinke's oedema in a smoker.
Dhingra, fig. 61.5

## 4.2 Pachydermia laryngis

A hypertrophic laryngitis of the posterior larynx: heaped-up red or grey granulation tissue in the interarytenoid area and posterior folds, bilateral and symmetrical, sometimes with contact ulcers on the vocal processes. Associated with alcohol, smoking, forceful speech and reflux. Not premalignant, but biopsy is essential to exclude carcinoma and tuberculosis. Microlaryngoscopic removal, anti-reflux treatment and voice therapy.

## 4.3 Atrophic laryngitis (laryngitis sicca)

Atrophy of the mucosa with foul crusts, usually in women with atrophic rhinitis and pharyngitis: hoarseness that improves after coughing up a crust, dry cough, and dyspnoea from obstructing crusts, which may reach the trachea. Humidification, sprays of glucose in glycerine, expectorants (ammonium chloride, iodides), and treatment of the nose.

# 5 CHAPTER FIVE Specific infections

The great mimics.

## 5.1 Tuberculosis

Almost always secondary to active pulmonary tuberculosis, mostly in middle-aged men, through infected sputum (bronchogenic) or the bloodstream. It favours the *posterior larynx*: the interarytenoid fold, false folds, true folds and epiglottis in that order.

- **Symptoms:** weakness of the voice (earliest), hoarseness, then severe pain from ulceration radiating to the ears and painful swallowing.
- **Signs:** redness of one fold with impaired adduction (the first sign); a *mamillated* swelling of the interarytenoid area; *mouse-nibbled* ulceration of the fold edge; shallow ulcers and granulations on the arytenoids; *turban epiglottis* (pseudoedema); swollen false folds and aryepiglottic folds; pallor of the surrounding mucosa.
- **Diagnosis:** chest radiograph, sputum, and laryngeal biopsy, essential to exclude carcinoma.
- **Treatment:** antituberculous chemotherapy and voice rest; lesions heal, sometimes with stenosis.

## 5.2 Lupus

An indolent tuberculous infection accompanying lupus of the nose and pharynx, attacking the *anterior larynx*: the epiglottis first, which may be destroyed, then the aryepiglottic and false folds. Painless and often found incidentally; no lung disease; responds well to antituberculous drugs.

## 5.3 Syphilis

Now rare; the tertiary gumma forms a smooth swelling that ulcerates and heals with stenosis. Never involved in the primary stage. Diagnosis by biopsy and serology.

## 5.4 Leprosy

With nasal and skin leprosy: nodular infiltration of the epiglottis, aryepiglottic folds and arytenoids, ulceration, and deformity and stenosis on healing. Dapsone and multidrug therapy.

## 5.5 Scleroma

*Klebsiella rhinoscleromatis*, endemic in parts of Africa, the Middle East, Asia and Latin America, usually with nasal disease: a smooth red subglottic swelling causing hoarseness, wheeze and dyspnoea, passing through catarrhal, granulomatous and sclerotic stages to subglottic stenosis. Long courses of tetracycline or ciprofloxacin with steroids; reconstruction for stenosis.

## 5.6 Fungal

Candidiasis in inhaled steroid users, after antibiotics and in immunosuppression: white plaques on red mucosa that can mimic leukoplakia; systemic antifungals (topical nystatin does not reach), and biopsy if they persist. Blastomycosis and histoplasmosis are rare and mimic carcinoma.

WHERE THE BOOKS DIFFER · WHERE THE BOOKS DIFFER

| Dhingra | Lee |
| --- | --- |
| Describes chronic laryngitis as an entity, in hyperaemic and hypertrophic forms, with treatment centred on voice rest, steam and removal of irritants. | Advises abandoning "chronic laryngitis" as a diagnosis, replacing it with "prolonged laryngeal inflammation" and a named cause, and gives reflux a central place. |

Both insist on biopsy for anything that might be cancer or tuberculosis.

# 6 CHAPTER SIX Systemic diseases

The larynx as a window.

- **Sarcoidosis:** pale, turban-like submucosal swelling of the supraglottis (epiglottis and aryepiglottic folds); non-caseating granulomas on biopsy; steroids systemically or by injection.
- **Rheumatoid arthritis:** *cricoarytenoid arthritis* (pain, hoarseness, referred otalgia, and when bilateral, stridor from fixed folds) and "bamboo nodules" in the folds; steroids and disease-modifying drugs; tracheostomy for obstruction.
- **Granulomatosis with polyangiitis:** subglottic stenosis, often the first sign; immunosuppression, steroid injection and dilatation.
- **Relapsing polychondritis:** inflammation of the cartilages of the ear, nose, larynx and trachea, with collapse of the airway.
- **Amyloidosis:** waxy yellow submucosal deposits, usually in the false folds and ventricles, with apple-green birefringence on Congo red staining; endoscopic excision and screening for systemic disease.
- **Pemphigus and pemphigoid:** blisters and scarring of the supraglottis.

# 7 CHAPTER SEVEN For the examination

What the examiner wants to hear.

## 7.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Causes of chronic laryngitis,** and why biopsy matters.
- **Laryngopharyngeal reflux:** symptoms, signs and treatment.
- **Reinke's oedema** and **pachydermia laryngis.**
- **Tuberculosis of the larynx:** sites, signs (mouse-nibbled, turban epiglottis, mamillated interarytenoid) and diagnosis; **lupus** compared.
- **Scleroma** and **cricoarytenoid arthritis.**

## 7.2 Common pitfalls

COMMON PITFALLS

- Treating persistent hoarseness in a smoker with antibiotics instead of laryngoscopy.
- Forgetting the chest radiograph in a patient with posterior laryngeal ulceration.
- Giving a short, once-daily course of proton pump inhibitor for laryngopharyngeal reflux and calling it a failure.

---

Written for this app from Oxford Textbook of Otolaryngology ch. 47, Oxford Specialist Handbook ch. 8, Oxford Handbook of ENT ch. 12, Dhingra ch. 58 and K. J. Lee ch. 50. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · LARYNX & AIRWAY

# Congenital lesions of the larynx

A newborn's airway is narrow, soft and quick to swell, and most of its congenital problems announce themselves as noise: a high-pitched crowing that is worse on feeding and crying and better lying prone. Most of these babies have laryngomalacia and will outgrow it. The art is to recognise the few who will not, the stenosis, the paralysed folds, the haemangioma, the cleft, and to look at the whole airway before deciding.

**AT A GLANCE**

- Laryngomalacia is the commonest; normal cry, better prone.
- Subglottic stenosis: under 4 mm in a term neonate.
- Subglottic haemangioma: propranolol.
- Palpate the interarytenoid area for a cleft.

## 1 CHAPTER ONE Overview

The congenital larynx.

### 1.1 Lesions

- Laryngomalacia (the commonest).
- Vocal fold paralysis (the second commonest).
- Congenital subglottic stenosis.
- Laryngeal web and atresia.
- Subglottic haemangioma.
- Laryngeal cysts and laryngocele.
- Laryngo-oesophageal cleft.
- Tracheomalacia and vascular rings (in the trachea, outside the larynx).

### 1.2 Clues from the cry

A *normal cry* with stridor points away from the glottis (laryngomalacia, subglottic stenosis, haemangioma); a *weak, hoarse or absent cry* points to the vocal folds (paralysis, web). Stridor eased by lying prone suggests laryngomalacia, micrognathia or macroglossia. Choking and cyanosis with feeds suggest a cleft, paralysis or a tracheo-oesophageal fistula.

---

CHAPTER TWO

# 2 Laryngomalacia

Floppy supraglottis.

### 2.1 Features

Excessive flaccidity of the supraglottis, which is drawn into the airway on inspiration. Stridor appears at birth or within the first weeks, is inspiratory and fluttering, worse with crying, feeding, excitement and lying supine, and better prone; the cry is normal. Symptoms usually peak at six to eight months and resolve by 12 to 24 months. Reflux often coexists and aggravates it.



**FIGURE 1** Laryngomalacia: the epiglottis folded longitudinally into an omega.
Dhingra, fig. 59.1

## 2.2 Appearances

On flexible endoscopy in the awake infant: an *omega-shaped*, elongated epiglottis curled on itself; short, tight aryepiglottic folds tethering it; and redundant mucosa over large arytenoids that prolapses into the glottis on inspiration. Vocal fold movement is normal. Lee classifies the collapse as type 1 (short aryepiglottic folds), type 2 (redundant arytenoid tissue) and type 3 (posterior collapse of the epiglottis, often with neuromuscular disease). It cannot be judged in a paralysed, anaesthetised child.

## 2.3 Treatment

- **Most:** reassurance and observation; positioning; anti-reflux treatment.
- **Severe disease** (failure to thrive, feeding difficulty, apnoeas, cyanosis, pulmonary hypertension): microlaryngoscopy and bronchoscopy to exclude a second lesion, then endoscopic *supraglottoplasty*: division of the tight aryepiglottic folds and trimming of redundant arytenoid mucosa with cold steel or laser.
- **Tracheostomy** rarely.

**OPERATIVE GUIDES IN THEATRE** 1 operation

Supraglottoplasty for laryngomalacia.

**Laryngomalacia**
Supraglottoplasty
›

# 3 CHAPTER THREE Vocal fold paralysis

The second commonest.

## 3.1 Causes

Birth trauma (stretching of the recurrent nerve in breech or forceps delivery), central nervous system anomalies (Arnold–Chiari malformation and hydrocephalus, usually bilateral), cardiac surgery and ligation of the ductus arteriosus, cardiomegaly and mediastinal lesions (usually left-sided), and idiopathic.

## 3.2 Presentation

- **Unilateral:** a weak, breathy cry, mild stridor, and choking or aspiration with feeds.
- **Bilateral:** a good cry but marked inspiratory stridor and respiratory distress, because both folds lie near the midline.

## 3.3 Assessment

Flexible endoscopy in the awake child; imaging of the brain, neck and chest; and a swallowing assessment.

## 3.4 Treatment

Observation for unilateral paralysis, with feeding support. Bilateral paralysis often needs a tracheostomy; definitive airway surgery is deferred, because many recover

within the first year or two, and the larynx grows. Shunting for Arnold–Chiari malformation may restore movement.

# 4 CHAPTER FOUR Subglottic stenosis

The narrow ring.

## 4.1 Congenital

Thickening of the cricoid cartilage (cartilaginous) or of the submucosa (soft tissue) below the vocal folds. Diagnosed when the subglottic lumen is under *4 mm in a term neonate* (normal 4.5 to 5.5 mm) or under 3 mm in a premature baby. The child may be well until a cold produces stridor, recurrent or prolonged croup; the cry is normal.

## 4.2 Acquired

Far commoner now: from prolonged or repeated *intubation* in premature and sick neonates, especially with oversized tubes and movement; presenting as failure to extubate or stridor after extubation.

## 4.3 Grading

Cotton–Myer: grade I up to 50% obstruction, II 51 to 70%, III 71 to 99%, IV no lumen; assessed by the largest endotracheal tube that allows a leak.

## 4.4 Treatment

Many mild congenital stenoses improve as the larynx grows. Options: observation; endoscopic balloon dilatation and radial incision; anterior cricoid split (in neonates failing extubation); laryngotracheal reconstruction with costal cartilage grafts; and cricotracheal resection for severe stenosis; tracheostomy where needed.

**OPERATIVE GUIDES IN THEATRE** 1 operation

Subglottic stenosis surgery.

**Subglottic Stenosis**
Subglottic stenosis
›

# 5 CHAPTER FIVE Webs, cysts and haemangioma

Space-occupying lesions.

## 5.1 Laryngeal web

Incomplete recanalisation of the larynx in the tenth week. Usually an anterior glottic web between the vocal folds with a concave posterior edge; associated with 22q11 deletion (velocardiofacial syndrome), which should be tested for. A weak or absent cry from birth, and airway obstruction when the web is large. Thin webs are divided endoscopically with knife or CO2 laser; thick webs often have subglottic extension and need excision through a laryngofissure with a silicone keel and later dilatations. *Laryngeal atresia* causes complete obstruction at birth and death unless an emergency airway is made (or a tracheo-oesophageal fistula provides one).

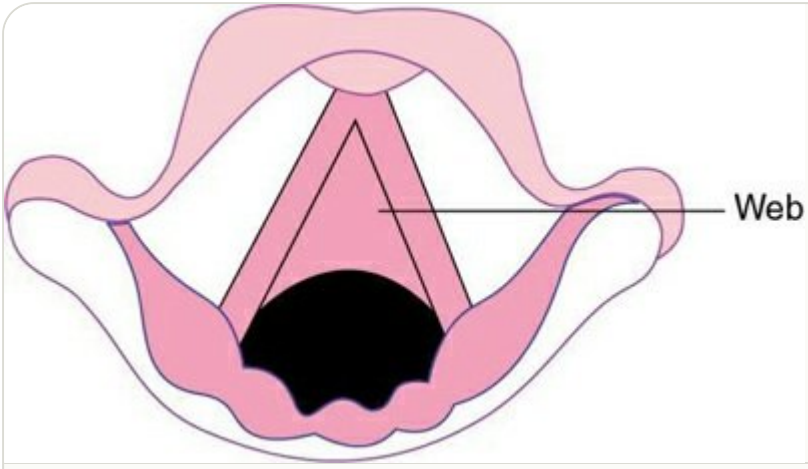



**FIGURE 2** A laryngeal web.
Dhingra, fig. 59.2

## 5.2 Subglottic haemangioma

Asymptomatic at birth; stridor appears at six weeks to six months as the haemangioma proliferates, often presenting as recurrent croup, with a normal cry;

worse with crying because of venous engorgement. About half have a cutaneous haemangioma, especially in the beard distribution; some have mediastinal lesions. Endoscopy shows a soft, compressible, reddish-blue swelling, usually posterolateral in the subglottis; biopsy is not needed and may bleed. MRI or CT shows the extent.

- **Propranolol** is now first-line and often produces dramatic involution.
- Systemic steroids (Dhingra describes dexamethasone, then prednisolone for many months) and intralesional steroid.
- CO2 or KTP laser or open excision for resistant lesions.
- Tracheostomy while awaiting involution, which is the natural history.

## 5.3 Laryngeal cysts

*Saccular cysts* (a mucus-filled saccule) and *ductal cysts* in the aryepiglottic fold, vallecula or subglottis appear as smooth, bluish, fluid-filled swellings that obstruct the airway. Needle aspiration can provide an emergency airway; endoscopic deroofing or excision with laser is definitive. Subglottic cysts follow intubation in premature infants.

## 5.4 Laryngocele

An air-filled dilatation of the saccule rising between the false fold and the thyroid lamina: *internal* (within the larynx, a supraglottic bulge), *external* (through the thyrohyoid membrane, a neck swelling that enlarges with Valsalva), or *combined*. Rare in infants; in adults linked to wind instrument players and glass blowers, and sometimes to a ventricular carcinoma blocking the saccule, which must be excluded. Infection forms a *laryngopyocele*. Endoscopic excision for internal ones, external excision for external and combined.

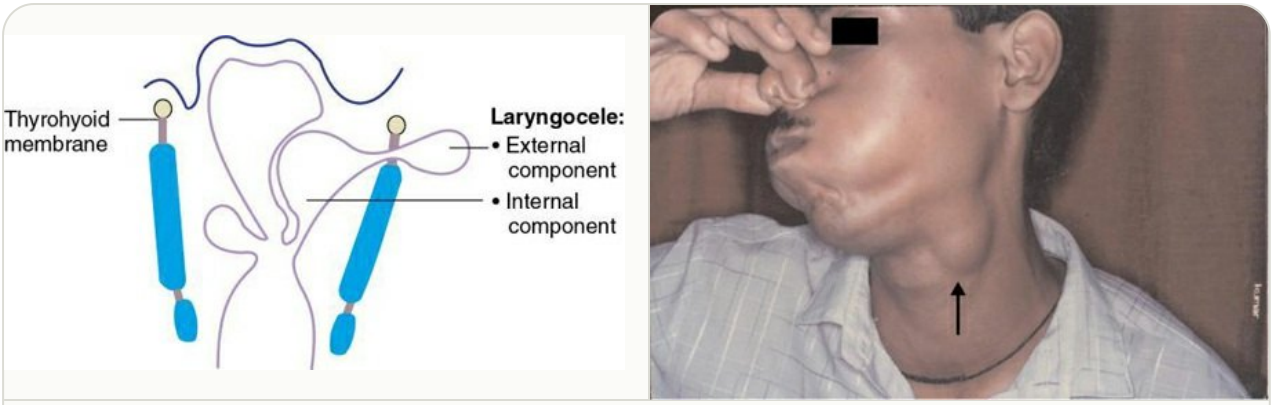



**FIGURE 3** First, a laryngocele of the mixed type with internal and external components. Second, a left laryngocele as seen on Valsalva (arrow).

Dhingra, figs 61.8 and 61.9

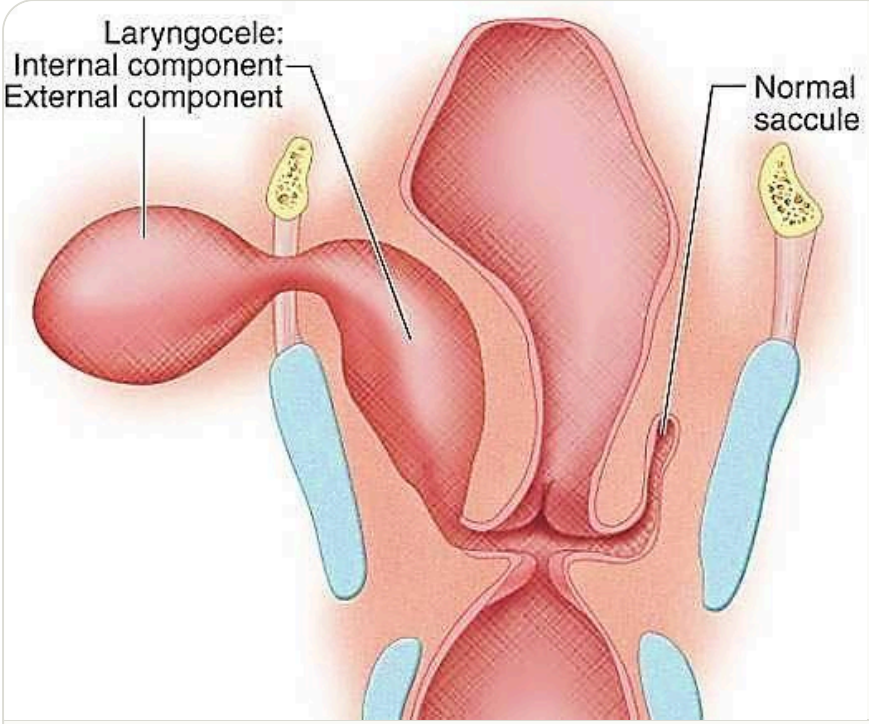
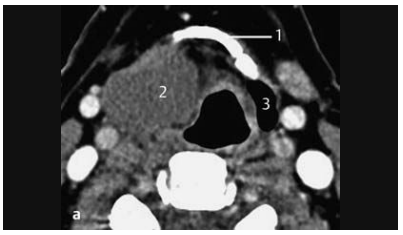



**FIGURE 4** Laryngocele.

Anatomy atlas · ENT Steps p. 240

**IMAGING IN THE APP** 1 case

Laryngocele on imaging.




**Laryngocele**

Infection & inflammation · 10 images

# 6 CHAPTER SIX Laryngeal cleft

A gap at the back.

Incomplete fusion of the posterior cricoid lamina and tracheo-oesophageal septum, leaving a communication between larynx and oesophagus.

### BENJAMIN–INGLIS CLASSIFICATION

| TYPE | EXTENT |
|---|---|
| I | Interarytenoid, above the vocal folds |
| II | Partly into the cricoid lamina |
| III | Through the whole cricoid, into the cervical trachea |
| IV | Into the thoracic trachea |

Choking, coughing and cyanosis with feeds, recurrent aspiration pneumonia, a hoarse cry and stridor. Diagnosed at microlaryngoscopy by *palpating* the interarytenoid area with a probe, since small clefts are hidden by redundant mucosa. Type I clefts may be managed by thickened feeds or endoscopic injection or repair; deeper clefts need endoscopic or open repair.

**OPERATIVE GUIDES IN THEATRE** 1 operation

Laryngeal cleft repair.

**Surgery for Aspiration: Laryngeal Cleft Repair and Laryngotracheal Separation**
Laryngeal cleft repair
›

# 7 CHAPTER SEVEN The trachea and beyond

Not always the larynx.

- **Tracheomalacia:** a soft, collapsing trachea, primary or after compression or tracheo-oesophageal fistula repair; expiratory stridor and a barking cough; most improve with growth; aortopexy, stenting or tracheostomy for severe cases.
- **Vascular rings and slings:** a double aortic arch, right arch with ligamentum, anomalous innominate artery or pulmonary artery sling compress the trachea and oesophagus, causing stridor and dysphagia. Contrast CT or MRI and a barium swallow; surgical division.
- **Tracheo-oesophageal fistula and oesophageal atresia.**
- **Cri-du-chat syndrome** (deletion of the short arm of chromosome 5): a cat-like cry with a narrow, diamond-shaped glottis, learning disability and hypotonia.

# 8 CHAPTER EIGHT For the examination

What the examiner wants to hear.

## 8.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Laryngomalacia:** presentation, endoscopic appearance, natural history, supraglottoplasty.
- **Normal versus hoarse cry** as a localising sign.
- **Subglottic stenosis:** the 4 mm rule, causes, grading and treatment.
- **Subglottic haemangioma:** age of onset, beard haemangioma, propranolol.
- **Laryngeal web** and its association with 22q11 deletion.
- **Laryngeal cleft:** palpate the interarytenoid area.

## 8.2 Common pitfalls

COMMON PITFALLS

- Labelling every noisy baby laryngomalacia without looking at the whole airway when symptoms are severe.
- Biopsying a subglottic haemangioma.
- Missing a cleft because the interarytenoid area was not palpated.

---

Written for this app from Oxford Textbook of Otolaryngology ch. 74, 79, Oxford Specialist Handbook ch. 23, Oxford Handbook of ENT ch. 17, Dhingra ch. 59 and K. J. Lee ch. 50, 53. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · LARYNX & AIRWAY

# Hoarseness, voice and speech disorders

A voice is made by three things working together: the lungs as a power supply, the vocal folds as a vibrator, and the throat, mouth and nose as resonators and articulators. Hoarseness is a disturbance of the vibrator; nasal speech a disturbance of the resonator; stuttering a disturbance of the rhythm. Most voice complaints are benign, many are functional, and a few are the first sign of cancer or neurological disease. Look at every voice that has been wrong for more than a few weeks, and listen to it as carefully as you look.

**AT A GLANCE**

- Functional aphonia: folds close on coughing.
- Puberphonia: Gutzmann pressure test.
- Spasmodic dysphonia: a dystonia; botulinum toxin.
- Hyponasality is blockage; hypernasality is velopharyngeal insufficiency.
- "Spasmodic dysphonia" is now laryngeal dystonia; inducible laryngeal obstruction often coexists with chronic cough.

## 1 CHAPTER ONE Hoarseness

The commonest voice complaint.

### 1.1 Definition

Hoarseness (dysphonia) is roughness of voice from irregularity of the period or intensity of successive sound waves. Normal voice needs folds that (1) approximate properly, (2) have the right size, mass and stiffness, and (3) vibrate regularly in the air stream.

- **Failed approximation:** paralysis, fixation, a mass between the folds, sulcus, bowing.
- **Altered size and mass:** oedema, nodules, polyps, tumours (more); surgical loss or fibrosis (less).
- **Altered stiffness:** paralysis (less), scar and spasm (more).
- **Irregular vibration:** congestion, haemorrhage, lesions on the edge.

## 1.2 Causes

### CAUSES OF HOARSENESS

| GROUP | EXAMPLES |
|---|---|
| Inflammation | Acute viral laryngitis, diphtheria, irritant gases; chronic laryngitis from smoking, reflux, occupation and inhaled steroids; tuberculosis, syphilis, leprosy, fungi |
| Benign lesions | Nodules, polyp, Reinke's oedema, cyst, contact granuloma, laryngocele, amyloid, sulcus |
| Neoplasms | Papilloma, haemangioma, chondroma, granular cell tumour; leukoplakia; carcinoma and sarcoma |
| Trauma | Shouting with fold haemorrhage, external injury, foreign body, intubation |
| Neurological | Recurrent, superior laryngeal or combined paralysis; spasmodic dysphonia; Parkinson disease, myasthenia, motor neurone disease, multiple sclerosis, essential tremor |
| Joint fixation | Cricoarytenoid arthritis, dislocation |
| Congenital | Web, cyst, laryngocele, paralysis, sulcus |
| Systemic and hormonal | Hypothyroidism, acromegaly, virilising hormones, sarcoidosis, granulomatosis with polyangiitis, amyloidosis |
| Functional | Muscle tension dysphonia, ventricular dysphonia, functional aphonia, puberphonia, phonasthenia |
| Ageing | Presbyphonia with bowed, thin folds |

## 1.3 Evaluation

1. **History:** onset and duration, voice use and occupation, smoking, alcohol, reflux, inhalers, systemic symptoms, surgery (thyroid, chest, neck), and red flags (pain, otalgia, dysphagia, stridor, haemoptysis, weight loss, neck mass).
2. **Listening:** breathy (incomplete closure), rough (irregular vibration), strained (hyperfunction or spasm), weak, diplophonic, tremulous.
3. **Laryngoscopy:** any hoarseness lasting more than two to three weeks, and sooner in a smoker or with red flags, needs the larynx examined; malignancy must be excluded over forty.

4. **Stroboscopy:** the mucosal wave, closure pattern and symmetry.
5. **Neck, chest, cardiovascular and neurological examination** for the cause of a paralysis; imaging, endoscopy and laboratory tests as indicated.
6. **Direct and microlaryngoscopy** for biopsy, palpation of the arytenoids and treatment.
7. **Voice measures:** maximum phonation time, acoustic analysis, the GRBAS rating, and patient-reported voice handicap indices.

CHAPTER TWO

# 2 Functional voice disorders

Normal larynx, abnormal use.

## 2.1 Muscle tension dysphonia

Excessive tension of the laryngeal and neck muscles during phonation: strained voice, vocal fatigue and neck ache, with supraglottic squeezing, a posterior gap or anteroposterior compression on laryngoscopy. It may be primary or compensatory. Voice therapy with laryngeal manual therapy is the treatment.

## 2.2 Ventricular dysphonia (dysphonia plicae ventricularis)

The false folds take over phonation: a rough, low-pitched, unpleasant voice; on phonation the false folds meet and hide the true folds.

- **Secondary:** compensating for true fold paralysis, fixation, excision or tumour; difficult to treat.
- **Functional:** in a normal larynx, often psychogenic; the voice starts normally and then roughens. Voice therapy and counselling.

## 2.3 Functional (psychogenic) aphonia

Sudden loss of voice, usually in emotionally stressed young women of fifteen to thirty, communicating in a whisper without other laryngeal symptoms. The folds stay apart on attempted phonation but *close normally on coughing*, and the cough is loud and clear. Reassure the patient that the larynx is normal, restore voice through

coughing, humming and throat clearing exercises, and address the stress; psychotherapy as needed.

## 2.4 Puberphonia (mutational falsetto)

Persistence of the high-pitched childhood voice in a physically normal adolescent boy or young man, often from emotional immaturity or social insecurity, while the larynx has matured. *Gutzmann's pressure test*: pressing the thyroid prominence backwards and downwards relaxes the overstretched folds and produces a low-pitched voice; the patient learns to reproduce it, first with pressure, then without, through syllables, words and sentences. The prognosis is good; type III thyroplasty is a last resort.

## 2.5 Phonasthenia

Weakness and fatigue of the phonatory muscles after misuse or laryngitis: a voice that tires quickly. Laryngoscopy shows an elliptical gap between the folds (thyroarytenoid weakness), a triangular posterior gap (interarytenoid weakness), or a *keyhole* glottis (both). Voice rest, vocal hygiene and therapy.

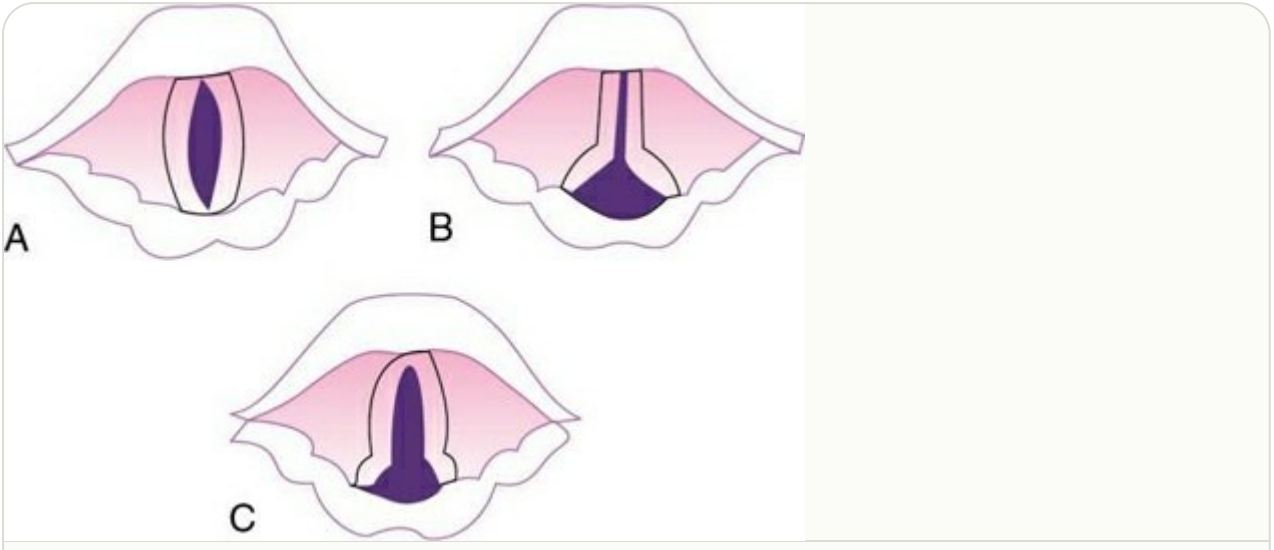



**FIGURE 1** The glottis in phonasthenia: (A) weakness of the thyroarytenoids, (B) of the interarytenoid, (C) of both, giving a keyhole glottis.
Dhingra, fig. 63.1

# 3 CHAPTER THREE Spasmodic dysphonia

A laryngeal dystonia.

## 3.1 Nature

A focal dystonia, thought to arise in the basal ganglia, with involuntary spasms of the laryngeal muscles during speech; the larynx is structurally normal and behaves normally for breathing, laughing, crying, whispering and often singing.

## 3.2 Types

- **Adductor (about 90%):** spasms of the thyroarytenoid and lateral cricoarytenoid; a strained, strangled, effortful voice with breaks, especially on words beginning with vowels.
- **Abductor:** spasms of the posterior cricoarytenoid; a breathy voice with breathy breaks after voiceless consonants (p, t, k), worse with stress and on the telephone.
- **Mixed.**

Voice tremor often coexists. Exclude Parkinson disease, essential tremor, multiple sclerosis, motor neurone disease and tardive dyskinesia.

## 3.3 Treatment

- **Botulinum toxin A** is the treatment of choice: small doses into the thyroarytenoid–lateral cricoarytenoid complex on one or both sides for adductor spasm, through the cricothyroid membrane under EMG guidance or through a flexible scope. Benefit lasts about three to four months; side effects are temporary breathiness and choking on fluids. For abductor spasm, injection into the posterior cricoarytenoid is harder and less effective, and is staged one side at a time to avoid airway obstruction.
- **Voice therapy** helps alongside toxin but not alone; psychotherapy is ineffective.
- **Surgery:** recurrent nerve section (abandoned because symptoms return); selective adductor denervation–reinnervation; type II (lateralisation) thyroplasty; thyroarytenoid myectomy.

**OPERATIVE GUIDES IN THEATRE** 1 operation

Office-based laryngeal procedures, including botulinum toxin injection.

**Office-Based Laryngeal Procedures**
Office procedures and botulinum toxin
›

**MOVEMENT DISORDERS OF THE LARYNX (OXFORD TEXTBOOK)**

**Laryngeal dystonia** is the term a recent multidisciplinary consensus proposed to replace "spasmodic dysphonia". It is task-specific: speaking is affected, but whispering, laughing and crying are not. *Adductor* dystonia, the commonest, gives strained-strangled breaks on voiced sounds; *abductor* dystonia gives breathy breaks on voiceless consonants; mixed forms occur. Newer variants are *singer's* dystonia, affecting only singing, and *adductor respiratory* dystonia, with adductor spasms on inspiration causing stridor. A simple bedside test is to count from 80 to 90 (voiced sounds, worse in adductor dystonia) and from 60 to 70 (voiceless sounds, worse in abductor dystonia).

**Vocal tremor** may be isolated or part of essential tremor of the head and hands, and is seen as rhythmic movement of the larynx and pharynx at rest phonation. **Parkinson's disease** gives a quiet, breathy, monotonous voice with bursts of rapid speech, usually after the limb signs; intensive loudness-focused voice therapy helps.

**Inducible laryngeal obstruction** (formerly paradoxical vocal fold movement or vocal cord dysfunction) is inappropriate adduction of the folds on inspiration, triggered by exercise, irritants or cough, and often mistaken for asthma. Laryngoscopy is usually normal unless done during symptoms or provocation, and many patients have sensations of tightness and altered breathing rather than visible closure. It often coexists with **chronic refractory cough** (cough over eight weeks, affecting 5 to 10% of adults), and both are best managed in a multidisciplinary upper airway clinic with breathing and laryngeal control techniques taught by speech and language therapists.

# 4 CHAPTER FOUR Resonance and fluency

Beyond the larynx.

## 4.1 Hyponasality (rhinolalia clausa)

Lack of nasal resonance on the nasal consonants m, n and ng ("my nose" sounds like "by dose"), from blockage of the nose or nasopharynx.

## 4.2 Hypernasality (rhinolalia aperta)

Too much nasal resonance on oral sounds, with nasal escape, because the nasopharynx is not sealed off from the oropharynx or there is a communication between mouth and nose.

CAUSES OF ALTERED NASAL RESONANCE

| HYPONASALITY | HYPERNASALITY |
|---|---|
| Common cold, allergic rhinitis | Velopharyngeal insufficiency |
| Nasal polyps and tumours | Cleft palate and submucous cleft |
| Adenoids, nasopharyngeal mass | Congenitally short palate, large nasopharynx |
| Choanal atresia | Palatal paralysis |
| Habitual or familial | After adenoidectomy, oronasal fistula; habitual or familial |

Hypernasality is assessed by perceptual rating, nasometry, nasendoscopy and videofluoroscopy of velopharyngeal closure, and treated by speech therapy, prosthetic palatal lifts, or surgery (pharyngeal flap, sphincter pharyngoplasty, palate repair, posterior wall augmentation).

**OPERATIVE GUIDES IN THEATRE** 1 operation

Cleft palate and velopharyngeal dysfunction.

**Cleft Palate and Velopharyngeal Dysfunction**
Velopharyngeal dysfunction
›

## 4.3 Stuttering

A disorder of fluency: hesitation, repetitions, prolongations and blocks, with secondary mannerisms such as grimacing, blinking and head jerks. Normal dysfluency is common between two and four years; parental overreaction may fix it. Prevention by educating parents; treatment by speech therapy and psychological support to reduce the fear of speaking.

## 4.4 Dysarthria and aphasia

Dysarthria is a motor disorder of speech production (flaccid, spastic, ataxic, hypokinetic, hyperkinetic); aphasia is a disorder of language (expressive, receptive). Both are neurological and need neurological assessment.

# 5 CHAPTER FIVE Voice care

Prevention and therapy.

VOCAL HYGIENE

- Hydrate well; humidify dry air.
- Avoid smoking, excess alcohol and caffeine; treat reflux.
- Avoid shouting, prolonged loud talking, whispering when hoarse, and habitual throat clearing.
- Rest the voice after heavy use; warm up before performing.
- Use amplification in noisy rooms and classrooms.

Voice therapy by a speech and language therapist retrains breathing, onset, pitch, resonance and muscle tension; it is the first treatment for nodules, muscle tension dysphonia, functional disorders and many benign lesions, and supports surgery.

# 6 CHAPTER SIX For the examination

What the examiner wants to hear.

## 6.1 Viva essentials

**WHAT THE EXAMINER IS LOOKING FOR**

- **Definition and causes** of hoarseness, and the two-to-three-week rule.
- **Functional aphonia:** the folds close on coughing.
- **Puberphonia** and Gutzmann's test.
- **Phonasthenia:** elliptical, triangular and keyhole glottis.
- **Spasmodic dysphonia:** types and botulinum toxin.
- **Hyponasality versus hypernasality** with causes.

## 6.2 Common pitfalls

**COMMON PITFALLS**

- Diagnosing functional aphonia without seeing the larynx.
- Calling spasmodic dysphonia psychogenic.
- Removing adenoids in a child with a submucous cleft and causing hypernasality.

Written for this app from Oxford Textbook of Otolaryngology ch. 47, 71, 72, Oxford Specialist Handbook ch. 8, 13, Oxford Handbook of ENT ch. 22, Dhingra ch. 63 and K. J. Lee ch. 50. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · LARYNX & AIRWAY

# Tracheostomy and airway endoscopy

The tracheostomy is one of the oldest operations in surgery and still one of the most important: it bypasses an obstructed upper airway, shortens the dead space, allows the chest to be cleared and ventilators to be weaned. Done calmly on an intubated patient it is a safe, orderly procedure; done in a hurry on a struggling, hypoxic patient it is one of the most dangerous in the specialty. Know the anatomy, know the alternatives that buy time, and know how to look after the tube, because most tracheostomy deaths happen afterwards, on the ward, from a blocked or displaced tube.

**AT A GLANCE**

- Never through the first ring.
- Do not change the tube before a track forms.
- Cricothyroidotomy for can't intubate, can't oxygenate.
- Rigid bronchoscopy for foreign bodies.
- NTSP bedside algorithms and capnography; tracheo-innominate fistula needs cuff overinflation and theatre.

## 1 CHAPTER ONE Tracheostomy

Purpose and indications.

### 1.1 Definitions

*Tracheotomy* is opening the trachea; *tracheostomy* is making that opening into a stoma at the skin. The terms are used interchangeably.

### 1.2 What a tracheostomy does

1. **Bypasses obstruction** anywhere from the lips to the stoma.
2. **Improves alveolar ventilation** by reducing dead space by 30 to 50% (normal dead space about 150 mL) and lowering airway resistance.
3. **Protects the airway** with a cuffed tube against aspiration of secretions (coma, bulbar palsy) and blood (maxillofacial and laryngeal trauma), and allows the pharynx

to be packed.

4. **Allows suction** of secretions when the patient cannot cough (coma, head injury, respiratory paralysis) or coughing hurts (chest and upper abdominal injury or surgery).
5. **Permits prolonged ventilation:** for intermittent positive pressure ventilation expected beyond a week or two (Dhingra suggests beyond 72 hours), tracheostomy is better tolerated than translaryngeal intubation and protects the larynx.
6. **Allows anaesthesia** when intubation is impossible (trismus, obstructing tumours).

## 1.3 Indications

INDICATIONS FOR TRACHEOSTOMY

| GROUP | EXAMPLES |
|---|---|
| Upper airway obstruction: infection | Epiglottitis, croup, diphtheria, Ludwig's angina, quinsy, retropharyngeal and parapharyngeal abscesses |
| Obstruction: trauma | Laryngotracheal injury, maxillofacial and mandibular fractures, endoscopic and intubation injury |
| Obstruction: other | Tumours of the larynx, pharynx, tongue and thyroid; foreign bodies; laryngeal oedema (burns, irritants, allergy, radiation); bilateral abductor paralysis; congenital lesions (webs, cysts, subglottic stenosis, haemangioma) |
| Retained secretions | Coma, head injury, stroke, overdose; respiratory muscle paralysis (spinal injury, Guillain–Barré, myasthenia, polio); tetanus; painful cough; aspiration |
| Respiratory insufficiency | Prolonged ventilation; chronic lung disease (emphysema, bronchitis, bronchiectasis) |
| Prophylactic | Major surgery of the tongue, floor of mouth, mandible, pharynx and larynx; obstructive sleep apnoea unresponsive to other treatment |

In children under a year most indications are congenital (subglottic stenosis, haemangioma, cysts, webs, bilateral paralysis); in older children inflammatory or traumatic (croup, epiglottitis, diphtheria, burns, prolonged intubation, papillomatosis).

# 2 CHAPTER TWO Types

When and where.

## 2.1 By urgency

- **Emergency:** obstruction is complete or nearly so and intubation is impossible; ideally replaced by cricothyroidotomy or by intubation followed by an orderly tracheostomy.
- **Elective (orderly):** planned, with the patient intubated where possible; therapeutic or prophylactic.
- **Permanent:** an end stoma after laryngectomy, or a long-term tube for bilateral paralysis or stenosis.
- **Percutaneous dilatational:** at the bedside in intensive care.
- **Minitracheostomy and cricothyroidotomy.**

## 2.2 By level

The thyroid isthmus lies over the second to fourth tracheal rings.

- **High** (above the isthmus, through the first ring): risks perichondritis of the cricoid and subglottic stenosis, and is avoided; it is acceptable only in laryngeal cancer, when the stoma will be removed at laryngectomy.
- **Mid** (through the second to fourth rings, dividing or retracting the isthmus): the preferred site.
- **Low** (below the isthmus): the trachea is deep, close to great vessels, and the tube presses on the suprasternal notch.

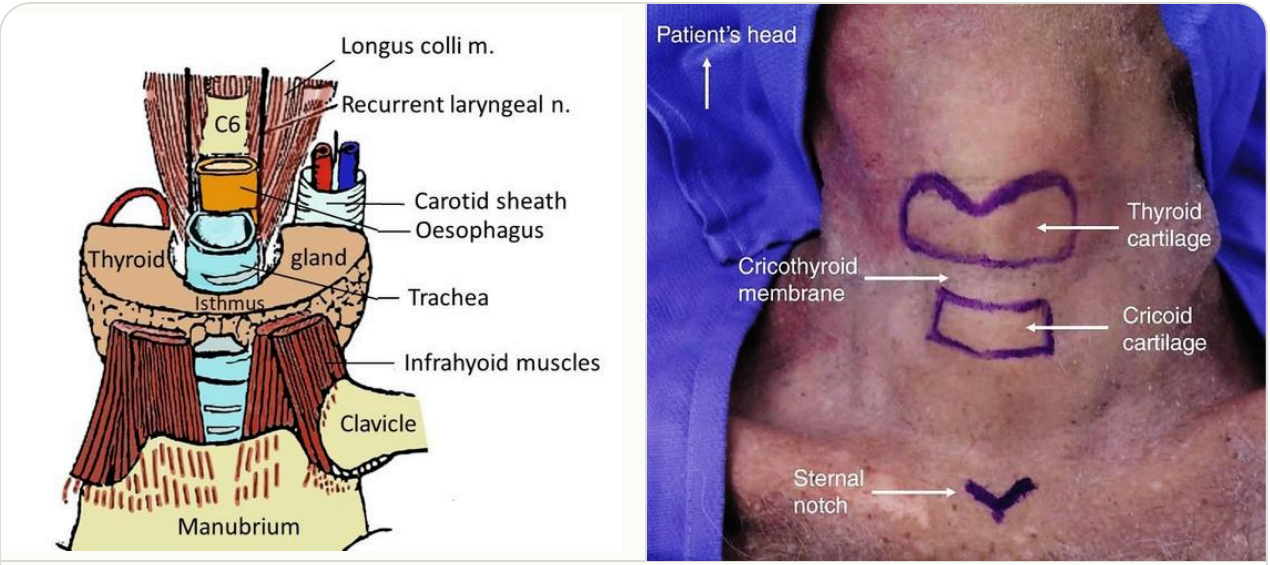



**FIGURE 1** The cervical trachea and recurrent laryngeal nerve, and surface marking of the thyroid and cricoid cartilages and cricothyroid membrane.

Anatomy atlas · Doss p. 176, ENT Steps p. 262

# 3 CHAPTER THREE Technique

The open operation.

## 3.1 Preparation

Intubate first whenever possible, especially in children. Supine with a shoulder roll to extend the neck and bring the trachea forward (unless the cervical spine is injured). Local infiltration with lidocaine and adrenaline in the awake patient; general anaesthesia when intubated.

## 3.2 Steps

1. **Incision:** a vertical midline incision from the cricoid to just above the sternal notch gives fast access with little bleeding and suits emergencies; a 4 to 5 cm horizontal incision two fingerbreadths above the notch gives a better scar in elective cases.
2. Dissect in the midline; retract or tie the anterior jugular veins.
3. Separate the strap muscles in the midline and retract them laterally.
4. Retract the thyroid isthmus, or divide it between clamps and suture-ligate it.

5. Instil a little lidocaine into the trachea to suppress coughing; steady the trachea with a cricoid hook.
6. Open the trachea at the second to fourth rings: a vertical slit, a window, an inverted U (Björk) flap sutured to the skin to ease reinsertion, or a slit with stay sutures; never through the first ring.
7. Withdraw the endotracheal tube to just above the opening under direct vision, insert the tracheostomy tube, inflate the cuff, confirm position (capnography), and only then remove the endotracheal tube.
8. Secure the tube with tapes (and sutures); do not close the skin tightly or pack the wound, which causes surgical emphysema.

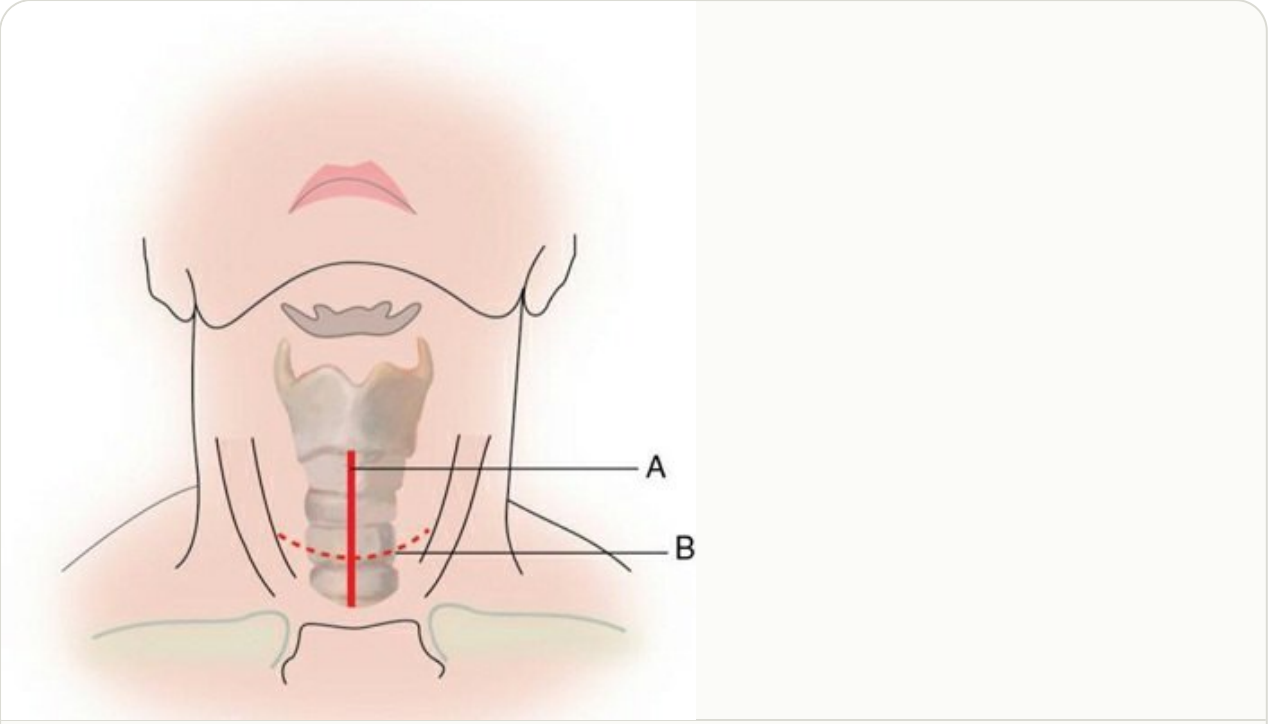



**FIGURE 2** Skin incisions for tracheostomy: (A) vertical midline, (B) transverse.
Dhingra, fig. 64.1

## 3.3 In infants and children

- Operate under general anaesthesia over an endotracheal tube or bronchoscope; the trachea is soft and easily lost, and straying laterally injures the recurrent nerve and carotid.
- Do not hyperextend the neck: it draws the pleura, innominate vessels and thymus into the field and invites too low an opening.
- Place *stay sutures* on either side of the midline before incising; they are the rescue if the tube comes out in the first days.

- Make a simple vertical incision; never remove cartilage.
- Do not push the blade deep: the posterior wall and oesophagus lie close.
- Choose a soft tube of the right diameter, length and curvature; confirm its position with a chest radiograph.

## 3.4 Percutaneous dilatational tracheostomy

In the intubated intensive care patient under sedation, through a small incision, a needle, guide wire and dilators (single tapered or serial) enter between the second and third rings under bronchoscopic guidance through the endotracheal tube. It avoids transfer to theatre and is quicker and cheaper, with fewer wound infections. Avoid in obesity, neck masses, difficult intubation, poor extension, impalpable landmarks, coagulopathy and in children. Complications: paratracheal placement, bleeding, posterior wall injury and emphysema.

**OPERATIVE GUIDES IN THEATRE** 2 operations

Open and percutaneous tracheostomy.

**Open Tracheostomy**
Open tracheostomy
›

**Percutaneous Tracheostomy**
Percutaneous tracheostomy
›

# 4 CHAPTER FOUR Tracheostomy tubes

Choosing the tube.

**FEATURES OF TRACHEOSTOMY TUBES**

| FEATURE | PURPOSE |
|---|---|
| Cuffed | Seals the airway for ventilation and against aspiration; keep pressure below about 25 to 30 cm of water to avoid mucosal necrosis |
| Uncuffed | For patients who no longer need ventilation or protection; children |

| FEATURE | PURPOSE |
|---|---|
| Inner cannula | Removed and cleaned to prevent blockage without changing the tube |
| Fenestrated | A window allows air to pass up through the larynx for speech when the inner cannula is removed and the tube occluded |
| Speaking (one-way) valve | Lets air in through the tube and out through the larynx (only with the cuff deflated) |
| Adjustable flange | For deep or obese necks |
| Metal (silver, Jackson) | Durable, for long-term permanent use; no cuff |




**FIGURE 3** A patient with a suction-aid tracheostomy tube receiving oxygen through a side port. Dhingra, fig. 64.3

# 5 CHAPTER FIVE Aftercare

Where safety is won or lost.

## 5.1 The first days

1. **Constant observation** by trained staff: bleeding, displacement, blockage and secretions; a call bell and writing pad.
2. **Suction** as often as needed with a sterile catheter, applying suction only while withdrawing it, to avoid mucosal injury.

3. **Humidification** of inspired air (heated humidifiers, nebulisers, heat-moisture exchangers) to prevent crusting and tracheitis; saline instillation and mucolytics for crusts.
4. **Inner cannula** cleaned regularly; the *outer tube is not changed for the first three to five days* until a track forms, after which tube changes become straightforward.
5. **Cuff** pressure checked; deflated when no longer needed.
6. Spare tubes (same size and smaller), tracheal dilator, obturator, suction and oxygen at the bedside.

**THE BLOCKED OR DISPLACED TUBE**

Sudden respiratory distress in a patient with a tracheostomy: remove the speaking valve and inner cannula, pass a suction catheter; if it will not pass, deflate the cuff and remove the tube; oxygenate through the stoma and the face; if the patient was not a laryngectomy, they can also be oxygenated and intubated from above. Follow a standard emergency algorithm and colour-coded bedhead signs.

## 5.2 Decannulation

Remove the tube as soon as it is not needed, because prolonged use brings infection, ulceration, granulation, stenosis and scarring. Deflate the cuff, change to a smaller uncuffed tube, then occlude (cap) it; if the patient breathes, speaks and clears secretions comfortably for 24 hours (longer in children), remove the tube and tape the stoma, which usually closes within a week; persistent tracheocutaneous fistulae are closed surgically. In infants, decannulate in hospital with full reintubation equipment and monitoring.

## 5.3 Difficult decannulation

Look for persistence of the original disease, granulations at the stoma or the tube tip (often a suprastomal granuloma), subglottic stenosis or oedema, collapse of the anterior wall above the stoma, tracheomalacia, and psychological dependence; endoscopy of the larynx and trachea decides.

# 6 CHAPTER SIX Complications

Immediate, early and late.

## 6.1 At operation

- Haemorrhage (anterior jugular veins, isthmus, aberrant vessels).
- *Apnoea* after relief of long-standing obstruction, when the high carbon dioxide drive is suddenly removed; ventilate.
- Pneumothorax and pneumomediastinum (especially in children).
- Recurrent laryngeal nerve injury; injury to the oesophagus with a tracheo-oesophageal fistula.
- Aspiration of blood; false passage; cardiac arrest.

## 6.2 Early (hours to days)

- Reactionary and secondary bleeding.
- *Displacement* and *blockage* of the tube: the commonest causes of death.
- Surgical emphysema, tracheitis and crusting, atelectasis and pneumonia, wound infection.

## 6.3 Late

- *Tracheo-innominate fistula* with massive haemorrhage from erosion by the tube tip or cuff, often heralded by a sentinel bleed.
- Subglottic and laryngeal stenosis (high tracheostomy), tracheal stenosis at the stoma or cuff site, tracheomalacia.
- Tracheo-oesophageal fistula from cuff pressure or the tube tip.
- Granulations, difficult decannulation, persistent tracheocutaneous fistula, ugly or keloid scar.
- Corrosion and fracture of old metal tubes with aspiration of fragments.

**OPERATIVE GUIDES IN THEATRE** 2 operations

Tracheocutaneous fistula and depressed scar, and stenosis of the tracheostoma after laryngectomy.

**Management of a Tracheocutaneous Fistula and Depressed Scar**
Tracheocutaneous fistula
›

**Stenosis of the Tracheostoma Following Total Laryngectomy**
Stomal stenosis
›

CHAPTER SEVEN
# 7 Emergency airway

Buying time.

When obstruction leaves no time for an orderly tracheostomy:

1. **Jaw thrust** and head tilt (not in spinal injury) lift the tongue off the pharyngeal wall.
2. **Oropharyngeal airway** with bag-and-mask ventilation.
3. **Nasopharyngeal airway:** better tolerated by an awake patient.
4. **Supraglottic airway (laryngeal mask):** a rescue when mask ventilation and intubation fail; it does not help obstruction at or below the glottis.
5. **Intubation:** the quickest definitive airway; then an orderly tracheostomy if needed.
6. **Cricothyroidotomy:** through the cricothyroid membrane, between the thyroid and cricoid cartilages, in the "can't intubate, can't oxygenate" situation. *Scalpel cricothyroidotomy* (a stab through the membrane, rotated, with a bougie and a 6 mm cuffed tube) is the recommended adult technique. Convert to a formal tracheostomy within about 72 hours, because perichondritis and subglottic stenosis follow prolonged cricothyroidotomy. It is avoided in children under about ten to twelve, whose membrane is tiny.
7. **Needle cricothyroidotomy** with a large cannula and *transtracheal jet ventilation*: a temporary measure, the choice in small children; ensure a path for expiration or barotrauma follows.
8. **Emergency tracheostomy:** with the trachea fixed between thumb and finger, a vertical incision from the thyroid to the notch, quickly through to the second and third rings; hazardous on a struggling patient without light, suction or help, so intubate first if at all possible.



**FIGURE 4** A laryngeal mask airway.
Dhingra, fig. 64.4

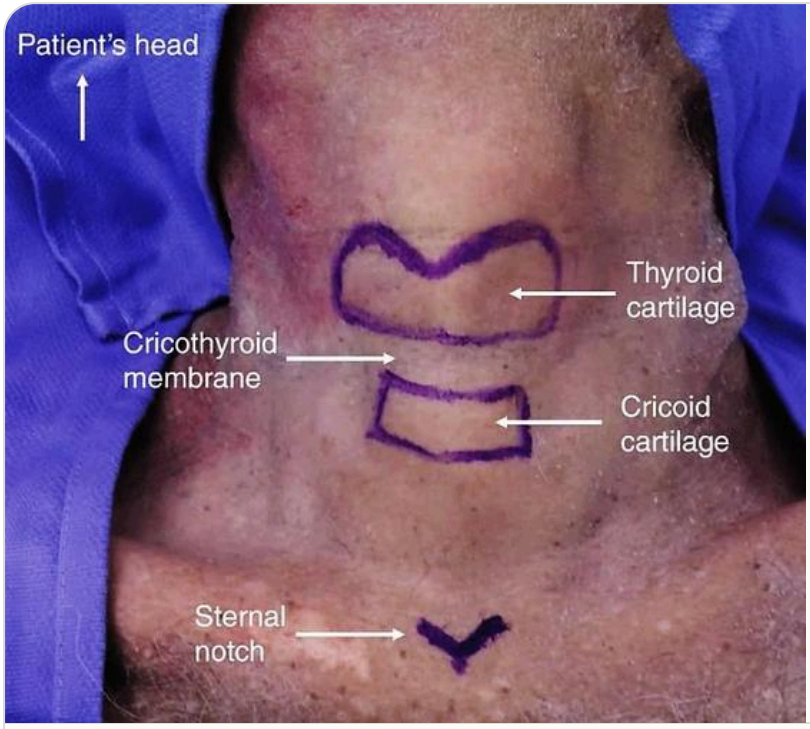



**FIGURE 5** Surface marking of the cricothyroid membrane.
Anatomy atlas · ENT Steps p. 262

# 8 CHAPTER EIGHT Airway endoscopy

Laryngoscopy and bronchoscopy.

## 8.1 Direct laryngoscopy

To see the larynx and hypopharynx directly: when indirect examination is impossible (children, a strong gag, an overhanging epiglottis), to examine the *hidden areas* (base of tongue, valleculae, lower piriform sinuses, infrahyoid epiglottis, anterior

commissure, ventricles and subglottis), to assess and biopsy tumours, and to treat (remove benign lesions and foreign bodies, dilate strictures). Contraindicated by cervical spine disease and by significant obstruction unless the airway has first been secured.

- **Position:** the *Boyce (sniffing, barking-dog) position*: head raised 10 to 15 cm on a pillow with the neck flexed on the chest and the head extended at the atlanto-occipital joint, aligning the oral, pharyngeal and laryngeal axes.
- **Technique:** protect the upper teeth; insert the laryngoscope along the right side of the tongue, move to the midline, lift the epiglottis, and examine every subsite systematically, lifting rather than levering on the teeth. Suspension and the operating microscope or angled telescopes allow *microlaryngoscopy*.
- **Complications:** injury to lips, teeth and tongue, bleeding, laryngeal oedema, laryngospasm.

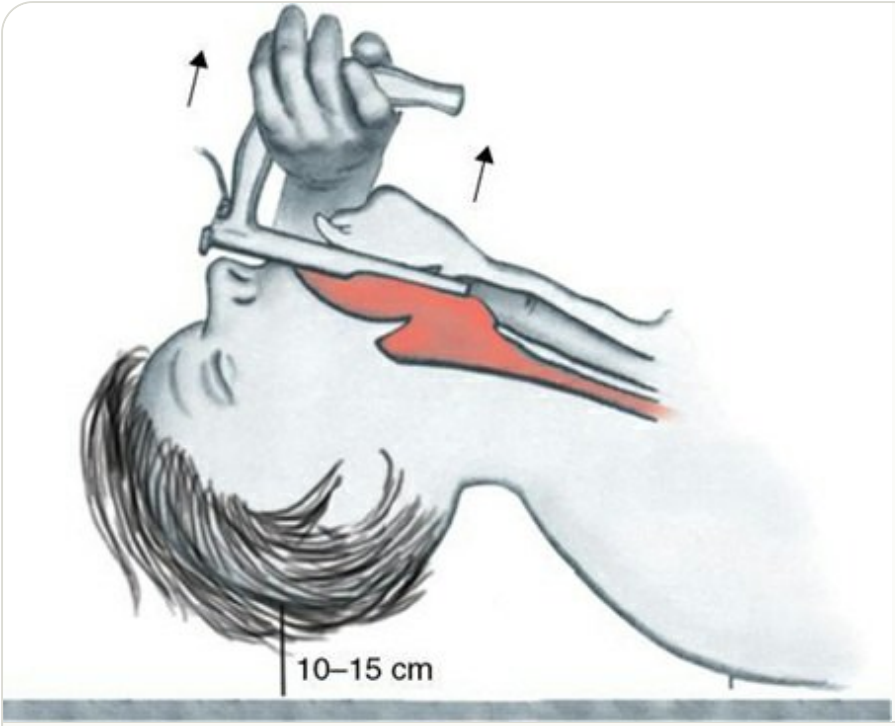



**FIGURE 6** Direct laryngoscopy in the sniffing position, lifting the laryngoscope without levering on the teeth.
Dhingra, fig. 91.1

## 8.2 Bronchoscopy

- **Indications:** unexplained wheeze, haemoptysis or cough over four weeks; radiographic collapse, persistent opacity, obstructive emphysema, hilar or mediastinal

shadows; vocal fold palsy; secretions for culture and cytology. Therapeutic: foreign bodies, retained secretions and mucus plugs.

- **Rigid bronchoscopy** under general anaesthesia allows ventilation through the scope, large biopsies, control of haemorrhage and is the instrument of choice for *foreign bodies*, especially in children. The bevel is turned to pass the glottis, and the head is moved to the left to enter the right bronchus and vice versa. Limit procedures in children to about twenty minutes to avoid subglottic oedema.
- **Flexible bronchoscopy** under topical anaesthesia (through the nose, an endotracheal tube or tracheostomy) reaches segmental and subsegmental bronchi, can be done at the bedside and in patients with neck or jaw problems, but controls ventilation and bleeding poorly and is not the tool for foreign bodies in children.

**OPERATIVE GUIDES IN THEATRE** 3 operations

Bronchoscopy, tracheoscopy and airway foreign bodies, and oesophagoscopy.

**Bronchoscopy**
Bronchoscopy
›

**Tracheoscopy, Bronchoscopy, and Airway Foreign Bodies**
›

**Esophagoscopy and Hypopharyngoscopy**
›

**SAFETY AND QUALITY (OXFORD TEXTBOOK)**

Tracheostomy problems account for up to half of airway-related hypoxic brain injuries and deaths in hospital, and most are preventable. The UK **National Tracheostomy Safety Project** emergency algorithms for a blocked or displaced tube (separate versions for patients with a patent upper airway and for laryngectomy "neck breathers") should be at every bedside, with bedside equipment, a sign showing the airway type, and staff trained to use them; the **Global Tracheostomy Collaborative** promotes the same team-based approach internationally. Waveform **capnography** confirms correct placement at surgery and after any tube change. In the first days, before a tract forms, displacement is an emergency: stay sutures and a Björk flap make reinsertion easier, and if the tube cannot be replaced the patient is oxygenated from above and intubated orally.

Preserving the thyroid isthmus where possible shortens surgery without adding complications. In a patient likely to need a laryngectomy, a **high** tracheostomy is preferred, so that the stoma can be excised with the specimen; the theoretical risk of tumour seeding is not clinically important. A thyroid ima artery is present in up to 5% and a high-riding innominate artery can be injured low in the neck. Tracheal stenosis is

promoted by cuff pressures above 30 $cmH_2O$, intubation beyond ten days, reflux, obesity, large tubes and percutaneous insertion. A **tracheo-innominate fistula** occurs in under 1% but kills over 80%; a sentinel bleed demands immediate assessment, and torrential bleeding is controlled by overinflating the cuff or by digital pressure through the stoma against the sternum while the patient goes to theatre.

# 9 CHAPTER NINE For the examination

What the examiner wants to hear.

## 9.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Functions and indications** of tracheostomy.
- **Why not through the first ring;** the levels.
- **Steps** of the operation, and differences in children.
- **Tubes,** aftercare, decannulation and causes of failed decannulation.
- **Complications,** especially displacement, blockage and tracheo-innominate fistula.
- **Emergency airway options:** cricothyroidotomy and jet ventilation.
- **Direct laryngoscopy** position and hidden areas; **rigid versus flexible bronchoscopy.**

## 9.2 Common pitfalls

COMMON PITFALLS

- Making a high tracheostomy through the first ring.
- Changing the tube on day one without a mature track or stay sutures.
- Suturing the skin tightly around the tube.
- Leaving a cricothyroidotomy in place for weeks.

Written for this app from Oxford Textbook of Otolaryngology ch. 70, Oxford Specialist Handbook ch. 8, Oxford Handbook of ENT ch. 20, Dhingra ch. 64, 91, 92 and K. J. Lee ch. 50. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · LARYNX & AIRWAY

# Lasers, radiofrequency, cryosurgery and hyperbaric oxygen

Lasers, radiofrequency, coblation, cryotherapy and hyperbaric oxygen are tools, and each works through one physical principle. Know which tissue absorbs which wavelength, how much collateral damage each tool leaves, and the safety rules. The airway fire is the disaster every ENT surgeon working with lasers must prevent.

**AT A GLANCE**

- CO2 (10,600 nm) is absorbed by water; KTP by haemoglobin; Nd:YAG coagulates deeply.
- CO2 cannot pass through a fibre.
- Airway fire: stop laser, stop gas, remove tube, saline, reintubate, bronchoscopy.
- Coblation works at 40 to 70 degrees through a saline plasma.

## 1 CHAPTER ONE Lasers

Light amplification by stimulated emission of radiation.

### 1.1 How a laser works

1. **Pumping:** an energy source (electric current or a flash lamp) excites the atoms of an *active medium*. The medium can be a gas (CO2, argon, helium–neon), a solid crystal (Nd:YAG, KTP, erbium, holmium), a liquid dye, or a semiconductor (diode).
2. **Spontaneous and stimulated emission:** excited electrons fall back and release photons. A photon striking another excited atom makes it release an identical photon, of the same wavelength, direction and phase. This is *stimulated emission*.
3. **Population inversion:** when more atoms are excited than not, amplification begins.
4. **Resonance:** two mirrors at the ends of a resonant cavity (one fully reflecting, one partly) bounce the light back and forth to amplify it. The beam leaves through the partly reflecting mirror.

The result is light that is *monochromatic*, *coherent* (in phase), *collimated* (parallel) and of high intensity, and that can be focused to a tiny spot.

## 1.2 Laser and tissue

A laser beam meeting tissue is *reflected*, *transmitted*, *scattered* or *absorbed*, and only the absorbed energy has any effect. Absorption depends on the **chromophore**, the target molecule that takes up a particular wavelength. *Water* absorbs infrared, which is why the CO2 and Er:YAG lasers cut so cleanly. *Haemoglobin* absorbs green and yellow light, with peaks at 418, 542 and 577 nm, the target of the KTP laser at 532 nm and the pulsed dye laser at 585 to 595 nm. *Melanin* absorbs across the spectrum, mainly in the visible and near-infrared, the target of ruby, alexandrite and Nd:YAG lasers.

The absorbed energy acts as heat. At about 60°C proteins denature and coagulate reversibly, around 80°C collagen breaks down, and at 100°C water boils and the tissue *vaporises*. A laser wound therefore has three zones:

1. A crater of vaporisation.
2. A zone of thermal necrosis, which seals small vessels, nerves and lymphatics.
3. A zone of thermal conductivity and repair, which is reversible.

Depth of penetration follows from absorption. Longer visible wavelengths penetrate further, but water-absorbed infrared light from CO2 and Er:YAG lasers penetrates only microns, giving precise cuts with little collateral damage. The Nd:YAG laser, by contrast, coagulates about 4 mm deep, which is good for haemostasis but imprecise. Beyond heat, lasers act photochemically in photodynamic therapy, photoacoustically in stone fragmentation, and by photoablation in the excimer laser.

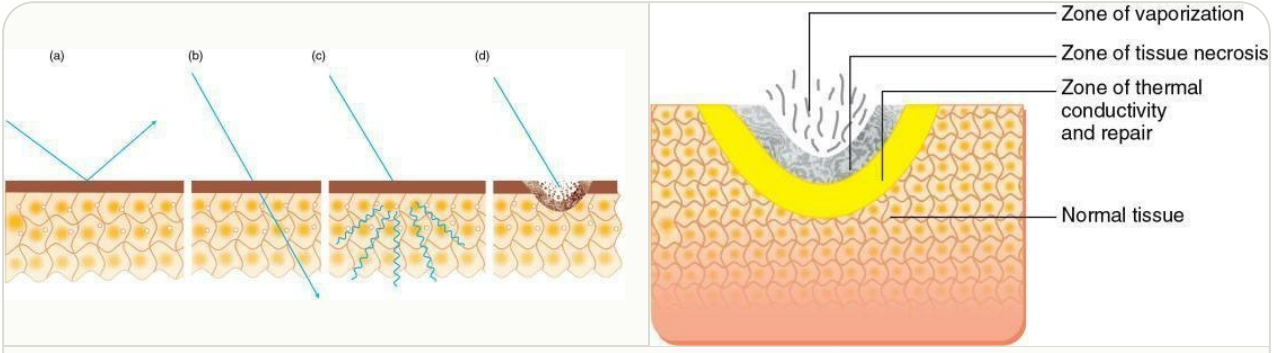



**FIGURE 1** First, the fates of a laser beam in tissue: (a) reflection, (b) transmission, © scatter, (d) absorption with tissue destruction. Second, the zones of tissue damage: vaporisation, thermal necrosis, and thermal conductivity and repair.

Dhingra, figs 71.1 and 71.2

## 1.3 Parameters the surgeon controls

The surgeon sets the **power** in watts and the **spot size**, which is smallest at the focal point: a focused beam cuts, while a defocused beam coagulates and ablates. Exposure can be **continuous**, **pulsed**, or **superpulsed and Q-switched**, in which very short pulses are separated by intervals longer than the tissue's *thermal relaxation time*, so that heat dissipates between pulses and collateral damage falls. **Power density** (W/cm²), the power divided by the spot area, sets the rate of cutting, and **fluence** (J/cm²), the power density multiplied by time, is the total energy delivered.

## 1.4 Delivery

Visible and near-infrared lasers, the argon, KTP, Nd:YAG, diode and holmium lasers, travel down *flexible optical fibres* and can be used through endoscopes and bronchoscopes. The CO2 laser cannot pass through ordinary fibres; it travels by an *articulated arm* with mirrors to a micromanipulator on the operating microscope, or through hollow flexible waveguides. Because CO2 and Nd:YAG beams are invisible, a red helium–neon *aiming beam* shows where they will strike.

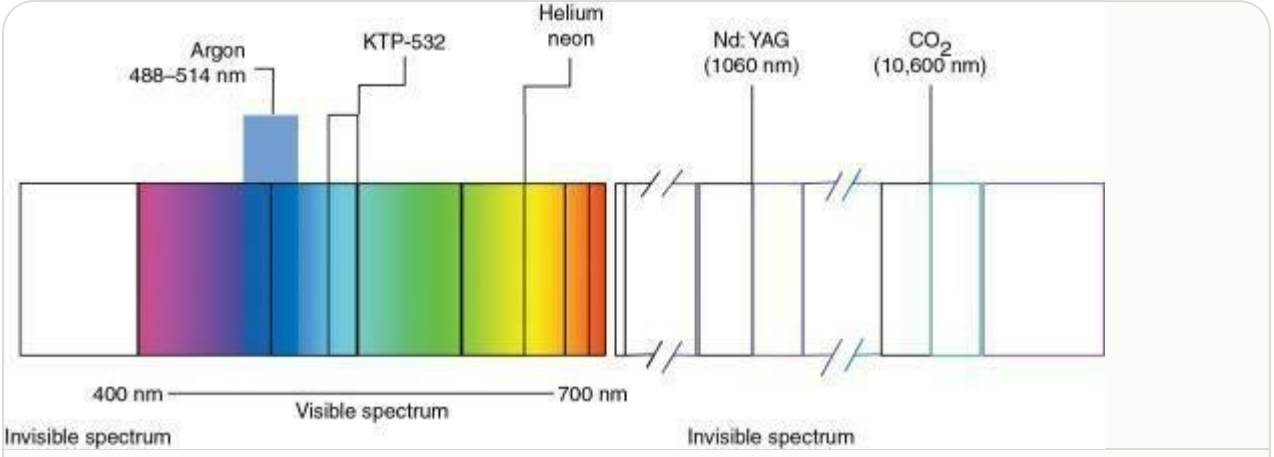



**FIGURE 2** The electromagnetic spectrum, with the wavelengths of the common surgical lasers. Dhingra, fig. 71.3

## COMMON LASERS IN ENT

| LASER | WAVELENGTH | ABSORBED BY | DELIVERY | TYPICAL USES |
|---|---|---|---|---|
| CO2 | 10,600 nm (far infrared, invisible) | Water | Articulated arm, micromanipulator, waveguide | Workhorse of laryngology: nodules, polyps, cysts, papillomas, Reinke oedema, leukoplakia, T1 glottic cancer (transoral laser microsurgery), cordotomy and arytenoidectomy, stapedotomy, skin resurfacing |
| KTP | 532 nm (green) | Haemoglobin, melanin | Fibre | Vascular lesions, papillomas (office angiolytic laser), stapedotomy, sinus surgery, HHT |
| Argon | 488 to 514 nm (blue–green) | Haemoglobin, melanin | Fibre | Vascular skin lesions, stapes surgery (historical) |
| Nd:YAG | 1064 nm (near infrared, invisible) | Pigment; deep penetration | Fibre | Deep coagulation: palliation of tracheobronchial and oesophageal tumours, vascular malformations, HHT |
| Diode | 800 to 980 nm | Haemoglobin, melanin | Fibre | Turbinate reduction, stapedotomy, dacryocystorhinostomy, vascular lesions |
| Er:YAG | 2940 nm | Water (peak) | Fibre or arm | Fine skin resurfacing, bone (stapes) |
| Ho:YAG | 2100 nm | Water | Fibre | Bone, sinus, stones |
| Pulsed dye | 585 to 595 nm (yellow) | Haemoglobin | Fibre | Port-wine stains, haemangiomas, telangiectasia |

## 1.5 Advantages and disadvantages

Lasers offer precision, haemostasis and a bloodless field, cause little postoperative oedema and pain, and reach targets through endoscopes. Against this are their cost, the training they demand, thermal injury, the lack of tactile feedback, and the hazards of *fire and eye injury*.

## 1.6 Laser safety

Safety starts with organisation: a laser safety officer, trained and certified staff, and a "*laser in use*" sign on a closed door.

Everyone in the room, including the patient, wears *wavelength-specific goggles*: clear glass or plastic for CO2, orange or amber for argon and KTP, and green-blue for Nd:YAG. The reason is anatomical. Visible and near-infrared lasers (400 to 1400 nm) pass through the transparent media and burn the *retina*, whereas CO2 and other far-infrared lasers are absorbed by water at the front of the eye and burn the *cornea*. The patient's eyes are covered with wet pads and exposed skin with wet towels. The smoke plume can carry viable viral particles, such as HPV from papillomas, and is mutagenic, so a separate smoke evacuator is used.

The **airway fire** needs the three elements of the *fire triangle*: heat from the laser, fuel from the tube, drapes and cottonoids, and an oxidiser, oxygen or nitrous oxide. It is prevented by removing each element. Use laser-safe tubes, metal or specially wrapped, or jet ventilation without a tube; fill the cuff with *saline coloured with methylene blue* so that a puncture shows at once; protect the cuff and tissues with wet cottonoids; keep oxygen at 30% or less, and never above 40%; use *no nitrous oxide*; use the lowest effective power; and direct the beam only at the target.

**AIRWAY FIRE DRILL**

1. Stop the laser.
2. Stop ventilation and disconnect oxygen.
3. Remove the tube.
4. Pour saline into the airway.
5. Ventilate by mask with air.
6. Reintubate.
7. Do bronchoscopy to assess the injury and remove debris.

8. Give steroids, antibiotics and close follow-up for stenosis.

**OPERATIVE GUIDES IN THEATRE** 2 operations

Transoral CO2 laser microsurgery of the larynx and the tongue base.

**Transoral CO2 Laser Microsurgery for Cancer of the Larynx**
Transoral laser microsurgery
›

**Transoral Laser Microsurgery of the Base of the Tongue**
TLM tongue base
›

## 1.7 Photodynamic therapy

In photodynamic therapy a photosensitiser, given systemically (porphyrins, temoporfin) or topically to the skin (aminolaevulinic acid), concentrates in tumour cells. Light of a matching wavelength, around 630 to 652 nm, then generates oxygen free radicals that kill them. It is used for superficial cancers of the skin, oral cavity and larynx, for field cancerisation, and to palliate recurrences, and patients must avoid sunlight for weeks because the skin is photosensitive.

# 2 CHAPTER TWO Radiofrequency and coblation

Lower temperatures.

## 2.1 Radiofrequency ablation

Radiofrequency ablation passes a high-frequency alternating current, about 350 kHz to 4 MHz and commonly 460 kHz, through a probe placed *within* the tissue. Ionic agitation heats the tissue to about 70 to 85°C, coagulating it without charring, and fibrosis over the following three to six weeks shrinks it.

It reduces the volume of the inferior turbinates, the soft palate and the tongue base in snoring and obstructive sleep apnoea, and is used for tonsil reduction, lingual thyroid, thyroid nodules and microcystic lymphatic malformations. At the tongue base the hypoglossal neurovascular bundle lies about 2.7 cm deep and 1.6 cm lateral

to the foramen caecum, and must be respected. The procedure is done as an outpatient, causes little pain and can be repeated.

## 2.2 Coblation (controlled ablation)

Coblation passes radiofrequency energy through *conductive saline* to create a thin *plasma field* of energised sodium ions. These break molecular bonds and dissolve tissue at only *40 to 70°C*, against 400 to 600°C for electrocautery, so there is little charring or collateral injury. A single wand ablates, suctions, irrigates and coagulates.

Coblation is used for tonsillectomy and adenoidectomy, including intracapsular tonsillotomy in children; tongue base and palatal surgery; turbinates and nasal polyps; laryngeal papillomas and cysts; cordotomy; subglottic and tracheal lesions; and juvenile angiofibroma. After coblation tonsillectomy, pain and haemorrhage are similar to or slightly better than other hot techniques, and intracapsular coblation causes less pain and bleeding.

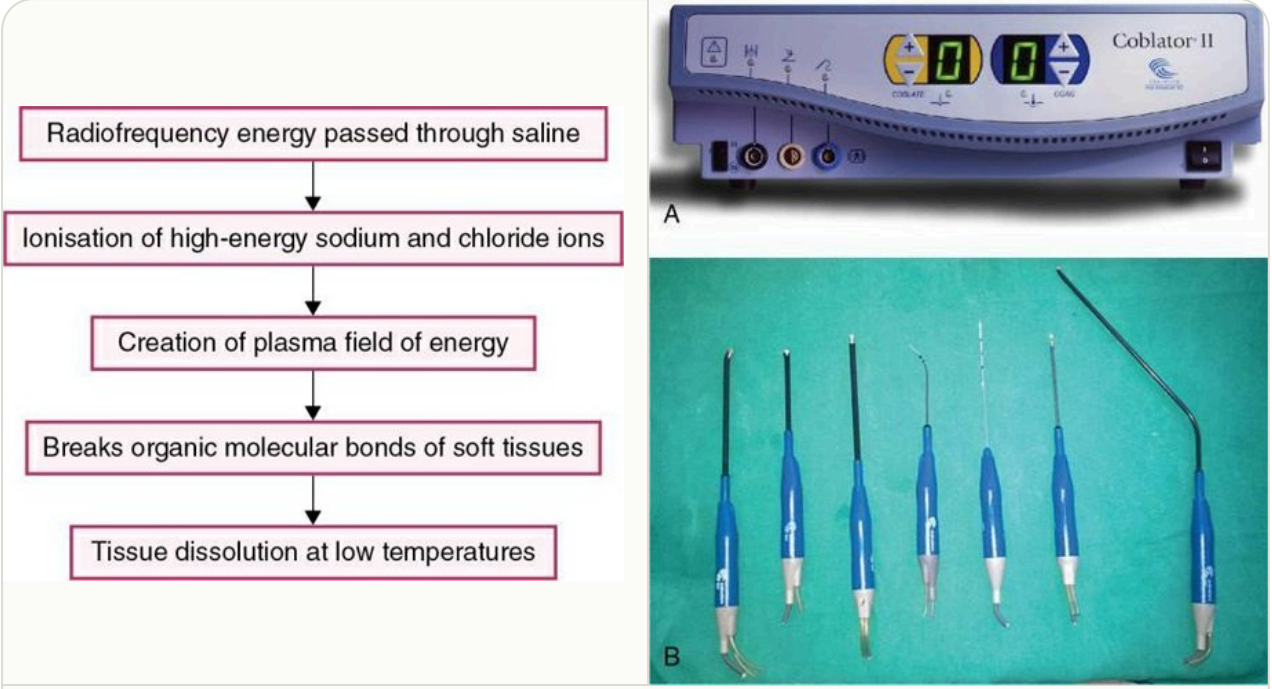



**FIGURE 3** First, the principle of coblation. Second, (A) a coblation generator and (B) the different wands.

Dhingra, figs 71.5 and 71.6

COBLATION COMPARED WITH ELECTROSURGERY

| FEATURE | COBLATION | ELECTROSURGERY (DIATHERMY) |
|---|---|---|
| Temperature | 40 to 70°C | 400 to 600°C |
| Mechanism | Chemical: plasma breaks molecular bonds | Thermal: heating, charring |
| Collateral damage | Superficial, minimal | Deeper |
| Field | Saline irrigated, suction built in | Dry |

# 3 CHAPTER THREE Cryosurgery

Destruction by freezing.

## 3.1 Principle

Cryosurgery uses liquid nitrogen, by spray or probe at down to −196°C, or nitrous oxide and CO2 probes, which rely on the *Joule–Thomson effect*: compressed gas expanding through a small orifice cools the tip to about −70°C or below. The tissue is frozen rapidly to −20 to −30°C or colder, *thawed slowly*, and the cycle repeated, with a margin of normal tissue and, if needed, a thermocouple to monitor temperature.

Cells die by several mechanisms: ice crystals and *dehydration* leave toxic concentrations of electrolytes; membrane lipoproteins are *denatured*; thermal shock damages the cells; and *vascular stasis* with microthrombosis causes ischaemic necrosis, which is why vascular lesions respond so well. There may also be an immune "cryoimmunisation" effect. The slough separates in three to six weeks and heals by secondary intention with little scarring.



**FIGURE 4** Cryotherapy for a haemangioma of the cheek.
Dhingra, fig. 72.1

## 3.2 Uses, advantages and limits

Cryotherapy treats haemangiomas and vascular malformations, premalignant lesions such as leukoplakia and actinic keratosis, and superficial skin cancers such as Bowen's disease and small basal cell carcinomas, with cure rates above 90%, including lesions over cartilage, which resists freezing. It also palliates advanced and recurrent tumours, reduces turbinates, and was formerly used for the tonsils of high-risk patients.

It can be done as an outpatient, often without anaesthesia, is safe in bleeding disorders and poor-risk patients, causes little pain, leaves minimal scarring and no keloid, and is cheap and repeatable. Its limits are that it provides no histology or margin control, its depth is imprecise, it leaves hypopigmentation and hair loss in skin, it causes swelling and slough, and it is unsuitable for recurrent or ill-defined cancers.

# 4 CHAPTER FOUR Hyperbaric oxygen

Pressure and oxygen.

## 4.1 Principle

Breathing 100% oxygen at more than one atmosphere absolute, usually 2.0 to 2.5 ATA for 90 to 120 minutes over a series of 10 to 40 sessions, increases the oxygen dissolved in plasma several-fold. This raises tissue oxygen tension, stimulates angiogenesis and fibroblast activity, improves leucocyte killing, reduces oedema and shrinks gas bubbles.




**FIGURE 5** A multiplace hyperbaric chamber.
Dhingra, fig. 71.4

## 4.2 ENT uses

For **osteoradionecrosis and soft-tissue radiation injury**, hyperbaric oxygen is an adjunct to debridement and reconstruction, but the evidence is mixed: Marx's older protocol supported giving it before dental extraction in irradiated jaws, whereas the randomised HOPON trial found no reduction in osteoradionecrosis, and pentoxifylline–tocopherol is an alternative. It is also used for refractory **necrotising**

**otitis externa** and skull base osteomyelitis, for compromised flaps and grafts, and for **sudden sensorineural hearing loss** as salvage or with steroids within two weeks of onset in moderate to profound loss. Outside ENT it treats decompression sickness, including inner ear decompression sickness in divers, air embolism, carbon monoxide poisoning and necrotising fasciitis.

### 4.3 Complications and contraindications

Middle ear barotrauma is the commonest complication and may need grommets; sinus barotrauma, oxygen toxicity seizures, temporary myopia and claustrophobia also occur. An untreated pneumothorax is an absolute contraindication. Eustachian tube dysfunction, upper respiratory infection, emphysema with bullae, uncontrolled epilepsy and certain chemotherapy drugs such as bleomycin and doxorubicin are relative contraindications.

## 5 CHAPTER FIVE Narrow band imaging

Seeing the vessels.

### 5.1 Principle and value

Narrow band imaging filters the endoscope's light into two bands, *415 nm (blue)* and *540 nm (green)*, which are absorbed by haemoglobin and penetrate only the superficial layers, so superficial capillaries appear brown and submucosal veins cyan. Abnormal *intraepithelial papillary capillary loops*, dilated and tortuous with brown dots, mark dysplasia and early cancer. This helps to detect small cancers under 1 cm and second primaries, to target biopsies, to define resection margins and to survey for recurrence after radiotherapy. Its accuracy is reduced by secretions, bleeding, keratosis and post-radiation change.

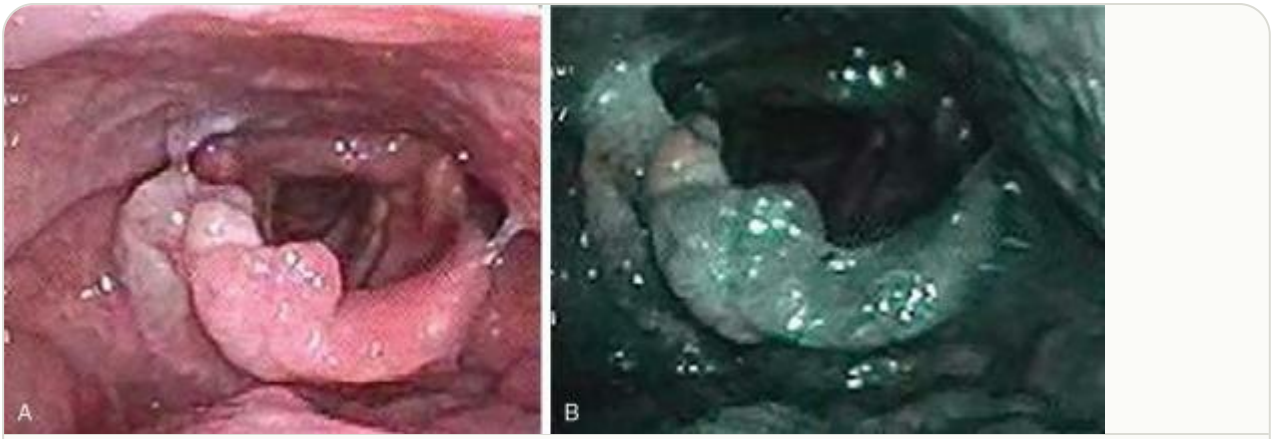



**FIGURE 6** Carcinoma of the epiglottis (A) under white light and (B) under narrow band imaging.
Dhingra, fig. 71.7

CHAPTER SIX

# 6 For the examination

What the examiner wants to hear.

## 6.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **What LASER stands for,** the properties of laser light, and its components.
- **Chromophores;** CO2 versus KTP versus Nd:YAG: wavelength, absorption, delivery, uses.
- **The three zones of a laser wound;** power density and fluence.
- **Laser safety:** eye protection by wavelength, laser-safe tubes and the blue-dyed saline cuff, low oxygen and no nitrous oxide, the airway fire drill.
- **Coblation versus diathermy;** radiofrequency for sleep surgery.
- **Mechanisms of cryotherapy;** freeze fast, thaw slowly.
- **Hyperbaric oxygen:** how it works, ENT indications, complications.

## 6.2 Common pitfalls

COMMON PITFALLS

- Using nitrous oxide or high oxygen during laser airway surgery.

- Giving the theatre team the wrong goggles for the wavelength.
- Treating an undiagnosed lesion with cryotherapy or laser vaporisation without taking a biopsy first.
- Forgetting ear barotrauma in patients starting hyperbaric oxygen.

---

Written for this app from Oxford Textbook of Otolaryngology ch. 71, Oxford Specialist Handbook ch. 8, Dhingra ch. 71, 72 and K. J. Lee ch. 3. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · LARYNX & AIRWAY

# The hoarse child and the child who cannot swallow

Children's voices and swallows fail for different reasons from adults', and they cannot always tell you what is wrong. A hoarse child is usually a shouting child with nodules, but the second commonest cause is recurrent respiratory papillomatosis, which can obstruct the airway. A child who coughs with feeds may have a neurological swallow, a laryngeal cleft, a tracheo-oesophageal fistula or a nose that is too narrow to breathe through while sucking. Both problems are assessed with the same tools, flexible endoscopy and instrumental swallow studies, and managed by the same partnership of ENT surgeon and speech and language therapist.

**AT A GLANCE**

- **Nodules** are the commonest cause of a hoarse child and respond to voice therapy; **RRP** (HPV 6 and 11) is second.
- **RRP:** debulking; cidofovir or bevacizumab for frequent recurrence; HPV vaccination.
- **Vocal fold paralysis** follows about 5% of aortic arch operations; injection or ansa–RLN reinnervation if persistent.
- **FEES and videofluoroscopy** show penetration (to the folds) and aspiration (below them), often silent in children.
- **Laryngeal clefts** (Benjamin–Inglis I–IV): palpate the interarytenoid area at every microlaryngoscopy.

## 1 CHAPTER ONE The developing larynx

Why children differ.

### 1.1 Anatomy with age

The infant larynx sits high, at about the level of C2 to C3, so the epiglottis can meet the soft palate and the infant can breathe while sucking. It descends through childhood to about C6 to C7 by fifteen. The infant larynx is about a third of adult size, the subglottis (bounded by the complete cricoid ring) is the narrowest point, and the vocal folds are short with an immature layered structure that matures through childhood and puberty. Babies breathe preferentially through the nose for

about the first six months, so nasal obstruction disrupts the coordinated suck, swallow and breathe cycle. Mature chewing and the ability to manage most textures develop by about two years, following motor development more than age.



**FIGURE 1** Laryngomalacia: an omega-shaped epiglottis, a reminder of how the infant larynx differs.

Dhingra, fig. 59.1

# 2 CHAPTER TWO The hoarse child

Voice disorders.

## 2.1 Causes

Dysphonia affects between 4 and 23% of children at some stage, and most causes are benign and self-limiting.

CAUSES OF DYSPHONIA IN CHILDREN

| GROUP | CAUSES | POINTS |
|---|---|---|
| Behavioural | **Vocal fold nodules** | Commonest cause; boys, loud voice use; voice therapy works well |

| GROUP | CAUSES | POINTS |
|---|---|---|
| Infective | **Recurrent respiratory papillomatosis** | Second commonest; HPV 6 and 11; may threaten the airway |
| Structural | Cysts, sulcus, polyps | Stroboscopy shows a disturbed mucosal wave |
| Congenital | Webs, laryngeal cleft, subglottic stenosis, cysts | From birth; often with stridor or feeding problems |
| Neurological | Vocal fold paralysis | Often after cardiac surgery; Arnold–Chiari malformation if bilateral |
| Inflammatory | Reflux, eosinophilic oesophagitis | Arytenoid oedema and erythema |
| Iatrogenic | Intubation injury, webs after repeated surgery | Granuloma, scar, posterior glottic stenosis |
| Functional | Puberphonia, functional dysphonia | Normal larynx, abnormal use |

## 2.2 Assessment

Listen to the voice for pitch, range, projection, breaks and resonance, and ask about voice use, noisy breathing, feeding, cardiac surgery and intubation. Look for syndromic features. A paediatric flexible nasendoscope (as small as 2.1 mm) shows mobility and gross lesions in most children; rigid stroboscopy is tolerated by mature children from about three or four years and shows the mucosal wave. **Laryngeal ultrasound** is a useful, non-invasive way to assess vocal fold movement in uncooperative children. A paralysed fold needs imaging of the nerve's course, and a contrast swallow or oesophagoscopy is considered when reflux or eosinophilic oesophagitis is suspected.

## 2.3 Vocal fold nodules

Nodules are bilateral swellings at the junction of the anterior and middle thirds, more common in boys, and caused by voice overuse. They respond well to **voice therapy**, often regress at puberty in boys, and are almost never operated on in children.

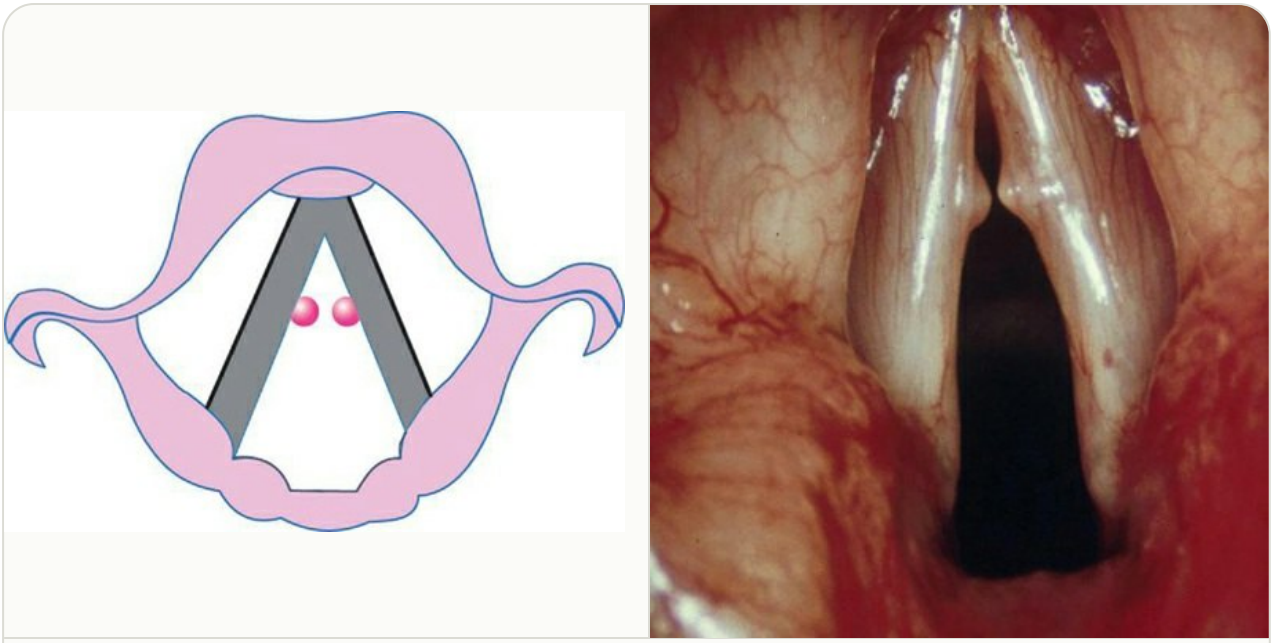


**FIGURE 2** The site of vocal nodules, and nodules on both folds.
Dhingra, figs 61.1 and 61.2

## 2.4 Recurrent respiratory papillomatosis

RRP is caused by HPV types 6 and 11, usually acquired at birth from the mother's genital tract; juvenile-onset disease typically appears before five years with progressive hoarseness and later stridor, and can be mistaken for asthma or croup. Treatment is repeated **microdebrider or cold-steel debulking**, preserving normal tissue and avoiding the anterior commissure to prevent webs; KTP laser is an alternative. Tracheostomy is avoided where possible because it may spread disease distally. Adjuvant treatments for aggressive disease (more than about four procedures a year) include intralesional **cidofovir**, whose small cancer risk must be discussed with parents, and systemic **bevacizumab**, which has given striking responses. **HPV vaccination** prevents the disease in the next generation and may reduce recurrences.

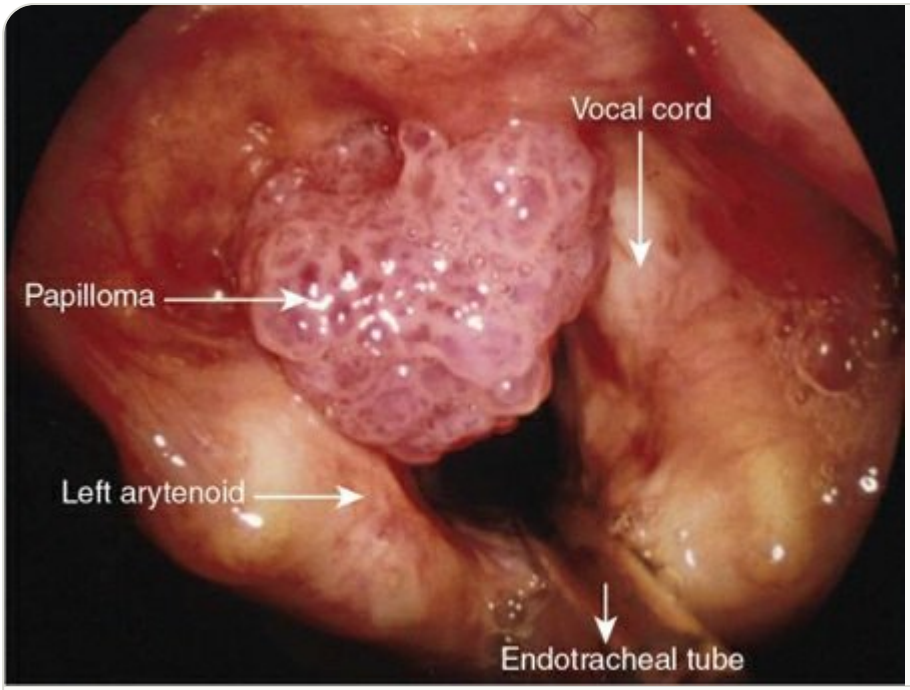



**FIGURE 3** Laryngeal papillomatosis.
Dhingra, fig. 61.11

## 2.5 Vocal fold paralysis

Paralysis is the second commonest congenital laryngeal anomaly. Unilateral paralysis, often left-sided after cardiac or aortic arch surgery (about 5% of such operations), causes a weak cry, feeding problems and aspiration; children having arch surgery should be screened before discharge. Bilateral paralysis causes stridor with a good cry and is associated with Arnold–Chiari malformation and hydrocephalus. Many idiopathic palsies recover, so observation is appropriate when breathing and feeding are safe. Persistent unilateral palsy is treated with **injection medialisation** (temporary materials such as Gelfoam for about six weeks or hyaluronic acid for months, and longer-lasting calcium hydroxylapatite or fat), which in children needs a general anaesthetic and overnight oximetry, or later with thyroplasty or **ansa cervicalis to recurrent laryngeal nerve reinnervation**, which gives good results in children.

## 2.6 Webs and other lesions

Anterior glottic webs may be congenital, often with a subglottic extension and associated with 22q11.2 deletion, or acquired after repeated microlaryngoscopy. Thin webs are divided endoscopically with a keel or mitomycin; thick congenital

webs may need an open repair. Phonosurgery in children follows adult principles but is more conservative: cold instruments, no laser on the true folds, and subepithelial infusion to protect the cover.

# 3 CHAPTER THREE The child who cannot swallow safely

Dysphagia.

## 3.1 Assessment

Swallowing problems affect about 1% of children and many more in clinical groups such as prematurity, neurodisability and cardiac disease. A feeding history (breast or bottle, textures, duration of meals, coughing, colour change, vomiting, chest infections, weight gain) and observation of a meal by a speech and language therapist come first. Instrumental tests are **videofluoroscopy**, which shows every phase including oesophageal transit, and **FEES**, which can be done at the bedside or during breastfeeding, shows secretions and laryngeal sensation, and avoids radiation; the two are comparable for the pharyngeal phase. They show *penetration* (material in the laryngeal vestibule down to the vocal folds) and *aspiration* (material below the folds), which in children is often silent.

## 3.2 Causes

CAUSES OF PAEDIATRIC DYSPHAGIA

| LEVEL | CAUSES |
|---|---|
| Nose and nasopharynx | Choanal atresia, piriform aperture stenosis, mid-nasal stenosis, craniosynostosis, adenoids |
| Mouth and oropharynx | Cleft palate, Pierre Robin sequence, macroglossia, tongue-tie (rarely), tonsillar hypertrophy |
| Larynx and pharynx | Laryngomalacia, vocal fold paralysis, laryngeal cleft, cricopharyngeal dysfunction |
| Oesophagus and trachea | Oesophageal atresia and tracheo-oesophageal fistula, vascular rings, strictures, eosinophilic oesophagitis |

| LEVEL | CAUSES |
|---|---|
| Neurological | Cerebral palsy, hypoxic brain injury, Chiari malformation, neuromuscular disease |
| Respiratory and cardiac | Chronic lung disease of prematurity, congenital heart disease (breathlessness, fatigue) |

## 3.3 Laryngeal cleft

A posterior laryngeal cleft is a failure of fusion between the larynx and oesophagus. In the **Benjamin–Inglis** classification, *type I* is confined to the interarytenoid area above the vocal folds; *type II* extends into the cricoid lamina; *type III* extends through the cricoid into the cervical trachea; and *type IV* extends into the thoracic trachea. Type I clefts present with coughing on feeds, chest infections and a weak or hoarse cry, and are easily missed unless the interarytenoid area is **palpated** at microlaryngoscopy. Thickened feeds help many; persistent aspiration is treated by **injection augmentation** of the interarytenoid area or endoscopic repair. Deeper clefts present earlier and more severely and need endoscopic or open repair, with tracheostomy and gastrostomy for type III and IV.

## 3.4 Other important causes

Oesophageal atresia with tracheo-oesophageal fistula presents at birth with drooling, choking and inability to pass a nasogastric tube, or later with an isolated H-type fistula and coughing on feeds. **Vascular rings** compress the trachea and oesophagus, causing stridor and dysphagia for solids. **Eosinophilic oesophagitis** causes food refusal, vomiting and food impaction in older children. **Tongue-tie** can interfere with breastfeeding and is divided by simple frenotomy when it clearly does, but it is not a common cause of feeding failure.

## 3.5 Management

Management is multidisciplinary: positioning, pacing, bottle teats and texture modification by the speech and language therapist, dietetic support, treatment of reflux, and non-oral feeding by nasogastric tube or **gastrostomy** when the swallow

is unsafe or intake inadequate, sometimes with fundoplication. The structural cause is corrected where possible.

### THE CHILD WITH A VOICE OR SWALLOWING PROBLEM

1. **Airway first**
   - → **stridor or distress** — airway assessment and microlaryngoscopy and bronchoscopy
   - → **safe** — step 2
2. **Voice**
   - → flexible endoscopy or ultrasound
   - → **nodules** — voice therapy
   - → **papillomas** — debulking, adjuvant therapy if frequent
   - → **paralysis** — image the nerve course, observe, inject or reinnervate
3. **Swallow**
   - → feeding history and observed meal
   - → videofluoroscopy or FEES
   - → **aspiration** — thickened feeds, non-oral feeding if unsafe

4 **Look for structure**

- → interarytenoid palpation for cleft
- → contrast study for fistula, ring or stricture
- → endoscopy and biopsy for eosinophilic oesophagitis
- → nasal endoscopy for nasal causes

# 4 CHAPTER FOUR For the examination

What the examiner wants to hear.

## 4.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Why the child larynx differs:** position, size, subglottis.
- **Nodules and papillomatosis:** the two commonest causes of a hoarse child.
- **RRP adjuvants:** cidofovir and bevacizumab; HPV vaccination.
- **Vocal fold paralysis after cardiac surgery;** injection medialisation and reinnervation in children.
- **Penetration versus aspiration;** FEES versus videofluoroscopy.
- **Benjamin–Inglis classification** of laryngeal clefts; palpate the interarytenoid area.

## 4.2 Common pitfalls

COMMON PITFALLS

- Treating papillomatosis as asthma or croup.
- Missing a type I laryngeal cleft by not palpating the interarytenoid area.
- Operating on vocal fold nodules in a child.

- Lasering the anterior commissure and creating a web.

---

Written for this app from Oxford Textbook of Otolaryngology ch. 79, 80, Oxford Specialist Handbook ch. 23, Oxford Handbook of ENT ch. 17. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

CLINICAL · LARYNX & AIRWAY

# Anaesthesia and the shared airway

In ENT surgery the surgeon and the anaesthetist share the airway, and many of the most dangerous moments in anaesthesia happen in our patients. The UK's national audit of major airway complications (NAP4) found that about two in five arose in disease of the head, neck or trachea, and the "can't intubate, can't oxygenate" scenario accounts for about a quarter of anaesthesia-related deaths. An ENT surgeon does not need to give anaesthetics, but must plan the airway with the anaesthetist, understand the options for sharing it, recognise when the surgeon must provide it, and know the routine pitfalls: the bleeding tonsil, the laser fire, the retained throat pack and the patient who cannot be reintubated.

AT A GLANCE

- **NAP4:** about two in five major airway complications arose in head, neck or tracheal disease.
- **Obstructed airway:** agree a primary and backup plan; awake fibreoptic intubation or awake tracheostomy; surgeon scrubbed before induction.
- **Sharing the airway:** microlaryngeal or laser tubes, jet ventilation (keep an outflow path), tubeless spontaneous ventilation, high-flow nasal oxygen.
- **Laser safety:** laser tube with saline cuff, oxygen under about 30%, no nitrous oxide; know the airway fire drill.
- **Bleeding tonsil:** resuscitate, rapid sequence induction by a senior anaesthetist, empty the stomach, extubate awake on the side.

## 1 CHAPTER ONE Planning together

Before the patient arrives.

### 1.1 Preoperative assessment

Preparation is part of **perioperative medicine**: stopping smoking even for a few weeks improves respiratory function and reduces complications; malnourished patients (weight loss over 10% or a BMI under 18.5) benefit from nutritional support before major head and neck surgery; anaemia, diabetes and cardiac disease are optimised; and treatment escalation plans (such as ReSPECT) are discussed. For any patient with airway pathology, the surgeon and anaesthetist review the history,

flexible endoscopy findings and imaging together and agree a **primary plan with backup plans** before induction, including who will provide a surgical airway and where the cricothyroid membrane is (marked beforehand, with ultrasound if the neck is difficult).

### 1.2 Children and the setting of care

Most paediatric ENT surgery, including tonsillectomy, is safely done in district hospitals for children from about two years (younger in high-acuity centres), with tertiary referral for children with major comorbidity, syndromes, severe obstructive sleep apnoea at a young age or extremes of weight. A reinforced laryngeal mask is suitable for most tonsillectomies in children over about three, and dexamethasone (about 0.15 mg/kg) reduces nausea and vomiting after tonsillectomy.

## 2 CHAPTER TWO The airway options

Who controls what.

### 2.1 The obstructed airway

There is no single proven method for a patient with upper airway obstruction. Options are **awake fibreoptic intubation** under topical anaesthesia (often with high-flow nasal oxygen and light sedation), **inhalational induction** keeping spontaneous breathing, **intravenous induction** with paralysis when intubation is judged straightforward, and **awake tracheostomy under local anaesthesia** when the larynx is likely to be impassable, which every head and neck surgeon must be able to perform. The surgeon is scrubbed, with the neck prepared, before induction in any patient at risk.

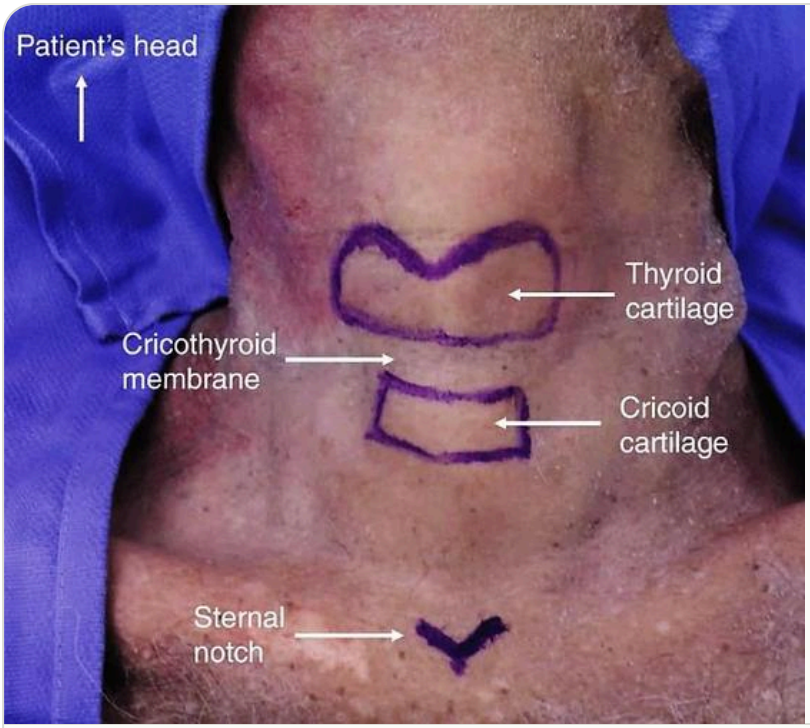



**FIGURE 1** Surface marking of the thyroid and cricoid cartilages and the cricothyroid membrane: mark it before induction.

Anatomy atlas · ENT Steps p. 262

## 2.2 Sharing the airway in endoscopic surgery

For microlaryngoscopy and pharyngeal surgery the options are:

### WAYS OF SHARING THE AIRWAY

| TECHNIQUE | HOW IT WORKS | POINTS |
|---|---|---|
| Microlaryngeal tube | Small cuffed tube (5.0 to 6.0 mm) | Secure airway; obscures the posterior glottis |
| Laser-resistant tube | Metal or wrapped tube, saline-filled cuff | Required for laser near the airway |
| Jet ventilation (supraglottic or transglottic) | High-pressure oxygen through a needle in the laryngoscope or a fine catheter through the glottis | Unobstructed view; risk of barotrauma and pneumothorax if outflow is blocked |
| Spontaneous ventilation (tubeless) | Total intravenous anaesthesia, topical anaesthesia, patient breathes | Good for dynamic airway assessment in children |

| TECHNIQUE | HOW IT WORKS | POINTS |
|---|---|---|
| High-flow nasal oxygen (apnoeic) | Humidified oxygen at up to about 70 L/min | Extends safe apnoea time; best in fit patients for short procedures |

Jet ventilation needs a clear expiratory path, because gas trapped below an obstruction causes barotrauma; laser-resistant transglottic catheters such as the Hunsaker Mon-Jet allow jetting through a tight lesion with care.

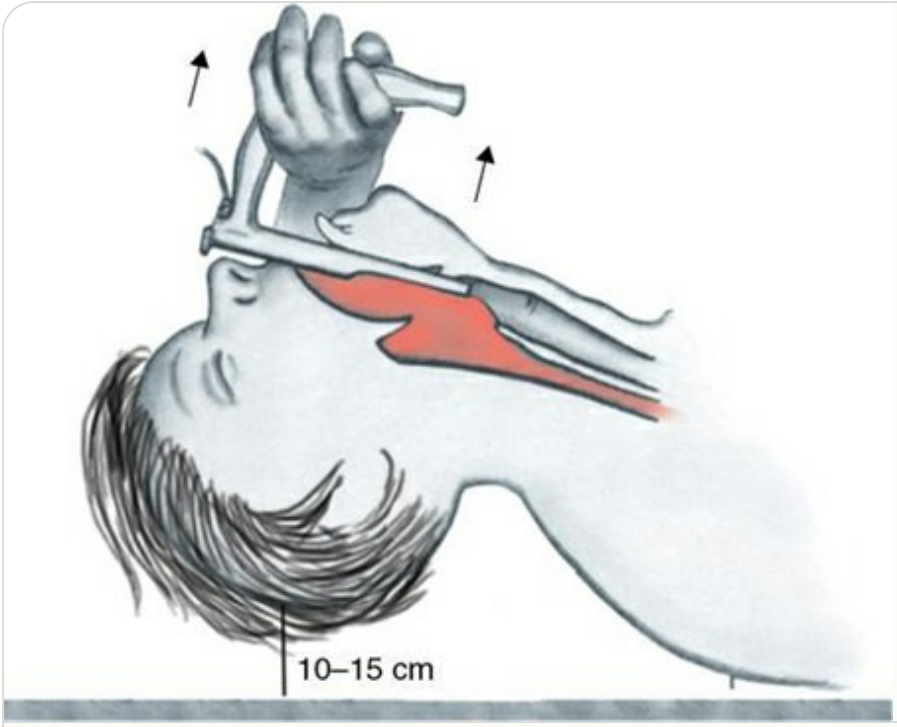



**FIGURE 2** Direct laryngoscopy in the sniffing position.
Dhingra, fig. 91.1

## 2.3 Laser safety

Airway fire needs fuel (the tube or swabs), oxygen and a spark (the laser). Prevention is a laser-safe tube with saline in the cuff (ideally coloured, so a breach is obvious), the lowest inspired oxygen possible (under about 30%), no nitrous oxide, wet swabs protecting tissues and the cuff, eye protection for everyone and signs on the doors. If a fire starts, stop the laser and ventilation, remove the tube, pour saline into the airway, ventilate with air by mask, then reintubate and inspect the airway with a bronchoscope.

# 3 CHAPTER THREE Common scenarios

Know the plan.

## 3.1 The bleeding tonsil

A child or adult who returns with a post-tonsillectomy bleed is hypovolaemic (often more than they look, especially children), has a stomach full of swallowed blood and an airway filling with blood. Resuscitate first with cannulas, fluids and blood, and correct clotting. In theatre, a senior anaesthetist uses a **rapid sequence induction** in the head-down, lateral or sitting position with two suctions ready and a smaller tube available; the surgeon is scrubbed. After haemostasis, the stomach is emptied with a wide-bore tube and the patient is extubated awake on the side.

## 3.2 Nasal and sinus surgery

Topical vasoconstriction gives a drier field: cocaine 4% (the often-quoted 200 mg limit has no firm scientific basis), Moffett's solution or adrenaline-soaked pledgets with co-phenylcaine. **Total intravenous anaesthesia** with propofol and remifentanil, a slightly head-up position and avoidance of tachycardia reduce bleeding, and a second-generation laryngeal mask with spontaneous breathing gives good airway protection and a smooth emergence.

## 3.3 Throat packs and extubation

A retained throat pack is a "never event". The joint Difficult Airway Society, ENT UK and maxillofacial review found little evidence for routine throat packs, so many anaesthetists no longer use them; if one is needed, it is part of the surgical swab count. After intraoral, pharyngeal or airway surgery the patient may have a patent airway but be very difficult to reintubate, so **extubation is planned** like intubation: awake, with suction, sometimes over an airway exchange catheter, or with a tracheostomy left in place when swelling is expected.

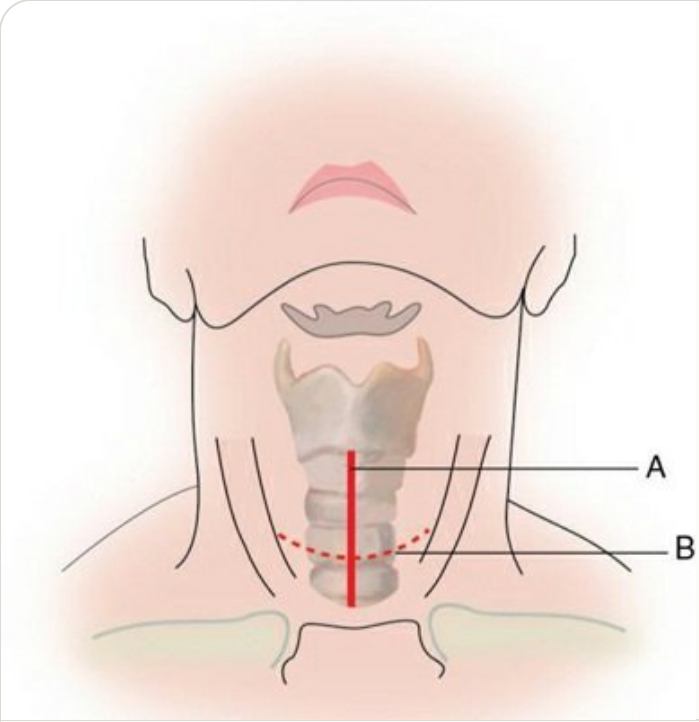



**FIGURE 3** Skin incisions for tracheostomy: vertical and transverse.
Dhingra, fig. 64.1

### 3.4 Environmental impact

Desflurane and nitrous oxide are potent greenhouse gases (desflurane accounts for most of the climate impact of volatile agents), so total intravenous anaesthesia and low-flow techniques are increasingly preferred.

# 4 CHAPTER FOUR For the examination

What the examiner wants to hear.

### 4.1 Viva essentials

**WHAT THE EXAMINER IS LOOKING FOR**

- **NAP4** and why ENT patients are at airway risk.
- **The obstructed airway:** awake fibreoptic intubation, inhalational induction and awake tracheostomy; front-of-neck access.

- **Ways of sharing the airway:** microlaryngeal tubes, jet ventilation, tubeless techniques and high-flow nasal oxygen.
- **Laser safety** and the management of an airway fire.
- **The bleeding tonsil:** resuscitation and rapid sequence induction.
- **Throat packs** and planned extubation.

## 4.2 Common pitfalls

COMMON PITFALLS

- Inducing anaesthesia in a patient with an obstructing tumour before the surgeon is scrubbed and the plan agreed.
- Using jet ventilation without a clear route for expiration.
- Running high oxygen concentrations during laser surgery.
- Underestimating blood loss in a child with a bleeding tonsil.

Written for this app from Oxford Textbook of Otolaryngology ch. 3, Oxford Specialist Handbook ch. 5, Oxford Handbook of ENT ch. 21. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

# Imaging for this region

*The core imaging conditions of this region, with their key facts and a representative scan. For the normal anatomy and a search pattern, read Lesson 11, The larynx and hypopharynx in the imaging volume.*

LARYNX & TRACHEA · TUMOURS

## Supraglottic Tumor

### IMAGING FINDINGS

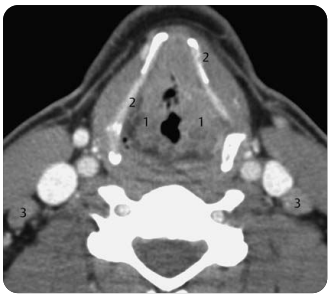
- Mass on the epiglottis, aryepiglottic folds or false cords
- Invasion of pre-epiglottic or paraglottic fat (T3)
- Nodes common, often bilateral

### TOP DIFFERENTIALS

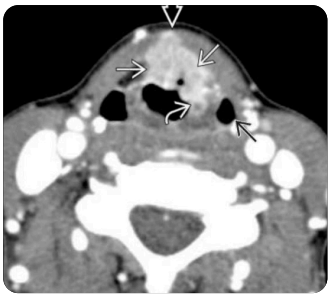
- Laryngocele
- Benign mucosal swelling
- Minor salivary tumour

### CHECKLIST

- Vallecula and tongue base; pyriform sinus
- Cartilage and extralaryngeal spread




CT with intravenous contrast enhancement, axial. The lesion extends bilaterally (1), highly indicative of a malignancy. The…



Axial CECT shows similar findings with invasion of bilateral paraglottic fat and left AE fold. Tumor bulges through thyroid cartilage notch. Pyriform sinu…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 526.

LARYNX & TRACHEA · TUMOURS

# Glottic Tumor

## IMAGING FINDINGS

- Thickening or mass of the true cord
- Anterior commissure thicker than 1 mm
- Paraglottic and subglottic spread

## TOP DIFFERENTIALS

- Benign cord lesion
- Chondrosarcoma (cricoid, chondroid matrix)
- Cord paralysis (medial arytenoid)

## CHECKLIST

- Cord movement on breath-hold images
- Thyroid cartilage erosion: inner cortex T3, through it T4a




CT with intravenous contrast enhancement, axial. Slight asymmetry of the vocal cords can be observed with slight enlargement of the vocal…



Axial CECT shows an enhancing right TVC exophytic mass. Anterior and posterior commissures are normal. Right arytenoid cartilage sclerosis is…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 530.

LARYNX & TRACHEA · TRAUMA

# Laryngeal Trauma

## IMAGING FINDINGS

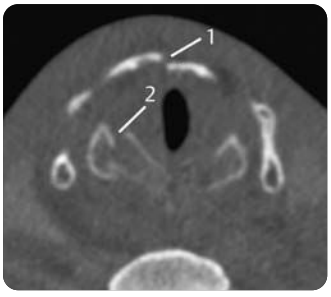
- Thyroid and cricoid cartilage fractures with displacement
- Arytenoid dislocation
- Haematoma and air narrowing the airway

## TOP DIFFERENTIALS

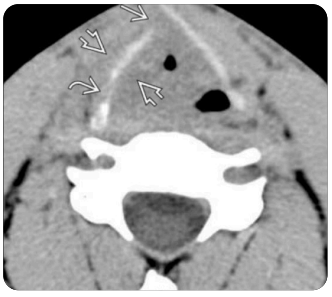
- Cord paralysis
- Unossified cartilage mimicking a fracture

## CHECKLIST

- Secure airway before scanning
- Cricoid fractures and displaced fragments usually need repair




CT, axial bone window. CT at the level of the glottis showed multiple fractures; this figure shows a midline fracture with slight inward…



Axial NECT through the supraglottic larynx shows step-off at the superior thyroid notch. There is a subtle fracture of the right thyroid lamina…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 278.

LARYNX & TRACHEA · INFECTION & INFLAMMATION

# Laryngocele

## IMAGING FINDINGS

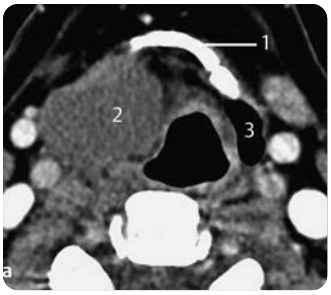
- Air or fluid-filled sac from the laryngeal ventricle
- Internal (paraglottic) or external through the thyrohyoid membrane
- Rim enhancement if infected (pyolaryngocele)

## TOP DIFFERENTIALS

- Thyroglossal duct cyst
- Second branchial cleft cyst
- Saccular cyst

## CHECKLIST

- Exclude a carcinoma at the ventricle in adults
- Airway compromise




CT with intravenous contrast enhancement, axial. The axial slice at the lower border of the hyoid (1) reveals a homogeneous, expanding…



Coronal CECT reformat shows a similar lesion. This is also known as a "mixed" laryngocele because it contains internal…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 292.

LARYNX & TRACHEA · INFECTION & INFLAMMATION

# Paralysis of the Recurrent Laryngeal Nerve

## IMAGING FINDINGS

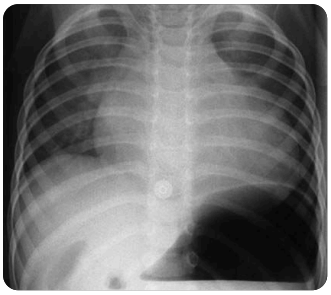
- Medially rotated arytenoid and dilated ventricle
- Dilated pyriform sinus; medial, thickened aryepiglottic fold
- Atrophy of the posterior cricoarytenoid and thyroarytenoid

## TOP DIFFERENTIALS

- Arytenoid dislocation after trauma
- Glottic carcinoma
- Laryngocele

## CHECKLIST

- Scan from skull base to the aortopulmonary window on the left
- Thyroid, oesophagus and mediastinal nodes



A 7-year-old child with hoarseness and left vocal cord paralysis. Chest radiograph, posteroanterior view. The grossly dilated heart is compressing…



CECT in a patient with new-onset hoarseness shows slightly medially rotated left arytenoid cartilage & a more prominent dilated left laryngeal…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 296.